
Noncanonical Equilibrium and Finite-Bath Corrections in Quantum Thermodynamics

In our previous discussion, I was given a brief introduction to the main research problem. In the last paper we studied, it was assumed that a small system weakly coupled to a large heat bath is described at equilibrium by a canonical Gibbs state with a well-defined temperature. However, this picture essentially relies on the idealization of an infinite bath. When the system-bath coupling is non-negligible and the bath has finite size, this description breaks down, and we need to reconsider what *equilibrium* means for our system.

In this report, two key papers will be reviewed: *Noncanonical statistics of a finite quantum system with non-negligible system-bath coupling* by Xu and Sun, and *Finite-bath corrections to the second law of thermodynamics* by Richens, Alhambra, and Masanes. The first paper examines how system-bath interactions deform the equilibrium state away from the standard Gibbs form, while the second explores how finite baths modify the second law and the free-energy bounds on work extraction. After understanding these results, the goal is to identify conditions under which interacting quantum subsystems can be in global equilibrium yet exhibit different effective temperatures, and to see how these effective temperatures depend on the underlying interaction parameters.

1 Noncanonical statistics of a finite quantum system with non-negligible system-bath coupling

1.1 Introduction

Statistical mechanics describes the average properties of a system without referring to all its microscopic states. In most situations, the validity of the canonical statistical description is guaranteed in the thermodynamic limit, which requires that, while the degrees of freedom of the heat bath is infinite, the system-bath coupling approaches infinitesimal. However, if the system only interacts with a small heat bath with finite degrees of freedom, the system-bath interaction cannot be ignored compared with the energy of the heat bath.

Consider a closed universe consisting of a small system S and a large bath B , described by a pure state $|\Psi\rangle$ constrained to a narrow energy shell. Results on canonical typicality show that for almost all such pure states, the reduced state $\rho_S = \text{Tr}_B(|\Psi\rangle\langle\Psi|)$ is extremely close to the canonical Gibbs state $e^{-\beta H_S}/Z$. In other words, thermal behavior of the subsystem does not require a mixed global ensemble; it arises generically from typical pure states of the total system. This is proven using Levys lemma (see [Popescu, Short, and Winter, Section IV and V](#)), which guarantees that local observables take values very close to their microcanonical averages for the overwhelming majority of pure global states. Therefore, for almost all pure

states of the universe, the reduced density matrix of the small system is (approximately) canonical.

Nevertheless, the conventional canonical statistics still cannot describe well the thermodynamics behavior of the finite system when the sufficiently large system-bath interaction is taken into consideration. In this paper, an effective system-bath coupling is generally considered by assuming the bath possesses a much more dense spectrum than that of the system, and then the system-bath interaction energy can be treated as the deformation of the energy shell for the total system. In this case, the system-bath interaction simply deforms the microcanonical energy shell of the total system, leading to a modified state that counts $\Omega_B(E - E_n - \Delta_n)$ for each system level and, consequently, to noncanonical equilibrium statistics for the system. This modified distribution obviously implies that corrections are necessary for the finite-system thermodynamic quantities in canonical statistics, such as average internal energy and its fluctuation.

The structure of this section is as follows: in Sec. 1.2, we outline how Xu and Sun derive an effective Hamiltonian for the total system using perturbation theory, and how this leads to a noncanonical statistical distribution that does not rely on any particular microscopic model. In Sec. 1.3, we then discuss their concrete example of coupled harmonic oscillators, where they use this framework to compute thermodynamic quantities such as internal energy, energy fluctuations, and the mutual information between two subsystems. Finally, Sec. 1.4 briefly summarizes the main results and their implications.

1.2 Finite system-bath coupling

In this paper, a composite coupled system is considered, consisting of a system S with Hamiltonian H_S and a heat bath B with Hamiltonian H_B . The interaction between the system and the bath is described by H_I , so that the total Hamiltonian is

$$H = H_S + H_B + H_I.$$

The heat bath is composed of N non-interacting particles. The system and bath Hamiltonians have the following spectral decompositions:

$$H_S = \sum_n E_n |n\rangle \langle n| \tag{1}$$

$$H_B = \sum_{j=1}^N \sum_{k_j} \epsilon_{k_j} |k_j\rangle \langle k_j| \otimes \prod_{i \neq j} I_i. \tag{2}$$

Here, $|n\rangle$ is the eigenstate of the system with corresponding eigenvalue E_n . For the bath, $|k_j\rangle$ denotes the eigenstate of the j -th particle with eigenenergy ϵ_{k_j} , and I_i is the identity

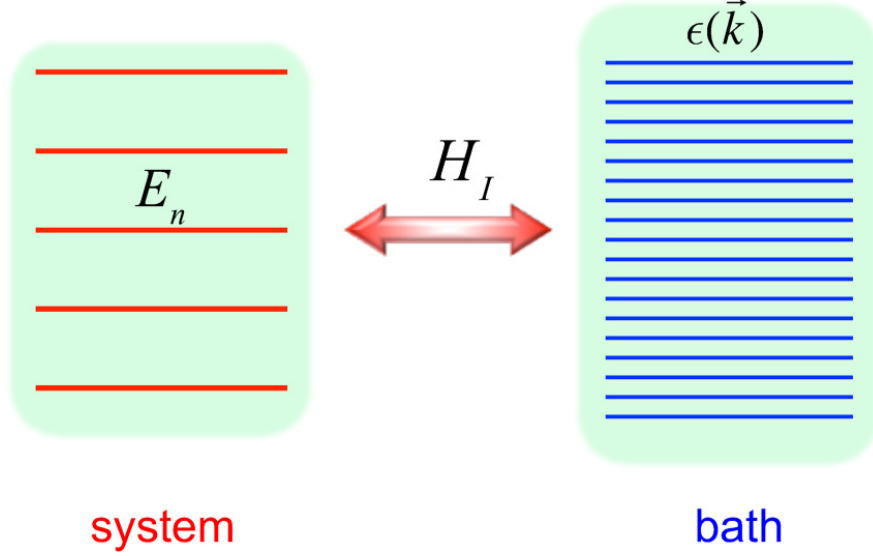


Figure 1: The energy spectra of the system and the heat bath.

operator on the Hilbert space of the i -th particle. For convenience, we introduce the notation

$$|\vec{k}\rangle \equiv \prod_{j=1}^N |k_j\rangle, \quad \vec{k} = (k_1, k_2, \dots, k_N),$$

to denote the eigenstates of H_B , and define their corresponding eigenenergies as

$$\epsilon(\vec{k}) = \sum_{j=1}^N \epsilon_{k_j}.$$

With these definitions, Eq. (2) can be rewritten in the usual compact form

$$H_B = \sum_{\vec{k}} \epsilon(\vec{k}) |\vec{k}\rangle \langle \vec{k}|.$$

Usually, the energy spectrum of the system is much sparser than the heat bath, i.e.,

$$\min |E_n - E_m| \gg \max |\epsilon(\vec{k}) - \epsilon(\vec{l})|, \quad (3)$$

where $\min |E_n - E_m|$ denotes the minimal spacing between neighboring energy levels of the system, and $\max |\epsilon(\vec{k}) - \epsilon(\vec{l})|$ is the maximal spacing between neighboring energy levels of the bath. This is illustrated in Fig. 1, taken from the original paper.

We assume that the interaction between the system and the bath can be written as a sum of terms, each coupling the system to a single bath particle at a time:

$$H_I = \sum_{j=1}^N V_j,$$

where each V_j acts nontrivially only on $\mathcal{H}_S \otimes \mathcal{H}_j$ and as the identity on all other bath degrees of freedom. Any operator on $\mathbb{H}_S \otimes \mathbb{H}_j$ can be expanded in the product basis $\{|n\rangle \otimes |k_j\rangle\}$.

$$V_j = \sum_{n,n',k_j,k'_j} g_{nk_j,n'k'_j} |n, k_j\rangle \langle n', k'_j|,$$

with $|n, k_j\rangle \equiv |n\rangle \otimes |k_j\rangle$ and $g_{nk_j,n'k'_j} = \langle n, k_j | V_j | n', k'_j \rangle$. Then embed with identities on the rest and we have

$$H_I = \sum_{j=1}^N \sum_{n,n',k_j,k'_j} g_{nk_j,n'k'_j} |n, k_j\rangle \langle n', k'_j| \otimes \prod_{i \neq j}^N I_i. \quad (4)$$

The system-bath interaction H_I is weak compared to H_0 , with

$$H_0 = H_S + H_B.$$

To obtain the canonical density matrix ρ_S of the system, we start from the microcanonical ensemble of the composite system (system+bath), neglecting H_I . The microcanonical state is

$$\rho_{\text{mic}} = \frac{P_{V(E,\delta)}}{\dim V(E,\delta)}, \quad (5)$$

where $V(E,\delta)$ is the subspace of $H_S + H_B$ associated with the energy shell $[E, E + \delta]$ (with $\delta \ll E$),

$$V(E,\delta) = \text{Span}\{|n, \vec{k}\rangle : E \leq E_n + \epsilon(\vec{k}) \leq E + \delta\}, \quad (6)$$

and the corresponding projector onto $V(E,\delta)$ has the explicit form

$$P_{V(E,\delta)} = \sum_{\substack{n, \vec{k} \\ E \leq E_n + \epsilon(\vec{k}) \leq E + \delta}} |n, \vec{k}\rangle \langle n, \vec{k}|. \quad (7)$$

The reduced state of the system is then

$$\rho_S = \text{Tr}_B (\rho_{\text{mic}}), \quad (8)$$

and the probability of finding the system in level $|n\rangle$ is

$$P_n(E) = \langle n | \rho_S | n \rangle = \frac{\Omega(E - E_n, \delta)}{\dim V(E, \delta)}, \quad (9)$$

where $\Omega(E - E_n, \delta) = \dim V_n(E - E_n, \delta)$ and

$$V_n(E - E_n, \delta) = \text{Span}\{|n, \vec{k}\rangle : E - E_n \leq \epsilon(\vec{k}) \leq E - E_n + \delta\}, \quad (10)$$

represents the subspace of $H_S + H_B$ associated with the subenergy shell $[E - E_n, E - E_n + \delta]$ when the system state is fixed at $|n\rangle$. Clearly,

$$V(E, \delta) = \sum_n \bigoplus V_n(E - E_n, \delta).$$

When the bath is finite and the system-bath interaction is small but not negligible, the reduced state of the system can be obtained by treating H_I as a perturbation to H_0 , i.e. by writing $H = H_0 + H_I$. In the basis $\{|n, k_j\rangle\}$, the off-diagonal matrix elements of H_I are $\langle n, k_j | H_I | n', k'_j \rangle = g_{nk_j, n'k'_j}$ with $n \neq n'$. From time-independent perturbation theory, the first-order correction to the eigenstate $|n, k_j\rangle$ contains contributions

$$c_{(n'k'_j)(nk_j)} = \frac{\langle n', k'_j | H_I | n, k_j \rangle}{(E_n + \epsilon_{k_j}) - (E_{n'} + \epsilon_{k'_j})} = \frac{g_{n'k'_j, nk_j}}{E_n - E_{n'} + \epsilon_{k_j} - \epsilon_{k'_j}}, \quad (11)$$

so that

$$|c_{(n'k'_j)(nk_j)}| = \left| \frac{g_{nk_j, n'k'_j}}{E_{n'} - E_n + \epsilon_{k'_j} - \epsilon_{k_j}} \right|. \quad (12)$$

The first-order effect of these off-diagonal terms can be neglected provided

$$\left| \frac{g_{nk_j, n'k'_j}}{E_{n'} - E_n + \epsilon_{k'_j} - \epsilon_{k_j}} \right| \ll 1, \quad (13)$$

which is satisfied for $n \neq n'$ because the system level spacings $|E_{n'} - E_n|$ are large (Eq. (3)). However, for $n = n'$ the denominator reduces to $\epsilon_{k'_j} - \epsilon_{k_j}$. Since the bath has a very dense spectrum, the difference $|\epsilon_{k'_j} - \epsilon_{k_j}|$ can be extremely small. In this case the ratio

$$\left| \frac{g_{nk_j, nk'_j}}{\epsilon_{k'_j} - \epsilon_{k_j}} \right|$$

is not necessarily $\ll 1$, and the diagonal part of H_I cannot be ignored. Retaining only these diagonal terms in n leads to the approximate form

$$H_I \approx \sum_{j=1}^N \sum_{n, k_j, k'_j} g_{nk_j, nk'_j} |n, k_j\rangle \langle n, k'_j| \otimes \prod_{i \neq j} I_i, \quad (14)$$

which is the starting point for constructing the effective Hamiltonian used in the noncanonical statistics.

We now express the effective Hamiltonian in the system eigenbasis $\{|n\rangle\}$, since we are primarily interested in how the bath shifts the system energy levels. The bath influences the system through these level shifts, so it is sufficient to track how they modify the system states, rather than the individual bath states. For each bath particle j , the Hamiltonian H_B

acts only on the bath part of the systembath state and leaves the system part unchanged. In the basis $\{|n\rangle\}$ we can therefore write H_B as

$$\begin{aligned} H_B &= \mathbb{I}_S \otimes \left(\sum_{j=1}^N \sum_{k_j} \epsilon_{k_j} |k_j\rangle\langle k_j| \otimes \prod_{i \neq j} I_i \right) \\ &= \sum_n |n\rangle\langle n| \otimes \left(\sum_{j=1}^N \sum_{k_j} \epsilon_{k_j} |k_j\rangle\langle k_j| \otimes \prod_{i \neq j} I_i \right). \end{aligned} \quad (15)$$

Moreover,

$$|n, k_j\rangle = |n\rangle \otimes |k_j\rangle \implies |n, k_j\rangle\langle n, k'_j| = (|n\rangle\langle n|) \otimes (|k_j\rangle\langle k'_j|).$$

Using this, Eq. (14) can be written as

$$H_I \approx \sum_n |n\rangle\langle n| \otimes \left(\sum_{j=1}^N \sum_{k_j, k'_j} g_{nk_j, nk'_j} |k_j\rangle\langle k'_j| \otimes \prod_{i \neq j} I_i \right). \quad (16)$$

Adding these contributions, the effective Hamiltonian becomes

$$\begin{aligned} H_{\text{eff}} &= H_0 + H_I = H_S + H_B + H_I \\ &= \sum_n |n\rangle\langle n| \otimes \left(E_n + \sum_{j=1}^N \sum_{k_j} \epsilon_{k_j} |k_j\rangle\langle k_j| + \sum_{j=1}^N \sum_{k_j, k'_j} g_{nk_j, nk'_j} |k_j\rangle\langle k'_j| \right) \otimes \prod_{i \neq j} I_i \\ &= \sum_n |n\rangle\langle n| \otimes \left(E_n + \sum_{j=1}^N \sum_{k_j, k'_j} \left[\epsilon_{k_j} \delta_{k_j, k'_j} + g_{nk_j, nk'_j} \right] |k_j\rangle\langle k'_j| \otimes \prod_{i \neq j} I_i \right). \end{aligned} \quad (17)$$

Thus the total effective Hamiltonian is diagonal with respect to the system eigenstates,

$$H_{\text{eff}} = \sum_n (E_n + h(n)) |n\rangle\langle n|, \quad (18)$$

where the operator $h(n)$ acts on the bath Hilbert space and is given by

$$h(n) = \sum_{j=1}^N \sum_{k_j, k'_j} \left[\epsilon_{k_j} \delta_{k_j, k'_j} + g_{nk_j, nk'_j} \right] |k_j\rangle\langle k'_j| \otimes \prod_{i \neq j} I_i. \quad (19)$$

Here $h(n)$ describes the effective bath Hamiltonian conditioned on the system being in the state $|n\rangle$. It can be further diagonalised as

$$h(n) = \sum_{j=1}^N \sum_{\alpha_j} \varepsilon_{\alpha_j}(n) |\alpha_j\rangle\langle \alpha_j| \otimes \prod_{i \neq j} I_i, \quad (20)$$

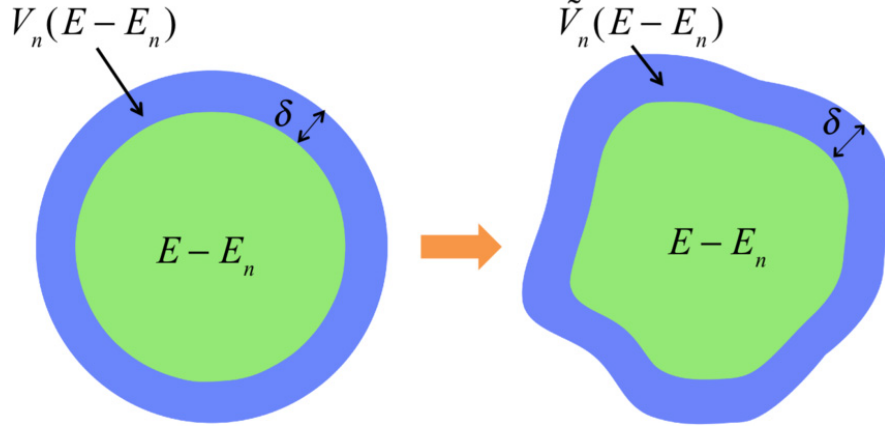


Figure 2: The subenergy shell without and with the consideration of the system-bath coupling are schematically shown in a manner analogous to classical phase space. This image was taken from the original paper.

with the eigenstate

$$|\vec{\alpha}\rangle = \prod_{j=1}^N |\alpha_j\rangle \quad (21)$$

and the corresponding eigenenergy

$$\varepsilon(\vec{\alpha}, n) = \sum_{j=1}^N \varepsilon_{\alpha_j}(n). \quad (22)$$

The new bath spectrum $\varepsilon(\vec{\alpha}, n)$ is therefore modified relative to the original spectrum $\epsilon(\vec{k})$ due to the systembath coupling.

This system-bath coupling is usually negligible when we study the thermalized state of the system in a large heat bath. However, in the situation of finite system statics, it is crucial to consider the system-bath interaction energy for the relatively small heat bath. The effective Hamiltonian Eq. (18) is already diagonalized with diagonal elements $E(n, \vec{\alpha}) = E_n + \varepsilon(\vec{\alpha}, n)$. Then it is straightforward to define the deformed subspace

$$\tilde{V}(E, \delta) = \text{Span}\{|n, \vec{\alpha}(n)\rangle : E \leq E_n + \varepsilon(\vec{\alpha}, n) \leq E + \delta\}, \quad (23)$$

which plays the same role as $V(E, \delta)$, with $|n, \vec{k}\rangle$ replaced by $|n, \vec{\alpha}(n)\rangle$ and $\epsilon(\vec{k})$ replaced by $\varepsilon(\vec{\alpha}, n)$. In the same way, we define

$$\tilde{V}(E - E_n, \delta) = \text{Span}\{|n, \vec{\alpha}(n) : E - E_n \leq \varepsilon(\vec{\alpha}, n) \leq E - E_n + \delta\}, \quad (24)$$

and $\tilde{\Omega}(E - E_n, \delta) = \dim \tilde{V}_n(E - E_n, \delta)$. The authors schematically illustrate the deformation of the subenergy shell in Fig. 2 (inserted from the original paper). By analogy with the

canonical construction, the deformed microcanonical state is

$$\tilde{\rho}_{\text{mic}} = \frac{P_{\tilde{V}(E,\delta)}}{\dim \tilde{V}(E,\delta)}, \quad \tilde{\rho}_S = \text{Tr}_B(\tilde{\rho}_{\text{mic}}). \quad (25)$$

where $P_{\tilde{V}(E,\delta)}$ is the projector onto the deformed energy shell,

$$P_{\tilde{V}(E,\delta)} = \sum_{\substack{n, \vec{\alpha}(n) \\ E \leq E_n + \varepsilon(\vec{\alpha}, n) \leq E + \delta}} |n, \vec{\alpha}(n)\rangle \langle n, \vec{\alpha}(n)|. \quad (26)$$

Substituting this into Eq. (25) yields

$$\begin{aligned} \tilde{\rho}_S &= \text{Tr}_B(\tilde{\rho}_{\text{mic}}) \\ &= \frac{1}{\dim \tilde{V}(E,\delta)} \sum_{\substack{n, \vec{\alpha}(n) \\ E \leq E_n + \varepsilon(\vec{\alpha}, n) \leq E + \delta}} \text{Tr}_B(|n, \vec{\alpha}(n)\rangle \langle n, \vec{\alpha}(n)|) \\ &= \sum_n \frac{\tilde{\Omega}(E - E_n, \delta)}{\dim \tilde{V}(E,\delta)} |n\rangle \langle n|. \end{aligned} \quad (27)$$

Thus, the probability of finding the system in state $|n\rangle$ is

$$\tilde{P}_n(E) = \langle n | \tilde{\rho}_S | n \rangle = \frac{\tilde{\Omega}(E - E_n, \delta)}{\dim \tilde{V}(E,\delta)}. \quad (28)$$

The matrix $\tilde{\rho}_S$ is therefore the noncanonical reduced density matrix of the system, and $\tilde{P}_n(E)$ is the corresponding noncanonical distribution that incorporates finite system-bath coupling.

To further obtain an explicit expression of $\tilde{P}_n(E)$, we introduce the entropy defined by $S(E) = k_B \ln(\Omega(E, s))$, where k_B is the Boltzmann constant. Note that the functional form of $\Omega(E, \delta)$ depends on the underlying Hamiltonian H . In particular, including or excluding the interaction H_I leads to different densities of states, so that the undeformed and deformed energy shells correspond to $\Omega(E, \delta)$ and $\tilde{\Omega}(E, \delta)$, respectively. Therefore, it is useful to regard $S(E)$ as a functional of the chosen energy shell $V(E, \delta)$, and write $S(E) \equiv S(V(E, \delta))$. When the system is in the state $|n\rangle$, the entropy can be expanded as,

$$\tilde{S}(E - E_n) = S(E - E_n) + \delta_n S(E - E_n) \quad (29)$$

$$\begin{aligned} &\approx S(E) - \frac{\partial S(E)}{\partial E} E_n + \delta_n S(E) - \frac{\partial[\delta_n S(E)]}{\partial E} E_n \\ &\approx S(E) - \frac{\partial S(E)}{\partial E} E_n + \phi_n, \end{aligned} \quad (30)$$

where

$$\phi_n = \delta_n S(E) - \frac{\partial[\delta_n S(E)]}{\partial E} E_n. \quad (31)$$

The term $\delta_n S(E - E_n)$ is the functional variation of $S(E - E_n)$, where the subscript n means that the functional variation also depends on the state of the system. In the above derivation, we implicitly assume that the deformation of the energy shell is small; that is, for large N the difference between $\Omega(E - E_n, \delta)$ and $\tilde{\Omega}(E - E_n, \delta)$ is taken to be small. To see why this is a reasonable assumption, it is helpful to sketch how this works in a concrete model. Xu and Sun point to the example of Dong *et al.*, “Quantum thermalization with couplings”¹, where the difference between the deformed and undeformed densities of states vanishes in the thermodynamic limit.

In the model of Dong *et al.*, the number of bath microstates compatible with the system level $|n\rangle$ scales as

$$\Omega_N(n) \propto [E - \epsilon_n(\kappa)]^{N-1}, \quad (32)$$

where

$$\epsilon_n(\kappa) = \epsilon_n - \kappa \lambda_n^2 \quad (33)$$

is the interactionshifted system energy. This can be written as

$$\Omega_N(n) \propto \exp \left[(N-1) \ln(E - \epsilon_n(\kappa)) \right]. \quad (34)$$

Substituting $\epsilon_n(\kappa)$ and rearranging gives

$$E - \epsilon_n(\kappa) = E + \kappa \lambda_n^2 - \epsilon_n, \quad (35)$$

so that

$$\ln(E - \epsilon_n(\kappa)) = \ln(E + \kappa \lambda_n^2 - \epsilon_n). \quad (36)$$

For a large bath with weak coupling, we have

$$\frac{\epsilon_n}{E + \kappa \lambda_n^2} \ll 1, \quad (37)$$

and we may expand the logarithm to first order:

$$\ln(E + \kappa \lambda_n^2 - \epsilon_n) \simeq \ln(E + \kappa \lambda_n^2) - \frac{\epsilon_n}{E + \kappa \lambda_n^2}. \quad (38)$$

Inserting this back into $\Omega_N(n)$ yields

$$\begin{aligned} \Omega_N(n) &\propto \exp \left[(N-1) \ln(E + \kappa \lambda_n^2) \right] \exp \left[-(N-1) \frac{\epsilon_n}{E + \kappa \lambda_n^2} \right] \\ &\equiv (\text{slowly varying prefactor}) \times \exp(-\beta_n \epsilon_n), \end{aligned} \quad (39)$$

¹H. Dong, S. Yang, X. F. Liu, and C. P. Sun, [Phys. Rev. A **76**, 044104 \(2007\)](#).

with

$$\beta_n = \frac{N-1}{E + \kappa \lambda_n^2}. \quad (40)$$

In the large-bath, weak-coupling regime (N large, $E \sim N$, $\kappa \lambda_n^2 \ll E$), the quantity β_n becomes effectively independent of both the level index n and the coupling, and the prefactor depends only weakly on n . The resulting diagonal distribution $\tilde{P}_n$ is then approximately canonical and essentially insensitive to the shell deformation. Xu and Sun summarise this robustness by stating that the difference between Ω and $\tilde{\Omega}$ vanishes in the thermodynamic limit, even though Dong *et al.* do not phrase their result explicitly in these terms.

The derivative terms of $S(E)$ in Eq. (30) can be evaluated by the idea from classical phase case: since $\Omega(E)$ is the dimension of the subspace $V(E, \delta)$, it is proportional to the volume of the energy shell in phase space. In an N -dimensional space, the volume confined by an isoenergetic surface of energy E can be considered as the volume of an N -dimensional polyhedron with effective radius $R(E)$, which is proportional to E^ζ , so that the volume is proportional to $E^{\zeta N}$, where ζ is a dimensionless real number independent of N and is usually related to the degeneracy of the system. The above analysis is from the original paper,² and is quite hand-wavy, so we will try to make sense of this. The phase space is N -dimensional. We pick the bath coordinates collectively as $\mathbf{X} \in \mathbb{R}^N$. The Hamiltonian $H(\mathbf{X})$ is an energy function on this space. In many physical cases (free particles, oscillators, polynomial potentials), the dominant part of H is homogeneous of some degree α , meaning

$$H(\lambda \mathbf{X}) = \lambda^\alpha H(\mathbf{X}).$$

Given this homogeneity, we look at the region

$$\mathcal{D}(E) = \{\mathbf{X} : H(\mathbf{X}) \leq E\}.$$

Now define a scaled coordinate

$$\mathbf{X} = E^{1/\alpha} \mathbf{Y}.$$

Then

$$H(\mathbf{X}) = H(E^{1/\alpha} \mathbf{Y}),$$

so the condition $H(\mathbf{X}) \leq E$ becomes $H(\mathbf{Y}) \leq 1$. That means the region $\mathcal{D}(E)$ is just the region $\mathcal{D}(1)$ scaled by a linear factor $E^{1/\alpha}$ in all directions. Therefore the “radius” of the region with energy E is

$$R(E) \propto E^{1/\alpha}.$$

For convenience, define $\zeta := 1/\alpha$, so that

$$R(E) \propto E^\zeta.$$

²Here the "original paper" refers to "[Noncanonical statistics of a finite quantum system with non-negligible system-bath coupling](#)" by Xu and Sun.

The corresponding density of states then behaves as

$$g(E) \sim \frac{d}{dE} D(E) \propto E^{\zeta N - 1}, \quad D(E) = \text{Vol}\{\mathbf{X} : H(\mathbf{X}) \leq E\}.$$

This scaling makes the connection between the exponent ζ and the growth of the degeneracy with energy more transparent.

We also know that in N dimensions, if we scale all the coordinates by a factor R , the volume is scaled by R^N . So the volume of the region below energy E is

$$D(E) = \int_{\mathcal{D}(E)} d^N \mathbf{X} = \int_{\mathcal{D}(1)} d^N (R(E) \mathbf{Y}) = R(E)^N \int_{\mathcal{D}(1)} d^N \mathbf{Y} = R(E)^N D(1).$$

Therefore, the volume of the energy shell defined for energies in $[E, E + \delta]$ is given by

$$\Omega(E) \propto (E + \delta)^{\zeta N} - E^{\zeta N}. \quad (41)$$

Now we assume $\delta \ll E$ (thin shell compared to the total energy, but still containing many states). Then we expand

$$\begin{aligned} f(E) = E^{\zeta N} &\rightarrow f(E + \delta) \approx f(E) + f'(E) \delta = E^{\zeta N} + (\zeta N) E^{\zeta N - 1} \delta \\ \implies (E + \delta)^{\zeta N} - E^{\zeta N} &\approx \zeta N \delta E^{\zeta N - 1}. \end{aligned} \quad (42)$$

Since ζN and δ are treated as constants, we can write

$$\Omega(E) \propto E^{\zeta N - 1}. \quad (43)$$

Using $S(E) = k_B \ln \Omega(E)$, this gives

$$\beta(E) = \frac{\partial S(E)}{\partial E} = \frac{\partial}{\partial E} [k_B (\zeta N - 1) \ln E] \approx \frac{k_B (\zeta N - 1)}{E}, \quad (44)$$

where we identify $\beta = (k_B T)^{-1}$ via the thermodynamic relation between entropy and temperature, with T the equilibrium temperature of the system. Obviously, Eq. (44) recovers the equipartition theorem $E \approx \zeta N k_B T$, i.e., each degree of freedom contributes $\zeta k_B T$ to the total energy. Using Eq. (30) we write

$$\frac{\tilde{S}(E - E_n)}{k_B} \simeq \frac{S(E)}{k_B} - \beta E_n + \frac{\phi_n}{k_B}.$$

Substituting this into Eq. (28) gives

$$\tilde{P}_n(E) = \frac{e^{S(E)/k_B} e^{-\beta E_n + \phi_n/k_B}}{\sum_m e^{S(E)/k_B} e^{-\beta E_m + \phi_m/k_B}}.$$

The common factor $e^{S(E)/k_B}$ cancels between numerator and denominator, so the noncanonical

distribution takes the form

$$\tilde{P}_n(E) = \frac{1}{\tilde{\mathcal{Z}}} e^{-\beta E_n + \phi_n/k_B}, \quad (45)$$

where $\tilde{\mathcal{Z}} = \sum \exp(-\beta E_n + \phi_n/k_B)$ is the partition function.

If the interaction energy could be neglected compared with the total energy E , the variation terms ϕ_n containing $\delta_n S(E)$ in Eq. (45) can be dropped and thus naturally lead to the usual canonical form,

$$P_n(E) = \frac{1}{\mathcal{Z}} e^{-\beta E_n}. \quad (46)$$

Otherwise, the deformation of the energy shell which leads to ϕ_n will shift the system energy level and thus modify $P_n(E)$. Usually ϕ_n cannot be calculated easily; however, for some special models (one of which will be discussed in the next section), we can find a simple expression for ϕ_n .

1.3 Illustration with harmonic oscillators system

In this section the authors consider a coupled harmonic oscillators system as an example to illustrate the statistical thermodynamic properties of a finite system. In what follows, we set $\hbar = 1$. The system S is a single harmonic oscillator with eigenfrequency ω described by the Hamiltonian

$$H_S = \omega a^\dagger a, \quad (47)$$

where $a^\dagger$ and a are the usual creation and annihilation operators. The zero-point energy term $1/2\omega$ has been omitted since it corresponds to an overall constant shift of the spectrum and does not affect the microcanonical state or the normalized probability distribution $\tilde{P}_n(E)$. The heat bath is generally modeled as a collection of harmonic oscillators with Hamiltonian

$$H_B = \sum_{j=1}^N \omega_j b_j^\dagger b_j. \quad (48)$$

Here $b_j^\dagger$ (b_j) is the creation (annihilation) operator of the oscillator with frequency ω_j . The zero-point term $1/2\omega_j$ has been omitted again for the same reason. We work in the weak-coupling regime, where the system-bath coupling constants λ_j are small compared to the characteristic energy scales of the uncoupled system and bath. Concretely, we assume $|\lambda_j| \ll \omega$ and $|\lambda_j| \ll \omega_j$, so that the interaction energy is much smaller than the level spacings of both H_S and H_B . In this regime the coupling primarily induces small level-dependent energy shifts and deformations of the energy shell, without strongly mixing different system

eigenstates. The authors assume the effective system-bath interaction

$$H_I = \sum_{j=1}^N \lambda_j a^\dagger a (b_j^\dagger + b_j), \quad (49)$$

which satisfies $[H_S, H_I] = 0$. Thus the system number states $|n\rangle$ remain eigenstates of the full Hamiltonian, and the interaction does not induce transitions between different system levels; it only shifts the energy of each level via the bath. This structure allows the full Hamiltonian to be diagonalized by a displacement of the bath modes.

In the product basis $|n\rangle \otimes |k_1, \dots, k_N\rangle$, the free Hamiltonians H_S and H_B are diagonal, whereas

$$H_I = \sum_{j=1}^N \lambda_j a^\dagger a (b_j^\dagger + b_j)$$

is off-diagonal in the bath occupation numbers k_j . Since $a^\dagger a |n\rangle = n |n\rangle$, we first fix the system in $|n\rangle$ and take the matrix element of $H_B + H_I$:

$$\begin{aligned} \langle n | H_I | n \rangle &= \sum_j \lambda_j \langle n | a^\dagger a (b_j^\dagger + b_j) | n \rangle \\ &= \sum_j \lambda_j n (b_j^\dagger + b_j), \end{aligned}$$

so the effective bath Hamiltonian in sector n is

$$H_{\text{bath}}^{(n)} = \langle n | (H_B + H_I) | n \rangle = \sum_{j=1}^N \left[\omega_j b_j^\dagger b_j + n \lambda_j (b_j^\dagger + b_j) \right], \quad (50)$$

where $a^\dagger a$ has been replaced by its eigenvalue n .

Each mode j is thus a harmonic oscillator with a linear term in $b_j^\dagger + b_j$. We diagonalize it by a displacement transformation. For each mode, define

$$c_j := b_j + \alpha_j, \quad c_j^\dagger := b_j^\dagger + \alpha_j^*. \quad (51)$$

Substituting $b_j = c_j - \alpha_j$ into $H_{\text{bath}}^{(n)}$, we get

$$\begin{aligned} \omega_j b_j^\dagger b_j &= \omega_j (c_j^\dagger - \alpha_j^*) (c_j - \alpha_j) = \omega_j c_j^\dagger c_j - \omega_j \alpha_j c_j^\dagger - \omega_j \alpha_j^* c_j + \omega_j |\alpha_j|^2, \\ n \lambda_j (b_j^\dagger + b_j) &= n \lambda_j (c_j^\dagger + c_j) - n \lambda_j (\alpha_j + \alpha_j^*). \end{aligned}$$

Therefore,

$$H_{\text{bath}}^{(n)} = \sum_{j=1}^N \left[\omega_j c_j^\dagger c_j + (-\omega_j \alpha_j + n \lambda_j) c_j^\dagger + (-\omega_j \alpha_j^* + n \lambda_j) c_j + \omega_j |\alpha_j|^2 - n \lambda_j (\alpha_j + \alpha_j^*) \right]. \quad (52)$$

We choose α_j so that the linear terms vanish:

$$-\omega_j \alpha_j + n \lambda_j = 0 \quad \Rightarrow \quad \alpha_j = \frac{n \lambda_j}{\omega_j}.$$

The second condition $-\omega_j \alpha_j^* + n \lambda_j = 0$ is then automatically satisfied, showing that α_j is real in this model. The constant term becomes

$$\omega_j |\alpha_j|^2 - n \lambda_j (\alpha_j + \alpha_j^*) = \frac{n^2 \lambda_j^2}{\omega_j} - \frac{2n^2 \lambda_j^2}{\omega_j} = -\frac{n^2 \lambda_j^2}{\omega_j}.$$

Hence

$$H_{\text{bath}}^{(n)} = \sum_{j=1}^N \left(\omega_j c_j^\dagger c_j - \frac{n^2 \lambda_j^2}{\omega_j} \right), \quad (53)$$

which is diagonal in the number basis $\{|k_j\rangle_c\}$ of the displaced oscillators.

Adding back the system Hamiltonian $H_S = \omega a^\dagger a$, we obtain

$$H^{(n)} = \omega n + \sum_{j=1}^N \left(\omega_j c_j^\dagger c_j - \frac{n^2 \lambda_j^2}{\omega_j} \right), \quad (54)$$

so the eigenenergies are

$$E_{n, \{k_j\}} = \omega n + \sum_{j=1}^N \omega_j k_j - \sum_{j=1}^N \frac{n^2 \lambda_j^2}{\omega_j}. \quad (55)$$

This result can be directly matched to the notation in the original paper. There, the bath modes are diagonalized by the displacement operator

$$D(\alpha_j) = \exp(\alpha_j b_j^\dagger - \alpha_j^* b_j), \quad (56)$$

leading to displaced Fock states

$$|m_j(n)\rangle = D\left(-\frac{\lambda_j n}{\omega_j}\right) |m_j\rangle. \quad (57)$$

This is equivalent to our approach; instead of introducing displaced operators $c_j = b_j + \alpha_j$, one can keep b_j fixed and act with the unitary displacement operator $D(-n\lambda_j/\omega_j)$ on the bath Fock states. Displacing the operators and displacing the states are two aspects of the same unitary transformation.

Matching our notation to that of the paper, Eq. (55) for the eigenvalues of the total Hamil-

tonian becomes

$$E(n, \vec{m}(n)) = n\omega + \Delta_n + \sum_{j=1}^N m_j \omega_j, \quad (58)$$

corresponding to the eigenstates

$$|n, \vec{m}(n)\rangle = |n\rangle \otimes \prod_{j=1}^N |m_j(n)\rangle. \quad (59)$$

Thus the level-dependent energy shift obtained above,

$$\Delta_n = - \sum_{j=1}^N \frac{n^2 \lambda_j^2}{\omega_j}, \quad (60)$$

coincides with Eq. (24) of the paper, where they write

$$\Delta_n = -\kappa n^2, \quad \kappa := \sum_{j=1}^N \frac{\lambda_j^2}{\omega_j}. \quad (61)$$

Therefore, in this model the deformation of each subenergy shell is only related to Δ_n , which is independent of the specific state of the heat bath. Then the deformed energy shell is simply related with the original energy shell without considering H_I by

$$\tilde{V}_n(E - E_n, \delta) = V_n(E - \Delta_n - E_n, \delta). \quad (62)$$

Furthermore, the functional form of $\tilde{S}(E - E_n)$ is the same as that of $S(E - E_n)$, with the replacement $E \rightarrow E - \Delta_n$:

$$\tilde{S}(E - E_n) = S(E - \Delta_n - E_n). \quad (63)$$

For small $|\Delta_n|$, a first-order Taylor expansion in Δ_n gives

$$S(E - \Delta_n - E_n) \approx S(E - E_n) - \left. \frac{\partial S}{\partial E} \right|_{E=E_n} \Delta_n. \quad (64)$$

Using Eq. (29), we have

$$\delta_n S(E - E_n) = S(E - \Delta_n - E_n) - S(E - E_n), \quad (65)$$

and therefore obtain

$$\delta_n S(E - E_n) \approx - \left. \frac{\partial S}{\partial E} \right|_{E=E_n} \Delta_n. \quad (66)$$

Since the bath is large, the entropy gradient varies slowly with energy, and one may approx-

imate $\left. \frac{\partial S}{\partial E} \right|_{E-E_n} \approx \left. \frac{\partial S}{\partial E} \right|_E \equiv \beta(E)$. Thus

$$\delta_n S(E) \approx - \left. \frac{\partial S}{\partial E} \right|_E \Delta_n. \quad (67)$$

(Note that Eq. (27) in the original paper appears to miss a minus sign.)

Differentiating $\delta_n S(E)$ with respect to E gives

$$\frac{\partial \delta_n S(E)}{\partial E} \approx - \frac{\partial^2 S(E)}{\partial E^2} \Delta_n.$$

Plugging this into $\phi_n(E)$,

$$\begin{aligned} \phi_n &= \delta_n S(E) - \frac{\partial \delta_n S(E)}{\partial E} E_n \\ &\approx - \frac{\partial S(E)}{\partial E} \Delta_n + \frac{\partial^2 S(E)}{\partial E^2} \Delta_n E_n = -\beta(E) \Delta_n + \xi \Delta_n E_n, \end{aligned} \quad (68)$$

where, using Eq. (44),

$$\xi = \frac{\partial^2 S(E)}{\partial E^2} = - \frac{k_B(\zeta N - 1)}{E^2}. \quad (69)$$

For the harmonic oscillator model we have

$$\Delta_n = -\kappa n^2, \quad E_n = n\omega,$$

so that

$$\phi_n = \beta(E) \kappa n^2 - \xi \omega \kappa n^3.$$

Substituting this into Eq. (45) for $\tilde{P}_n(E)$ yields

$$\tilde{P}_n(E) = \frac{1}{\mathcal{Z}} e^{-\beta n \omega + \beta \kappa n^2 - \xi \omega \kappa n^3} = \frac{1}{\mathcal{Z}} e^{-\beta(n\omega - n^2 \kappa) - \xi \omega \kappa n^3}. \quad (70)$$

The square and cubic terms in n in the exponent of $\tilde{P}_n(E)$ strongly modify the distribution compared with the canonical form $P_n(E) = \exp(-\beta n \omega) / \mathcal{Z}$, especially for large n . However, Eq. (70) does not apply to arbitrarily large N for the following reason: the total energy E of the system plus bath is conserved, and the bath, being a collection of harmonic modes, has energy $\sum_{\{m_j\}} m_j \omega_j \geq 0$, so its energy must be non-negative, $E_{\text{bath}} \geq 0$. On the deformed energy shell for level n , the bath energy is

$$E_{\text{bath}} = E - E_n - \Delta_n = E - n\omega + \kappa n^2,$$

where $E_n = n\omega$ and $\Delta_n = -\kappa n^2$. Thus

$$E - n\omega + \kappa n^2 \geq 0 \implies \kappa n^2 - \omega n + E \geq 0. \quad (71)$$

Solving the corresponding quadratic equation

$$\kappa n^2 - \omega n + E = 0$$

gives

$$n_{\pm} = \frac{\omega \pm \sqrt{\omega^2 - 4\kappa E}}{2\kappa}. \quad (72)$$

For an upward parabola ($\kappa > 0$), the inequality is satisfied outside the interval $[n_-, n_+]$. In the regime where the noncanonical distribution is only a small deformation of the canonical one, physically relevant levels are low-lying, so we restrict to $n \leq n_-$. This yields an upper bound on the occupied system levels,

$$n < \frac{\omega - \sqrt{\omega^2 - 4\kappa E}}{2\kappa}. \quad (73)$$

Therefore, for such a coupled system, the maximum system level is

$$n_{\max} = \left\lfloor \frac{\omega - \sqrt{\omega^2 - 4\kappa E}}{2\kappa} \right\rfloor, \quad (74)$$

where $\lfloor x \rfloor$ denotes the largest integer not exceeding x . Hence, Eq. (70) cannot be extended to arbitrarily large n beyond this constraint.

We can check the weak-coupling limit to see that this bound is sensible. Expanding for small κ , we find

$$\sqrt{\omega^2 - 4\kappa E} = \omega \sqrt{1 - \frac{4\kappa E}{\omega^2}} \approx \omega \left(1 - \frac{2\kappa E}{\omega^2} \right) = \omega - \frac{2\kappa E}{\omega}.$$

Then

$$n_{\max} \approx \frac{\omega - (\omega - \frac{2\kappa E}{\omega})}{2\kappa} = \frac{E}{\omega}. \quad (75)$$

In the weak-coupling limit we thus recover the obvious bound that the system cannot carry more energy than the total energy E :

$$E_n = n\omega \leq E \implies n \leq \frac{E}{\omega}.$$

For low energy levels ($n \lesssim 20$), the noncanonical distribution $\tilde{P}_n(E)$ does not differ much from the canonical one, as shown in Fig. 3, taken from the paper.

The figure, generated with $\omega = 1$, the total energy $E = 100$, and $\kappa = 0.00235$, shows a system where the highest energy level is $n_{\max} = 160$ according to Eq. (74). The canonical distribution is plotted by setting $\kappa = 0$ in $\tilde{P}_n(E)$.

From Eq. (43),

$$S(E) = \ln(\Omega(E)) = (\zeta N - 1) \ln(E), \quad (76)$$

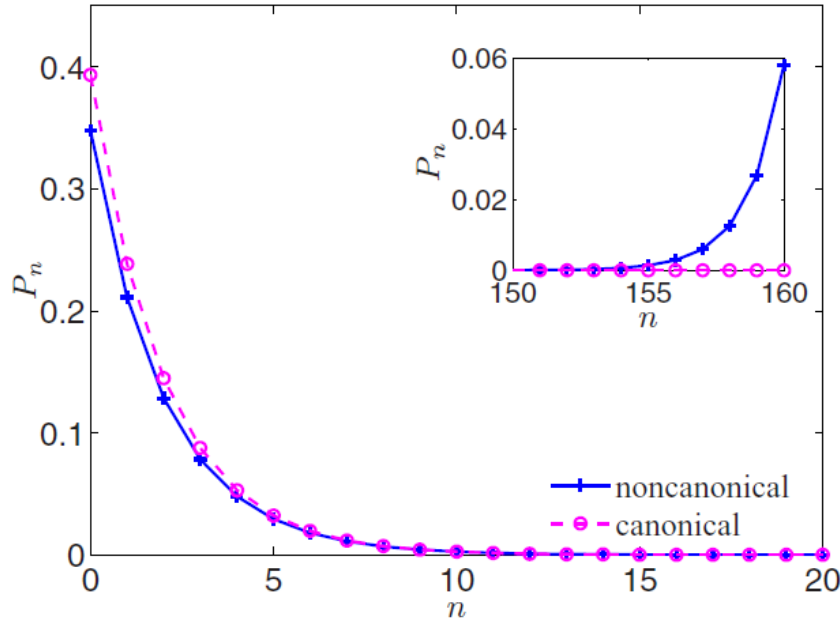


Figure 3: Noncanonical (blue solid line) and canonical (pink dashed line) distribution functions for the coupled oscillator system. The authors choose $\omega = 1$, $E = 100$, $\beta = 0.5$, and $\kappa = 0.00235$, for which the highest energy level is $n_{\max} = 160$. The noncanonical and canonical distributions are very similar for low energy levels, but differ significantly for high energy levels.

it follows that

$$\beta = \frac{\partial S(E)}{\partial E} > 0 \quad \rightarrow \quad \xi = \frac{\partial^2 S(E)}{\partial E^2} < 0.$$

Consequently, the extra quadratic and cubic terms in the exponent of (70) become positive at large n . These noncanonical corrections therefore push the probability upward, partially canceling the usual $-\beta\omega n$ decay. Since $|\Delta_n|$ grows with n^2 , the high energy levels gain more population than in the canonical ensemble, as visible in the inset of Fig. 3.

The internal energy of the system is defined as

$$U = \langle H_S \rangle = \text{Tr}(\tilde{\rho}_S H_S). \quad (77)$$

Putting the noncanonical density matrix from Eq. (29),

$$\tilde{\rho}_S = \sum_n \tilde{P}_n(E) |n\rangle \langle n|,$$

and the harmonic oscillator Hamiltonian

$$H_S = \sum_n E_n |n\rangle \langle n|, \quad E_n = n\omega,$$

together we obtain

$$\begin{aligned}
U &= \text{Tr} \left(\sum_n \tilde{P}_n(E) |n\rangle \langle n| \cdot \sum_m E_m |m\rangle \langle m| \right) = \sum_{n,m} \tilde{P}_n(E) E_m \text{Tr} (|n\rangle \langle n| m\rangle \langle m|) \\
&= \sum_n \tilde{P}_n(E) E_n = \sum_n n\omega \tilde{P}_n(E).
\end{aligned} \tag{78}$$

Because the high-energy states have relatively larger populations, the internal energy of the system under the noncanonical statistics is larger than that under the canonical one. This fact is illustrated in Fig. 4, where the internal energy U is plotted with respect to β . The distinction between noncanonical and canonical results for U evidently appears when the

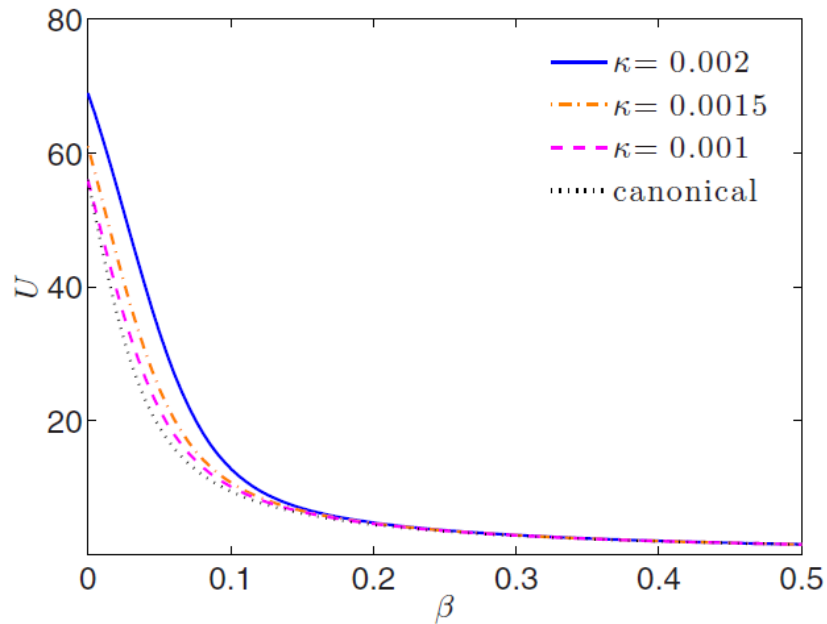


Figure 4: The authors plot the internal energy U versus β for both noncanonical and canonical statistics, with $\omega = 1$ and $E = 100$. Noncanonical cases are shown for several coupling strengths: $\kappa = 0.002$ ($n_{\max} = 138$), $\kappa = 0.0015$ ($n_{\max} = 122$), and $\kappa = 0.001$ ($n_{\max} = 112$). These are compared to the canonical case ($\kappa = 0$, $n_{\max} = 122$). The plot reveals that for small β , the internal energy U in the noncanonical statistics is significantly larger than that predicted by the canonical distribution.

inverse temperature β decreases and the interaction energy strength κ grows. As β approaches zero, the terms proportional to β in (70) vanish. Additionally, $\xi = \partial^2 S / \partial E^2 \propto 1/E^2$, and since the total energy E of the system and the bath becomes large at high temperature, ξ tend to zero. Thus, the entire exponent approaches zero, so:

$$\tilde{P}_n(E) \propto 1. \tag{79}$$

The partition function $\mathcal{Z}$ then simply counts the number of allowed states,

$$\begin{aligned}\mathcal{Z} &= \sum_{n=0}^{n_{\max}} 1 = n_{\max} + 1, \\ \Rightarrow \tilde{P}_n(E) &= \frac{1}{n_{\max} + 1}.\end{aligned}\tag{80}$$

Substituting this into Eq. (78) gives

$$U = \frac{\omega}{n_{\max} + 1} \sum_{n=0}^{n_{\max}} n = \frac{\omega}{n_{\max} + 1} \times \frac{n_{\max}(n_{\max} + 1)}{2} = \frac{\omega n_{\max}}{2}.\tag{81}$$

Therefore, the high-temperature limit of the internal energy U is $n_{\max}\omega/2$, which is finite as the total energy is upper-bounded by E for a small heat bath. This differs substantially from the case in the thermodynamics limit: the average energy of a harmonic oscillator which is in contact with an infinite heat bath will diverge when β decreases to zero.

Fig. 4 illustrates this behavior for a fixed $\omega = 1$ and $E = 100$. Different coupling strength κ yield different maximum levels $n_{\max}$:

- $\kappa = 0.002$:

$$\begin{aligned}n_{\max} &= \left\lfloor \frac{1 - \sqrt{1 - 4 \times 0.002 \times 100}}{2 \times 0.002} \right\rfloor = 138 \\ \Rightarrow U_{\max} &= \frac{1 \times 138}{2} = 69\end{aligned}$$

- $\kappa = 0.0015$:

$$\begin{aligned}n_{\max} &= \left\lfloor \frac{1 - \sqrt{1 - 4 \times 0.0015 \times 100}}{2 \times 0.0015} \right\rfloor = 122 \\ \Rightarrow U_{\max} &= \frac{1 \times 122}{2} = 61\end{aligned}$$

- $\kappa = 0.001$:

$$\begin{aligned}n_{\max} &= \left\lfloor \frac{1 - \sqrt{1 - 4 \times 0.001 \times 100}}{2 \times 0.001} \right\rfloor = 112 \\ \Rightarrow U_{\max} &= \frac{1 \times 112}{2} = 56\end{aligned}$$

As shown, the noncanonical internal energy is significantly larger at small β .

Another feature reflecting the nonmonotony of the noncanonical distribution is the relative fluctuation of the system's internal energy. The variance of the system energy is

$$\text{Var}(U) = \langle (E_n - U)^2 \rangle = \sum_{n=0}^{n_{\max}} (E_n - U)^2 \tilde{P}_n(E) = \sum_{n=0}^{n_{\max}} (n\omega - U)^2 \tilde{P}_n(E).$$

and the relative fluctuation is defined as

$$\Delta U = \frac{\sqrt{\text{Var}(U)}}{U},$$

so that

$$(\Delta U)^2 = \frac{1}{U^2} \sum_{n=0}^{n_{\max}} (n\omega - U)^2 \tilde{P}_n(E). \quad (82)$$

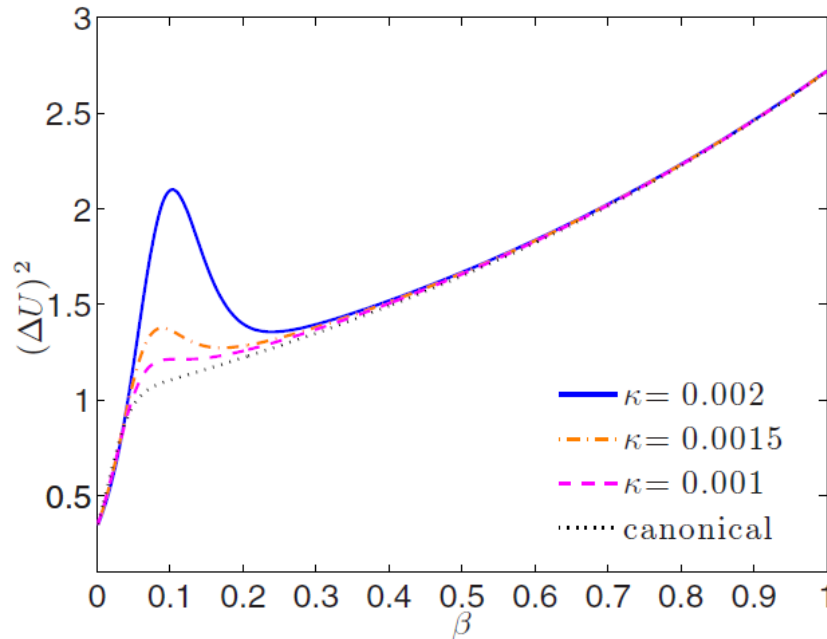


Figure 5: The internal energy relative fluctuation $(\Delta U)^2$ of the system with respect to β for noncanonical and canonical statistics. The parameters are chosen the same as Fig. 4. A local maximum appears in the energy fluctuation for small β which results of the noncanonical statistics.

As shown in Fig. 5, at both the low- and high-temperature limits, the noncanonical and canonical statistics of the system present similar fluctuation behavior. To characterize these two limits, we consider the system as a harmonic oscillator with truncated energy levels (the highest energy level is labeled by $n_{\max}$) under canonical statistics ($\kappa = 0$), whose energy fluctuation is denoted as $(\Delta U_C)^2$.

The limit $n_{\max} \rightarrow \infty$ corresponds to the standard canonical ensemble. Comparing the trun-

cated system (finite bath) with the infinite one, we can identify when finite-bath effects become relevant.

At very low temperature ($\beta\omega \gg 1$), the Boltzmann factor $e^{-\beta n\omega}$ decays rapidly. The ground state ($n = 0$) dominates, the first excited state ($n = 1$) is suppressed by a factor $e^{-\beta\omega}$, and higher levels are negligible. Thus, only the lowest one or two energy levels contribute significantly, and the system behaves like a two-level system. In this approximation, the mean energy is

$$U_C = \sum_{n=0}^1 E_n \cdot P_n = 0 \cdot P_0 + \omega \cdot P_1 = \omega P_1,$$

where the probabilities are

$$P_0 = \frac{1}{1 + e^{-\beta\omega}},$$

$$P_1 = \frac{e^{-\beta\omega}}{1 + e^{-\beta\omega}} \approx e^{-\beta\omega} \quad (\text{since } e^{-\beta\omega} \ll 1).$$

The variance is

$$\begin{aligned} \text{Var}(U_C) &= \sum_{n=0}^1 (E_n - U)^2 P_n = (0 - \omega P_1)^2 P_0 + (\omega - \omega P_1)^2 P_1 \\ &= \omega^2 [P_1^2 P_0 + (1 - P_1)^2 P_1] = \omega^2 [P_1^2 (1 - P_1) + (1 - P_1)^2 P_1] \\ &= \omega^2 P_1 (1 - P_1) [P_1 + (1 - P_1)] = \omega^2 P_1 (1 - P_1) \\ &= \omega^2 e^{-\beta\omega} (1 - e^{-\beta\omega}) \approx \omega^2 e^{-\beta\omega} \end{aligned}$$

The relative fluctuation then becomes

$$(\Delta U_C)^2 = \frac{\text{Var}(U_C)}{U_C^2} \approx \frac{\omega^2 P_1 (1 - P_1)}{\omega^2 P_1^2} \approx e^{\beta\omega}. \quad (83)$$

To confirm that this result aligns with the canonical low-temperature limit for $n_{\max} \rightarrow \infty$, we evaluate the full canonical partition function. Since $\beta\omega > 0$, then exponential satisfies

$$e^{\beta\omega} > 1 \implies e^{-\beta\omega} < 1.$$

$e^{-\beta\omega}$ is a positive real number, its absolute value is itself, so $|e^{-\beta\omega}| = e^{-\beta\omega} < 1$. With this condition we can use the geometric series

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1 - x}, \quad (84)$$

for $n_{\max} \rightarrow \infty$:

$$\mathcal{Z} = \sum_{n=0}^{\infty} e^{-\beta\omega n} = \frac{1}{1 - e^{-\beta\omega}}. \quad (85)$$

The mean energy is

$$\langle E \rangle = \frac{1}{\mathcal{Z}} \sum_{n=0}^{\infty} n\omega e^{-\beta\omega n}.$$

Using

$$\sum_{n=0}^{\infty} nx^n = \frac{x}{(1-x)^2}, \quad (86)$$

with $x = e^{-\beta\omega}$, we obtain

$$\begin{aligned} \sum_{n=0}^{\infty} ne^{-\beta\omega n} &= \frac{e^{-\beta\omega}}{(1 - e^{-\beta\omega})^2}, \\ \langle E \rangle &= \frac{\omega}{\mathcal{Z}} \times \frac{e^{-\beta\omega}}{(1 - e^{-\beta\omega})^2} = \frac{\omega e^{-\beta\omega}}{1 - e^{-\beta\omega}}. \end{aligned} \quad (87)$$

For the second moment,

$$\langle E^2 \rangle = \frac{1}{\mathcal{Z}} \sum_{n=0}^{\infty} (n\omega)^2 e^{-\beta\omega n} = \frac{\omega^2}{\mathcal{Z}} \sum_{n=0}^{\infty} n^2 e^{-\beta\omega n},$$

and

$$\sum_{n=0}^{\infty} n^2 x^n = \frac{x(1+x)}{(1-x)^3}, \quad (88)$$

gives

$$\langle E^2 \rangle = \frac{\omega^2}{\mathcal{Z}} \times \frac{e^{-\beta\omega}(1 + e^{-\beta\omega})}{(1 - e^{-\beta\omega})^3} = \frac{\omega^2 e^{-\beta\omega}(1 + e^{-\beta\omega})}{(1 - e^{-\beta\omega})^2}. \quad (89)$$

The variance is therefore

$$\begin{aligned} \text{Var}(E) &= \langle E^2 \rangle - \langle E \rangle^2 = \frac{\omega^2}{(1 - e^{-\beta\omega})^2} [e^{-\beta\omega}(1 + e^{-\beta\omega}) - e^{-2\beta\omega}] \\ &= \frac{\omega^2}{(1 - e^{-\beta\omega})^2} e^{-\beta\omega} \end{aligned}$$

Finally, the relative fluctuation is

$$(\Delta U_C)^2 = \frac{\text{Var}(E)}{\langle E \rangle^2} = \frac{\frac{\omega^2 e^{-\beta\omega}}{(1 - e^{-\beta\omega})^2}}{\frac{\omega^2 e^{-2\beta\omega}}{(1 - e^{-\beta\omega})^2}} = \frac{e^{-\beta\omega}}{e^{-2\beta\omega}} = e^{\beta\omega} \quad (90)$$

This analytic result confirms that in the low-temperature limit, the fluctuation $(\Delta U_C)^2 \approx \exp(\beta\omega)$ is the same whether the spectrum is truncated or infinite. This is very important as it tells us that for sufficiently cold systems, finite baths behave exactly like infinite baths.

Now consider the high-temperature limit. For a harmonic oscillator (setting $\omega = 1$ for simplicity) truncated at $n_{\max}$, the finite geometric series

$$\sum_{k=0}^n x^k = \frac{x^{n+1} - 1}{x - 1}, \quad (91)$$

gives

$$\mathcal{Z} = \sum_{n=0}^{n_{\max}} e^{-\beta n} = \frac{1 - e^{-\beta(n_{\max}+1)}}{1 - e^{-\beta}}. \quad (92)$$

For small β we can expand the numerator and the denominator of $\mathcal{Z}$. Using

$$1 - e^{-ax} = ax - \frac{a^2 x^2}{2} + \frac{a^3 x^3}{6} + O(x^4), \quad (93)$$

$$\frac{1}{1 - e^{-x}} = \frac{1}{x} + \frac{1}{2} + \frac{x}{12} + O(x^3), \quad (94)$$

we obtain

$$1 - e^{-\beta(n_{\max}+1)} = \beta(n_{\max}+1) - \frac{\beta^2(n_{\max}+1)^2}{2} + \frac{\beta^3(n_{\max}+1)^3}{6} + O(\beta^4), \quad (95)$$

$$\frac{1}{1 - e^{-\beta}} = \frac{1}{\beta} + \frac{1}{2} + \frac{\beta}{12} + O(\beta^3). \quad (96)$$

Multiplying,

$$\begin{aligned} \mathcal{Z} &= \left(\beta(n_{\max}+1) - \frac{\beta^2(n_{\max}+1)^2}{2} + \frac{\beta^3(n_{\max}+1)^3}{6} \right) \left(\frac{1}{\beta} + \frac{1}{2} + \frac{\beta}{12} \right) \\ &= (n_{\max}+1) - \frac{n_{\max}(n_{\max}+1)}{2}\beta + \frac{n_{\max}(n_{\max}+1)(2n_{\max}+1)}{12}\beta^2 + O(\beta^3) \\ &= (n_{\max}+1) \left(1 - \frac{n_{\max}}{2}\beta + \frac{n_{\max}(2n_{\max}+1)}{12}\beta^2 \right). \end{aligned} \quad (97)$$

Taking the logarithm and using $\ln(1+x) = x - x^2/2 + O(x^3)$ with $x = -\frac{n_{\max}}{2}\beta + \frac{n_{\max}(2n_{\max}+1)}{12}\beta^2$,

we find

$$\begin{aligned}\ln(\mathcal{Z}) &= \ln(n_{\max} + 1) + \ln\left(1 - \frac{n_{\max}}{2}\beta + \frac{n_{\max}(2n_{\max} + 1)}{12}\beta^2\right) \\ &= \ln(n_{\max} + 1) - \frac{n_{\max}}{2}\beta + \frac{n_{\max}(n_{\max} + 2)}{24}\beta^2 + O(\beta^3).\end{aligned}\quad (98)$$

The mean energy $\langle E \rangle = -\partial \ln(\mathcal{Z})/\partial \beta$ is

$$\langle E \rangle = \frac{n_{\max}}{2} - \frac{n_{\max}(n_{\max} + 2)}{12}\beta + O(\beta^2), \quad (99)$$

and the variance $\text{Var}(E) = \partial^2 \ln(\mathcal{Z})/\partial \beta^2$ is

$$\text{Var}(E) = \frac{n_{\max}(n_{\max} + 2)}{12} + O(\beta^2). \quad (100)$$

Using $1/(1-x)^2 \approx 1 + 2x + 3x^2$, we find

$$\begin{aligned}(\Delta U_C)^2 &= \frac{\frac{n_{\max}(n_{\max} + 2)}{12}}{\left(\frac{n_{\max}}{2} - \frac{n_{\max}(n_{\max} + 2)}{12}\beta\right)^2} \approx \frac{n_{\max} + 2}{3n_{\max}} \left(1 + \frac{n_{\max} + 2}{3}\beta + O(\beta^2)\right) \\ &= \frac{1}{3} + \frac{2}{3n_{\max}} + \frac{(n_{\max} + 2)^2}{9n_{\max}}\beta + O(\beta^2).\end{aligned}\quad (101)$$

for large $n_{\max}$, $(n_{\max} + 2)^2 \approx n_{\max}^2$. Thus, in the high-temperature limit, the energy fluctuation behaves as

$$(\Delta U_C)^2 \approx \frac{1}{3} + \frac{2}{3n_{\max}} + \frac{n_{\max}}{9}\beta, \quad (102)$$

which is a linear function of β . Note that the high-temperature limit and thermodynamic limit do not commute with each other, as

$$\lim_{n_{\max} \rightarrow \infty} \lim_{\beta \rightarrow 0} (\Delta U_C)^2 = \frac{1}{3}, \quad (103)$$

while

$$\lim_{\beta \rightarrow 0} \lim_{n_{\max} \rightarrow \infty} (\Delta U_C)^2 = \lim_{\beta \rightarrow 0} e^{\beta\omega} = 1. \quad (104)$$

Physically, at high temperature a finite bath cannot supply many excited levels; the system hits the ceiling at $n_{\max}$, and the fluctuations saturate differently than in the true infinite-bath limit.

However, in the intermediate range of β , a local maximum in energy fluctuation distinguishes the noncanonical distribution from the canonical one, especially for strong system-

bath interaction κ . This maximum can be qualitatively understood as follows: we can write the noncanonical distribution as

$$\tilde{P}_n(E) = \mathcal{Z}^{-1} \exp(-\beta \eta_n n \omega), \quad (105)$$

where

$$\eta_n = 1 - \kappa n / \omega - \kappa n^2 / E, \quad \xi = -\frac{\beta}{E}, \quad (106)$$

is a positive factor for $n \leq n_{\max}$. As $\eta_n \leq 1$, the factor η_n effectively reduces the inverse temperature β appearing in the Boltzmann weight, so that the system behaves as if it were at a higher effective temperature,

$$\beta_{\text{eff}}(n) = \beta \eta_n \leq \beta. \quad (107)$$

thus the linear region for β is enlarged in the noncanonical statistics. In other words, the coupled system behaves as if it were effectively hotter than a canonical oscillator at the same β . As a result, the range of β over which thermodynamic quantities vary approximately linearly with β (the high-temperature linear regime) becomes larger in the noncanonical case. On the fluctuation curves in Fig. 5, this shows up as a broader interval of β over which the noncanonical curve remains almost linear, whereas the canonical curve already bends away. These effects are most pronounced when the interaction strength κ is large and the temperature is high, because then many highly excited levels with $\eta_n \ll 1$ are populated and the deviations from canonical statistics become significant.

Besides the high distribution tail for a single system, the noncanonical statistics provides other new characters when the system is a composite of two subsystems l_1 and l_2 . Even if these two subsystems do not directly interact with each other, the deformation of the energy shell can effectively result in a correlation between them. Here we still use harmonic oscillator (HO) systems for illustration. The system consists of two single-mode HOs with Hamiltonian

$$H_S = \sum_{k=1,2} \omega_k a_k^\dagger a_k. \quad (108)$$

The system interacts with a common small heat bath, which can be modeled by the Hamiltonian

$$H_B = \sum_{j=1}^N \omega_j b_j^\dagger b_j \quad (109)$$

The interaction term reads

$$H_I = \sum_{k=1,2} \sum_{j=1}^N \lambda_{kj} a_k^\dagger a_k (b_j^\dagger + b_j). \quad (110)$$

Following the same discussion about the energy shell deformation for a single system, we can

obtain the joint distribution of the composite system. Given that the system is in level (n, m) , its effective energy is

$$E_{\text{sys}}(n, m) = n\omega_1 + m\omega_2 + \Delta_{n,m} = E_{nm} + \Delta_{n,m}, \quad (111)$$

where $E_{nm} = n\omega_1 + m\omega_2$. Then the energy available to the bath is

$$E_B = E - E_{\text{sys}}(n, m) = E - E_{nm} - \Delta_{n,m}. \quad (112)$$

We now restrict the bath+interaction to the system state $|n, m\rangle$. Using

$$a_1^\dagger a_1 |n, m\rangle = n |n, m\rangle, \quad a_2^\dagger a_2 |n, m\rangle = m |n, m\rangle,$$

we obtain

$$\begin{aligned} H_B + H_I \Big|_{(n,m)} &= \sum_j \left[\omega_j b_j^\dagger b_j + (\lambda_{1j}n + \lambda_{2j}m)(b_j^\dagger + b_j) \right] \\ &= \sum_j \left[\omega_j b_j^\dagger b_j + Q_j(n, m)(b_j^\dagger + b_j) \right], \end{aligned} \quad (113)$$

where $Q_j(n, m) \equiv \lambda_{1j}n + \lambda_{2j}m$. For a fixed pair (n, m) we introduce shifted operators

$$\tilde{b}_j = b_j + \alpha_j, \quad \tilde{b}_j^\dagger = b_j^\dagger + \alpha_j, \quad (114)$$

so that $b_j = \tilde{b}_j - \alpha_j$, $b_j^\dagger = \tilde{b}_j^\dagger - \alpha_j$. Substituting into the Hamiltonian gives

$$\omega_j b_j^\dagger b_j + Q_j(b_j^\dagger + b_j) = \omega_j \tilde{b}_j^\dagger \tilde{b}_j - \omega_j \alpha_j (\tilde{b}_j^\dagger + \tilde{b}_j) + \omega_j \alpha_j^2 + Q_j(\tilde{b}_j^\dagger + \tilde{b}_j - 2\alpha_j) \quad (115)$$

As before, we choose α_j so that the linear terms vanish:

$$-\omega_j \alpha_j + Q_j = 0 \implies \alpha_j = \frac{Q_j}{\omega_j}. \quad (116)$$

With this choice, the total Hamiltonian in the (n, m) sector becomes

$$H \Big|_{(n,m)} = (\omega_1 n + \omega_2 m) + \sum_j \omega_j \tilde{b}_j^\dagger \tilde{b}_j - \sum_j \frac{Q_j(n, m)^2}{\omega_j}. \quad (117)$$

The last term is a system-dependent constant shift, which we denote by

$$\Delta_{nm} = - \sum_j \frac{Q_j(n, m)^2}{\omega_j} = - \sum_{j=1}^N \frac{(\lambda_{1j}n + \lambda_{2j}m)^2}{\omega_j}. \quad (118)$$

Using Eqs. (45) and (68), we get

$$\phi_{nm} = -\beta\Delta_{nm} + \xi\Delta_{nm}E_{nm}, \quad (119)$$

$$\implies \tilde{P}(E, E_n^{(1)}, E_m^{(2)}) = \frac{1}{\mathcal{Z}_t} e^{-\beta E_{nm} - \beta\Delta_{nm} + \xi\Delta_{nm}E_{nm}} = \frac{1}{\mathcal{Z}_t} e^{-\beta(E_{nm} + \Delta_{nm}) + \xi\Delta_{nm}E_{nm}}, \quad (120)$$

where

$$\Delta_{nm} = -\sum_{j=1}^N \frac{(\lambda_{1j}n + \lambda_{2j}m)^2}{\omega_j}, \quad (121)$$

$$\mathcal{Z}_t = \sum'_{n,m} e^{-\beta(E_{nm} + \Delta_{nm}) + \xi\Delta_{nm}E_{nm}}. \quad (122)$$

Here $\sum'_{n,m}$ means the summation of the system energy levels should satisfy the constraint $0 \leq E_{nm} + \Delta_{nm} \leq E$. It can be seen from (120) that the statistics of two subsystems are not independent with each other due to the cross-term in Δ_{nm} . This statistical correlation can be described by mutual information defined as

$$S(l_1 : l_2) = S(l_1) + S(l_2) - S(l_1 + l_2), \quad (123)$$

where the marginal entropies are

$$S(l_k) = -\sum_{n=0}^{n_{\max}} \tilde{P}(E, E_n^{(k)}) \ln \left(\tilde{P}(E, E_n^{(k)}) \right), \quad (124)$$

and the joint entropy is

$$S(l_1 + l_2) = -\sum'_{n,m} \tilde{P}(E, E_n^{(1)}, E_m^{(2)}) \ln \left(\tilde{P}(E, E_n^{(1)}, E_m^{(2)}) \right). \quad (125)$$

Not that

$$\tilde{P}_n^{(1)} = \sum'_m \tilde{P}_{nm}(E) = \frac{1}{\mathcal{Z}_t} \sum'_m e^{-\beta(E_{nm} + \Delta_{nm}) + \xi\Delta_{nm}E_{nm}}, \quad (126)$$

$$\tilde{P}_n^{(2)} = \sum'_n \tilde{P}_{nm}(E) = \frac{1}{\mathcal{Z}_t} \sum'_n e^{-\beta(E_{nm} + \Delta_{nm}) + \xi\Delta_{nm}E_{nm}}. \quad (127)$$

For simplicity, we assume the two HOs are identical with $\omega_1 = \omega_2 \equiv \omega$, $\lambda_{1j} = \lambda_{2j} \equiv \lambda_j$, and κ is defined the same as Eq. (61). As shown in Fig. 6, there appears to be a nonzero mutual entropy if we use noncanonical statistics to describe the composite system in a common small heat bath. In contrast, when the interaction energy is so small that it can be neglected compared with the system and bath energies, the total state is well described by the canonical distribution without coupling. In this limit the joint density matrix can be written as a simple product of two independent Gibbs states, and the mutual entropy (mutual information)

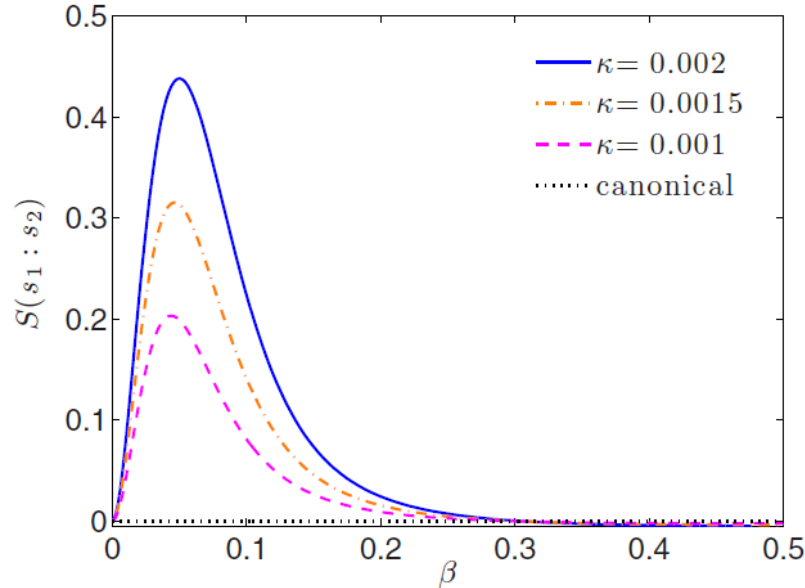


Figure 6: The mutual entropy of two identical harmonic oscillators (HOs) by noncanonical and canonical statistics. The eigenfrequency of two identical HOs is $\omega = 1$, and $\kappa = 0.002$, $n_{\max} = 138$ (the blue solid line); $\kappa = 0.0015$, $n_{\max} = 122$ (orange dotted-dash line); $\kappa = 0.001$, $n_{\max} = 112$ (pink dashed line); and the canonical case for $\kappa = 0$, $n_{\max} = 112$.

becomes

$$S(l_1 : l_2) = 0,$$

which means that the two subsystems are completely uncorrelated.

1.4 Conclusion

In this report we have examined how the usual canonical description of a small quantum system in contact with a heat bath is modified when the bath is finite and the system-bath coupling cannot be neglected. Starting from the microcanonical ensemble of the full "universe" (system+bath), we reviewed how the standard Gibbs state emerges in the thermodynamic limit, and then followed Xu and Sun's construction of an effective Hamiltonian that encodes finite-bath and interaction effects through a deformation of the energy shell. This leads naturally to a noncanonical reduced state for the system, in which the probabilities are no longer simple Boltzmann weights.

For the single harmonic oscillator model, the key ingredient is the level-dependent energy shift $\Delta_n = -\kappa n^2$ generated by a diagonal system-bath coupling. Treating the bath entropy $S(E)$ as a smooth function of the total energy and expanding $S(E - E_n)$ and its interaction-induced correction $\delta_n S(E - E_n)$, one arrives at the noncanonical equilibrium distribution

$$\tilde{P}_n(E) \propto \exp[-\beta(n\omega - \kappa n^2) - \xi \omega \kappa n^3],$$

with $\beta = \partial_E S$ and $\xi = \partial_E^2 S$. The quadratic and cubic terms in n lead to a pronounced high-energy tail compared to the canonical distribution $P_n \propto e^{-\beta n \omega}$, and hence to an enhanced internal energy and larger fluctuations when the coupling κ is significant. At the same time, the bath energy must remain non-negative, and this imposes a finite upper cutoff $n_{\max}$ on the allowed system levels. This makes all thermodynamic quantities bounded in the high-temperature limit, and explains why the limits $\beta \rightarrow 0$ and $n_{\max} \rightarrow \infty$ do not commute.

The analysis of energy fluctuations shows how finite-bath effects behave in the low- and high-temperature limits. For very low temperatures ($\beta \omega \gg 1$), the oscillator effectively becomes a two-level system. In this regime the relative fluctuation $(\Delta U_C)^2 \simeq e^{\beta \omega}$ is the same whether the spectrum is truncated or infinite, so a finite bath and an infinite bath are indistinguishable. For high temperatures and a finite bath, however, the internal energy U and $(\Delta U)^2$ approach finite values set by $n_{\max}$, and the fluctuation curve develops a characteristic shape that cannot be reproduced by the canonical distribution. In an intermediate range of β , the noncanonical distribution can be written in terms of an effective inverse temperature $\beta_{\text{eff}}(n) = \beta \eta_n$ with $\eta_n < 1$ for highly excited levels. This makes the system behave as if it were at a higher temperature than the canonical prediction, which explains the enhanced high-energy tail and the local maximum seen in the fluctuation plots.

The noncanonical framework becomes even more interesting when the system is a composite of two subsystems. For two harmonic oscillators coupled to a common finite bath, the deformed energy shell produces a joint distribution $\tilde{P}_{nm}(E)$ that depends on the combination $(\lambda_{1j}n + \lambda_{2j}m)^2$. Even though the oscillators do not interact directly, their statistics are no longer independent: the joint distribution does not factorize into a product of two marginals, and the mutual information $S(l_1 : l_2)$ is strictly positive. This mutual information measures the correlation induced purely by the shared finite bath. In the weak-coupling, large-bath limit the deformation disappears, the joint state factorizes, and the mutual information tends to zero, recovering the usual picture of two independent subsystems in contact with an ideal reservoir.

2 Finite-bath corrections to the second law of thermodynamics

2.1 Introduction

As we discussed, in the microscopic regime some of the standard assumptions of macroscopic thermodynamics no longer hold. Fluctuations, especially in quantities like work, can be comparable to, or even exceed, the characteristic energy scales of the system, so treating them as negligible becomes unjustified. A second assumption that can fail is that the environment acts as an ideal, infinitely large heat reservoir. At microscopic scales the bath may be finite, so energy exchange can noticeably disturb it, which is exactly the situation we analyzed in the previous section. In particular, when the time scales involved in a process are small, the effective heat bath is necessarily finite, as implied by Lieb-Robinson bounds. We will go through the definition and the implications of this bound in the following sections.

The (macroscopic) second law of thermodynamics can be states as follows. Consider a system in an arbitrary state, with average energy U and entropy S . Suppose that, in order to transform the system, we can make use of a heat bath at temperature T , and extract an average amount of work $\langle W \rangle$. A necessary condition for the possibility of a transformation is

$$\langle W \rangle \leq -\Delta U + T\Delta S = -\Delta F, \quad (128)$$

where ΔU and ΔS are the changes in energy and entropy, and the free energy is defined by $F = U - TS$. Eq. (128) follows by combining the first law, $\Delta U = Q - W$, with the standard second-law (Clausius) bound for an ideal infinite bath at fixed temperature T . A detailed derivation will be given in later sections. Eq. (128) implies that attempting to extract more work than $-\Delta F$ is impossible: no protocol that respects energy conservation and uses a thermal environment can implement such a transformation.

Remarkably, when the bath has infinite heat capacity, the same inequality also becomes sufficient: satisfying $\langle W \rangle \leq -\Delta F$ guarantees that there exists an allowed protocol that achieves the desired transformation with that (or arbitrarily close to that) work value. Therefore, there are no additional hidden constraints coming from disturbing the bath. In this case the temperature stays relatively constant during the process, so one can implement a quasi-static isothermal protocol: change the system in many infinitesimal steps and reequilibrate with the bath at the same β each step. As the step size goes to zero, entropy production can be made arbitrarily small, and the extracted work approaches the reversible value $\langle W \rangle \rightarrow -\Delta F$ and the bath can be left uncorrelated with the system at the end of the process. A bound is called tight if it can be saturated (achieved as an equality) for some optimal protocol. Consequently, the inequality is tight: one can, in principle, implement transformations that saturate the bound and extract $\langle W \rangle = -\Delta F$.

However, in the microscopic scale, the assumption that the bath is infinite might not always be a reasonable approximation. Recall that, within a finite time, a system can only interact with a finite region of its environment. Hence for finite-time processes one should consider an effective bath with a finite heat capacity.

A reasonable question then would be how does this fundamental bound (128) change when we take into account the finiteness of the heat bath? Consider a bath with heat capacity C , which is not necessarily infinite. We find that the necessary and sufficient condition for the possibility of a transformation is

$$\langle W \rangle \leq -\Delta F - \frac{1}{\beta} [I(E' : S') + D(P(E') \| p_G(E'))], \quad (129)$$

where D is the relative entropy defined as

$$D(P(E') \| p_G(E')) = \text{Tr}(P(E') (\ln(P(E')) - \ln(p_G(E')))), \quad (130)$$

$P(E')$ is the probability distribution for the final energy of the bath E' , $p_G(E')$ is the Gibbs distribution at temperature β , and $I(E' : S')$ is the mutual information of the final state of

the system and bath. Note that $P(E')$ is the actual final distribution of bath energies, while $p_G(E')$ is the Gibbs distribution at the bath temperature β . The relative entropy term measures how far the bath has been pushed away from being thermal at temperature β . If the bath is large enough to remain effectively unchanged, you can have $P(E') = p_G(E')$, making the relative entropy term zero. Therefore, this bound has two different contributions: one from the correlations created between system and bath, and another for the change of state of the bath itself. This is a necessary condition simply because if a protocol is a thermal operation, then its work output must obey this inequality. Sufficiency is proven by showing that this bound is tight. (we will show both in the next chapter.)

From (129) we derive a necessary condition that depends only on the initial and final states of the system and the heat capacity C and temperature $T = 1/\beta$ of the bath, which reads

$$\langle W \rangle \leq -\Delta F - \frac{1}{2C\beta} \Delta S^2. \quad (131)$$

The key step is to lower-bound the total irreversibility penalty $I + D$ using only macroscopic bath information, its temperature T and finite heat capacity C , together with the entropy change ΔS of the system. To leading order in $1/C$, this yields a universal minimum correction proportional to $(\Delta S)^2/C$, and substituting this bound into Eq. (129) produces Eq. (131). Compared to Eq. (129), Eq. (131) trades exact tightness for usability: it no longer requires knowledge of the final bath state, but provides a stronger than macroscopic necessary condition whenever the bath has finite heat capacity. Keep in mind that we are just going through what will be covered in this report, and all the detailed derivations will be in the following sections. Evidently, this expression converges to the macroscopic second law (128) in the limit where the heat capacity of the bath C is large and provides a stronger condition when C is finite.

Also condition (131) becomes sufficient when the final state is maximally mixed. When the final system state is maximally mixed, the finite-bath bound (Eq. (131)) becomes achievable; the target state is completely structureless, so one can implement the transformation without leaving extra information in the bath beyond the unavoidable finite heat capacity effect. In that case Eq. (131) is not merely an upper bound; it equals the maximum extractable work. Moreover, because the finite-bath correction depends on $(\Delta S)^2$, it penalizes both directions of a cycle equally. As a result, preparing the initial state from the maximally mixed state requires at least $\frac{1}{2\beta C}(\Delta S)^2$ extra work, while extracting work back to the maximally mixed state loses the same amount, implying $W_{\min}^{\text{prepare}} > W_{\max}^{\text{extract}}$. Therefore, we conclude that thermodynamic reversibility requires an infinite heat bath, as we concluded in the last paper also.

There are situations where stochastic fluctuations of work may be undesirable and one would like to manipulate only definite amounts of useful energy. To address this, the framework of single-shot thermodynamics was established in *Truly work-like work extraction via a single-shot analysis* By Johan Åberg and *Fundamental limitations for quantum and nanoscale thermodynamics* by Micha Horodecki & Jonathan Oppenheim, where the concept of deter-

ministic work was introduced. Single-shot thermodynamics studies thermodynamic state transformations beyond the many-run, average work regime, focusing instead on what can be guaranteed in an individual run (or except for a small failure probability). In this framework one introduces deterministic work, where work is represented by a work storage system (weight) that undergoes a definite energy shift: $\rho \otimes |0\rangle\langle 0| \rightarrow \sigma \otimes |W\rangle\langle W|$, with $|0\rangle$ and $|W\rangle$ energy eigenstates of the weight. They also gave expressions for how much one can extract from any given state and for how much one needs to create it, assuming access to an infinite heat bath. This contrasts with fluctuating-work scenarios where the weight’s final energy (and hence the work) is random. The single-shot framework therefore replaces average-work bounds with quantities suited to guaranteed performance.

In the paper we’re discussing they explore how that amount of deterministic work content and work cost of forming a state changes when one is limited to a finite bath and we find that achieving these processes with arbitrary accuracy becomes impossible. This may not be the case, however, if one allows for additional small error probability during the processes, given by the tails of the distribution of energy of the heat bath. For such a case, we give expressions for the deterministic work for a given probability of failure.

Finite-bath thermodynamics has recently been studied in several settings (e.g., corrections to Carnot efficiency, Landauer erasure, and fluctuation relations). Variants of the work bound used here were already derived in earlier papers, some obtained essentially the same inequality, and others recovered only part of the correction (e.g., the term $D(P(E')\|p_G(E'))$). The key difference is that in this paper they also show that the bound is achievable. In other words, if your process satisfies the inequality, then there really is a protocol (within their model) that can reach that amount of work.

Some papers model an infinite bath in a very specific way: as infinitely many noninteracting copies of a small thermal system, and then study finite bath effects by taking only a finite number of copies. Those finite copy results lead to corrections that are qualitatively similar to the corrections discussed in this paper. The main difference is that such approaches rely on a particular microscopic model of the bath (many identical thermal subsystems). In contrast, Richens-Alhambra-Masanes do not assume any detailed bath structure; they characterize the bath only through macroscopic properties, most importantly its finite heat capacity, which makes their bounds more broadly applicable across different physical baths.

We begin in Sec. 2.2 by introducing a model-independent description of a finite heat bath. We then move to Sec. 2.3, where we set up the general framework for thermodynamic system-bath interactions and study work-extraction protocols in the fluctuating-work regime; this leads us to the main bound, which we then illustrate with the example of Landauer erasure. Finally, in Sec. 2.4 we turn to interactions between system and bath that do not involve an explicit work storage device, and use this setting to derive fundamental limits on deterministic work extraction and work expenditure.

2.2 General characterization of a finite bath

We consider a bath to be a large (but in this case not infinitely so) system with a density of states given by $\Omega(E, V) = e^{S(E, V)}$, where $S(E, V)$ is the entropy in the microcanonical ensemble for a given energy E and volume V . We shall make three assumptions about it.

- (1) The entropy $S(E, V)$ is extensive:

$$S(kE, kV) = kS(E, V), \quad \text{for all } k > 0.$$

- (2) The dimensionless volume V is large.

- (3) The bath is in a Gibbs state with a given temperature β , such that a microstate of energy E has probability $\frac{1}{\mathcal{Z}}e^{-\beta E}$, where $\mathcal{Z} = \text{Tr}(e^{\beta E})$.

Throughout, we work in units where Boltzmann's constant is set to unity, $k_B = 1$. Assumption (1) implies that we can write the entropy $S(E, V)$ as

$$S(E, V) = Vf(u), \tag{132}$$

for some function $f(u)$ of the energy density $u = E/V$. If the energy E has degeneracy $\Omega(E)$ (i.e., there are $\Omega(E)$ different microstates with that same energy), then

$$p(E) = \sum_{i:E_i=E} p(E_i) = \sum_{i:E_i=E} \frac{e^{-\beta E}}{\mathcal{Z}} = \frac{\Omega(E)e^{-\beta E}}{\mathcal{Z}}. \tag{133}$$

We are not interested in the overall normalization, so we write the distribution as

$$p(E) \propto \Omega(E)e^{-\beta E}. \tag{134}$$

Using the definition of the microcanonical entropy,

$$S = \ln(\Omega(E)) \implies \Omega(E) = e^{S(E)} = e^{Vf(u)}, \tag{135}$$

where we used extensivity to write $S(E, V) = Vf(u)$ with the energy density $u = E/V$. Substituting $E = Vu$ then gives

$$p(u) \propto e^{V[f(u) - \beta u]}. \tag{136}$$

Define the exponent function

$$g(u) := f(u) - \beta u. \tag{137}$$

For large V , the dominant contribution comes from where the exponent is maximal. Hence,

$$g'(u) = f'(u) - \beta = 0 \implies f'(u_\beta) = \beta, \tag{138}$$

where we name u_β the absolute maximum of $f(u) - \beta u$. Also

$$g''(u) = f''(u), \quad (139)$$

and since we want to have a maximum, we set

$$g''(u_\beta) = f''(u_\beta) < 0. \quad (140)$$

Now we do a Taylor expansion:

$$g(u) = g(u_\beta) + g'(u_\beta)(u - u_\beta) + \frac{1}{2}g''(u_\beta)(u - u_\beta)^2 + \dots$$

The linear term vanishes since $g'(u_\beta) = 0$. Therefore,

$$p(u) \propto e^{V(g(u_\beta) + \frac{1}{2}f''(u_\beta)(u - u_\beta)^2)} = e^{V(f(u_\beta) - \beta u_\beta + \frac{1}{2}f''(u_\beta)(u - u_\beta)^2)}. \quad (141)$$

Once β is fixed, u_β is a constant, so terms like $f(u_\beta)$ only affect normalization and can be absorbed into the proportionality constant. Using Eq. (140), we write

$$f''(u_\beta) = -|f''(u_\beta)| \implies p(u) \propto \exp\left(-\frac{V}{2}|f''(u_\beta)|(u - u_\beta)^2\right). \quad (142)$$

This expression has the form of a normal distribution with average

$$\langle u \rangle = u_\beta, \quad (143)$$

and variance

$$\frac{1}{2\sigma^2} = \frac{V}{2}|f''(u_\beta)| \implies \sigma^2 = \langle (u - \langle u \rangle)^2 \rangle = |Vf''(u_\beta)|^{-1}. \quad (144)$$

Now, let us relate $f''(u_\beta)$ to the heat capacity, defined as

$$C = \frac{d\langle U \rangle}{dT} = V \frac{d\langle u \rangle}{dT} = -\frac{V}{T^2} \frac{d\langle u \rangle}{d\beta} = -\frac{V}{T^2} \frac{du_\beta}{d\beta}. \quad (145)$$

Differentiating (138) with respect to β gives

$$f''(u_\beta) \frac{du_\beta}{d\beta} = 1 \implies \frac{du_\beta}{d\beta} = \frac{1}{f''(u_\beta)}, \quad (146)$$

and substituting in (145) gives

$$C = -\frac{V}{T^2} \frac{1}{f''(u_\beta)}. \quad (147)$$

Note that Eq. (140) implies that the heat capacity is positive, as is always the case in ordinary matter. Also note that $C \propto V$, because $f(u_\beta)$ is independent of V . Using this and

Eq. (144) implies that the fluctuations of u are

$$\langle (u - \langle u \rangle)^2 \rangle^{1/2} = \frac{T\sqrt{C}}{V} \propto V^{-1/2}, \quad (148)$$

which are small when V is large.

We rescale the energy so that the mean is at zero, $E \rightarrow E - \langle E \rangle$. Now we do a Taylor expansion of $S(E, V)$ around $E = 0$:

$$S(E, V) \approx S(0, V) + S'(0, V)E + \frac{1}{2}S''(0, V)E^2. \quad (149)$$

For a thermal bath, the usual thermodynamic identity is

$$\frac{\partial S}{\partial E} = \beta = \frac{1}{T}.$$

Differentiate this with respect to E and use Eq. (145) to get

$$\frac{\partial^2 S}{\partial E^2} = \frac{d\beta}{dE} = \frac{d\beta}{d\langle U \rangle} = -\frac{1}{CT^2}. \quad (150)$$

Define

$$\gamma := \frac{1}{CT^2}. \quad (151)$$

Plug the expansion into $\Omega = e^S$,

$$\Omega(E, V) \approx \exp\left(S(0, V) + \beta E - \frac{\gamma}{2}E^2\right). \quad (152)$$

$S(0, V)$ is just a constant and independent of E . Thus, we are finally able to approximate the density of states of the bath as

$$\Omega(E, V) \propto \exp\left(\beta E - \frac{\gamma}{2}E^2\right). \quad (153)$$

2.3 Fluctuating work

2.3.1 Thermal operations with fluctuating work

Next, the authors introduce a standard framework for describing thermodynamic transformations. The setting includes a system with Hamiltonian H_S , a bath with Hamiltonian H_B initially prepared in the thermal state (as described in the previous section), and an ideal weight with Hamiltonian

$$H_W = \int_{\mathbb{R}} dx \, x |x\rangle\langle x|, \quad (154)$$

where the orthonormal basis $|x\rangle : x \in \mathbb{R}$ labels the weights position. Note the assumption that the initial state of system, bath, and weight is uncorrelated. The weight acts as an ideal battery with a continuous energy spectrum: a transition $|x\rangle \rightarrow |x + W\rangle$ corresponds to storing a definite amount of work W as an increase in the weights energy. This device lets one cleanly distinguish fluctuating work (the weight ends in a distribution over energies) from deterministic work (the weight ends in a single energy eigenstate).

By contrast, XuSun study equilibrium properties of a strongly coupled system with a finite bath; since they do not consider explicit work extraction protocols, an explicit work storage system is unnecessary in their setup. Intuitively, without a weight, a decrease in the systems energy cannot be unambiguously interpreted as work extracted, because the same energy could simply have been transferred as heat to the bath. With a weight present, work is identified operationally as the energy gained by the weight.

In quantum mechanics, any physically implementable operation on SBW can be modeled by coupling SBW to additional degrees of freedom E (representing control or an environment), evolving the closed composite $SBWE$ unitarily, and then tracing out E . The resulting reduced map on SBW is therefore of the form

$$\Gamma_{SBW}(\rho) = \text{Tr}_E \left(U(\rho_{SBW} \otimes \sigma_E) U^\dagger \right),$$

which is CPTP. Conversely, by Stinespring dilation³, every CPTP map admits such a realization, so CPTP maps capture the most general class of physically valid transformations on SBW under the assumption of no initial system-environment correlations. This map satisfies the following conditions.

- *Microscopic reversibility (second law)*. It has an (CPTP) inverse Γ_{SBW}^{-1} , which implies unitarity $\Gamma_{SBW}(\rho_{SBW}) = U \rho_{SBW} U^\dagger$.

A standard statement of the second law for a process involving a heat bath at inverse temperature $\beta = 1/T$ is the Clausius inequality

$$\Delta S_S - \beta Q \geq 0, \tag{155}$$

where $\Delta S_S := S(\rho'_S) - S(\rho_S)$ is the change in the system's von Neumann entropy and Q is the heat absorbed by the system from the bath.

Assume the initial global state is uncorrelated, $\rho_{SB} = \rho_S \otimes \tau_B$, and that the joint evolution on SBW is unitary. Since unitaries preserve total entropy,

$$S(\rho'_{SB}) = S(\rho_{SB}), \quad \text{and since } \rho_{SB} = \rho_S \otimes \tau_B, \quad S(\rho_{SB}) = S(\rho_S) + S(\tau_B).$$

Consider the mutual information after the process,

$$I(S:B)' := S(\rho'_S) + S(\rho'_B) - S(\rho'_{SB}).$$

³J. Watrous, *The Theory of Quantum Information* (Cambridge University Press, 2018), Sec. 2.2.

Because mutual information is nonnegative, $I(S : B)' \geq 0$, and using $S(\rho'_{SB}) = S(\rho_{SB})$ we obtain

$$\begin{aligned} I(S : B)' &= S(\rho'_S) + S(\rho'_B) - S(\rho_{SB}) \\ &= (S(\rho'_S) - S(\rho_S)) + (S(\rho'_B) - S(\tau_B)) \\ &= \Delta S_S + \Delta S_B \geq 0. \end{aligned} \tag{156}$$

This is where microscopic reversibility enters: if the global dynamics were not unitary, entropy could flow to unmodelled degrees of freedom, and one could not identify the change in local entropies purely with correlation-building within the model. Requiring a CPTP inverse guarantees unitarity and enables the clean entropy-balance relation (156).

To connect (156) to Clausius, use the fact that the bath begins in the Gibbs state $\tau_B = e^{-\beta H_B} / Z_B$. By definition of relative entropy,

$$D(\rho'_B \| \tau_B) = \text{Tr}(\rho'_B \log \rho'_B) - \text{Tr}(\rho'_B \log \tau_B).$$

Using $S(\rho) = -\text{Tr}(\rho \log \rho)$ and $\log \tau_B = -\beta H_B - \log Z_B$, we get

$$D(\rho'_B \| \tau_B) = -S(\rho'_B) + \beta \text{Tr}(H_B \rho'_B) + \log Z_B. \tag{157}$$

Since $D(\tau_B \| \tau_B) = 0$, applying (157) to $\rho'_B = \tau_B$ yields

$$0 = -S(\tau_B) + \beta \text{Tr}(H_B \tau_B) + \log Z_B \quad \Rightarrow \quad \log Z_B = S(\tau_B) - \beta \text{Tr}(H_B \tau_B).$$

Substituting this into (157) gives

$$\begin{aligned} D(\rho'_B \| \tau_B) &= \beta \left(\text{Tr}(H_B \rho'_B) - \text{Tr}(H_B \tau_B) \right) - \left(S(\rho'_B) - S(\tau_B) \right) \\ &= \beta \Delta E_B - \Delta S_B \geq 0, \end{aligned} \tag{158}$$

where $\Delta E_B := \text{Tr}(H_B \rho'_B) - \text{Tr}(H_B \tau_B)$ and $\Delta S_B := S(\rho'_B) - S(\tau_B)$.

Define the heat absorbed by the system as $Q := -\Delta E_B$. Then (158) becomes

$$\Delta S_B \leq -\beta Q \quad \Rightarrow \quad -\Delta S_B \geq \beta Q.$$

Combining with (156) yields

$$\Delta S_S \geq -\Delta S_B \geq \beta Q,$$

and therefore

$$\Delta S_S - \beta Q \geq 0, \tag{159}$$

which is precisely the Clausius form of the second law.

Finally, we can relate this to the macroscopic work bound (128). Using the first law in the convention $\Delta U = Q - W$ (i.e. $W = Q - \Delta U$), Clausius implies $Q \leq T \Delta S_S$, and hence

$$W = Q - \Delta U \leq T \Delta S_S - \Delta U = -\Delta F,$$

where $\Delta F := \Delta U - T \Delta S_S$. This reproduces the macroscopic second-law inequality stated in (128).

We went through this derivation to make explicit how the microscopic reversibility assumption (i.e. global unitarity, ensured here by the existence of a CPTP inverse) leads to the Clausius inequality and hence to the macroscopic work bound (128). This is why the authors label this requirement as second-law: it is not itself the second law, but the microscopic postulate that enables one to derive second-law constraints within their operational framework.

- *Energy conservation (first law).* $[U, H_S + H_B + H_W] = 0$.
- *Independence from the "position" of the weight.* The unitary commutes with the translations on the weight $[U, \Delta_W] = 0$. The generator of the translations Δ_W is canonically conjugated to the position (or energy) of the weight $[H_W, \Delta_W] = i$.

Δ_W is defined as the generator of translations of the weight. A translation on the weight is a family of unitaries $T(a)$ that shift the weight's position or energy label:

$$T(a) |x\rangle = |x + a\rangle.$$

Equivalently, on wavefunctions $\psi(x)$ it acts as

$$(T(a)\psi)(x) = \psi(x - a),$$

i.e. the whole wavefunction is shifted. This is exactly the same content as $T(a) |x\rangle = |x + a\rangle$, just written in function language.⁴ if $a \rightarrow T(a)$ is a continuous one-parameter unitary group, then Stone's theorem⁵ says there exists a unique self-adjoint operator (the generator) Δ_W such that

$$T(a) = e^{-ia\Delta_W}.$$

So Δ_W can be defined as

$$\Delta_W = i \frac{d}{da} T(a) \Big|_{a=0}, \tag{160}$$

on its domain. Since the authors take H_W and Δ_W to be conjugate variables, the situation is analogous to the canonical commutation relation $[X, P] = i$ for position X and momentum P . In this representation, H_W effectively plays the role of the position operator X , because it is diagonal in the $|x\rangle$ basis with eigenvalue x :

$$H_W |x\rangle = x |x\rangle.$$

⁴Wikipedia, Translation operator (quantum mechanics)

⁵Wikipedia, Stone's theorem on one-parameter unitary groups.

To determine how $T(a)$ acts on H_W , we use Hadamard's lemma.⁶

$$e^A B e^{-A} = B + [A, B] + \frac{1}{2!} [A, [A, B]] + \dots \quad (161)$$

With $T(a) = e^{-ia\Delta_W}$, we obtain

$$\begin{aligned} T(a)^\dagger H_W T(a) &= e^{ia\Delta_W} H_W e^{-ia\Delta_W} \\ &= H_W + [ia\Delta_W, H_W] + \frac{1}{2!} [ia\Delta_W, [ia\Delta_W, H_W]] + \dots \end{aligned} \quad (162)$$

Using the conjugacy relation $[H_W, \Delta_W] = i$ (equivalently $[\Delta_W, H_W] = -i$), all higher nested commutators vanish and we get

$$T(a)^\dagger H_W T(a) = H_W + a. \quad (163)$$

This tells us that a translation of the weight by a simply adds a constant a to the weight Hamiltonian. Equivalently, $T(a)$ shifts the spectrum as

$$H_W \mapsto H_W + a,$$

so the weight acts as an ideal work storage device: translating its position/energy by a increases its energy by exactly a , independent of where it started.

The condition $[U, \Delta_W] = 0$ is equivalent to invariance under weight translations. Indeed, since

$$T(a) = e^{-ia\Delta_W},$$

we have $[U, \Delta_W] = 0 \iff [U, I_{SB} \otimes T(a)] = 0$ for all a , i.e.

$$U (I_{SB} \otimes T(a)) = (I_{SB} \otimes T(a)) U \iff U = (I_{SB} \otimes T(a)) U (I_{SB} \otimes T(a)^\dagger). \quad (164)$$

In words, the global protocol does not depend on the absolute position of the weight: translating the weight and then applying U is equivalent to applying U and then translating.

Because U commutes with Δ_W , it preserves the spectral subspaces of Δ_W . In the discrete case, $\Delta_W = \sum_p p |p\rangle\langle p|$, this implies a block-diagonal form

$$U = \sum_p |p\rangle\langle p|_W \otimes U_{SB}(p).$$

For the ideal weight, Δ_W typically has continuous spectrum, and the sum is replaced by a direct integral,

$$U = \int_{\mathbb{R}} dp |p\rangle\langle p|_W \otimes U_{SB}(p),$$

meaning that for each spectral value p , the unitary acts within that sector and applies some

⁶Physicspages, *Hadamard's Lemma*, [PDF document](#).

unitary $U_{SB}(p)$ on SB .

Substituting this decomposition into the reduced map on SB ,

$$\begin{aligned}\Phi_{SB}(\rho_{SB}) &:= \text{Tr}_W \left(U(\rho_{SB} \otimes \rho_W) U^\dagger \right) \\ &= \text{Tr}_W \left(\left[\int dp |p\rangle\langle p| \otimes U_{SB}(p) \right] (\rho_{SB} \otimes \rho_W) \left[\int dp' |p'\rangle\langle p'| \otimes U_{SB}(p')^\dagger \right] \right),\end{aligned}\quad (165)$$

one finds that if the weight state is diagonal in the $\{|p\rangle\}$ basis (i.e. $\rho_W = \int dp \mu(p) |p\rangle\langle p|$), then only the diagonal contributions remain after tracing out W . In that case,

$$\Phi_{SB}(\rho_{SB}) = \int_{\mathbb{R}} dp \mu(p) U_{SB}(p) \rho_{SB} U_{SB}(p)^\dagger, \quad (166)$$

where $\mu(p) := \langle p | \rho_W | p \rangle \geq 0$ and $\int dp \mu(p) = 1$. Hence, under $[U, \Delta_W] = 0$ (and for a weight state diagonal in the Δ_W basis), the reduced dynamics on SB is a convex mixture of unitaries.

- *Classicality of work.* Before and after applying the global map Γ_{SBW} the position of the weight is measured, obtaining outcomes $|x\rangle$ and $|x + W\rangle$, respectively. In general, the work W is a fluctuating random variable.

The *classicality of work* assumption provides an operational definition of work in the weight framework: the position of the weight is measured in the $\{|x\rangle\}$ basis before and after applying the global map (or global unitary), yielding outcomes x_i and x_f (equivalently, $|x_i\rangle$ and $|x_f\rangle$). The work assigned to that run is then

$$W := x_f - x_i.$$

Because x_i and x_f are classical measurement records, W is a classical random variable and generally fluctuates from run to run.

To describe these measurements within the quantum formalism, let

$$\Pi_x := |x\rangle\langle x|$$

denote the projector onto the weight eigenstate $|x\rangle$. In a single run, conditioned on observing outcome x , the (selective) post-measurement state is

$$\rho_{W|x} = \frac{\Pi_x \rho_W \Pi_x}{\text{tr}(\Pi_x \rho_W)}.$$

If instead one does *not* condition the quantum state on the observed value of x (i.e. one averages over outcomes, or equivalently discards/traces out the classical record), the measurement induces a *non-selective* update. This non-selective update is described by a *pinching channel* (also called the Lüders channel), defined as follows.

Let $\mathcal{X}$ be a complex Euclidean space. A channel $\Phi \in \mathcal{C}(\mathcal{X})$ is said to be a *pinching channel* (or simply a *pinching*) if there exists a collection $\{\Pi_a : a \in \Sigma\}$ of projection operators satisfying

$$\sum_{a \in \Sigma} \Pi_a = \mathbb{I}_{\mathcal{X}},$$

(i.e. $\{\Pi_a\}$ represents a projective measurement) such that

$$\Phi(X) = \sum_{a \in \Sigma} \Pi_a X \Pi_a,$$

for all $X \in L(\mathcal{X})$.

-J. Watrous, *The Theory of Quantum Information*, Chapter 4 ([PDF](#)).

Specializing to the weight position basis, the corresponding pinching (dephasing) map is

$$\mathcal{P}_x(\rho_W) = \sum_x \Pi_x \rho_W \Pi_x$$

in the discrete case, and formally

$$\mathcal{P}_x(\rho_W) = \int_{\mathbb{R}} dx \Pi_x \rho_W \Pi_x$$

in the continuous case. In either case, $\mathcal{P}_x$ removes all coherences between different $|x\rangle$ sectors while preserving the classical distribution

$$p(x) = \langle x | \rho_W | x \rangle.$$

To make the role of discarding the outcome explicit, one may introduce a classical register X that stores the measurement outcome. After the first measurement, the joint classical-quantum state can be written as

$$\rho_{XW} = \int_{\mathbb{R}} dx p(x) |x\rangle\langle x|_X \otimes \rho_{W|x}.$$

Keeping the record corresponds to keeping ρ_{XW} . If instead one ignores the record in the quantum description (i.e. traces it out), one obtains

$$\text{tr}_X(\rho_{XW}) = \int_{\mathbb{R}} dx \Pi_x \rho_W \Pi_x = \mathcal{P}_x(\rho_W).$$

Thus, applying the pinching map $\mathcal{P}_x$ corresponds precisely to discarding the classical measurement register (not conditioning on the recorded outcome), even though the outcomes x_i and x_f may still be used to form the classical random variable $W = x_f - x_i$.

Equivalently, since the initial x -measurement is part of the operational definition of work and the dynamics is not conditioned on the specific value of x_i , one may assume without loss of

generality that the input weight state is already x -diagonal, i.e.

$$\rho_W = \mathcal{P}_x(\rho_W).$$

Finally, since

$$H_W = \int_{\mathbb{R}} dx \, x |x\rangle\langle x|$$

is diagonal in the same basis, it follows that the (unconditioned) post-measurement state commutes with the weight Hamiltonian:

$$[\mathcal{P}_x(\rho_W), H_W] = 0.$$

In particular, if $\rho_W = \mathcal{P}_x(\rho_W)$, then $[\rho_W, H_W] = 0$.

A key property of the von Neumann entropy is concavity:

$$S\left(\sum_k p_k \rho_k\right) \geq \sum_k p_k S(\rho_k),$$

and similarly for integrals. Therefore, applying this property to (166) yields

$$\begin{aligned} S(\Phi_{SB}(\rho_{SB})) &= S\left(\int dp \, \mu(p) U_{SB}(p) \rho_{SB} U_{SB}(p)^\dagger\right) \\ &\geq \int dp \, \mu(p) S(U_{SB}(p) \rho_{SB} U_{SB}(p)^\dagger) = S(\rho_{SB}), \end{aligned} \quad (167)$$

where we used unitary invariance of entropy in the final step. Thus, these transformations cannot decrease the entropy of the joint system and bath, which prevents the weight from being exploited as a hidden source of free energy.

Let us define the (unnormalized) coherence-mode projection map

$$\Theta_\alpha[\rho_S] := \int_{-\infty}^{\infty} dt \, e^{i\alpha t} e^{iH_S t} \rho_S e^{-iH_S t}. \quad (168)$$

This is the α -Fourier component of the Heisenberg-evolved state $\rho_S(t) := e^{iH_S t} \rho_S e^{-iH_S t}$, and it extracts precisely the part of ρ_S corresponding to the energy gap α .

To see this explicitly, expand ρ_S in the energy eigenbasis $H_S|n\rangle = E_n|n\rangle$:

$$\rho_S = \sum_{m,n} \rho_{mn} |m\rangle\langle n|. \quad (169)$$

Then

$$e^{iH_S t} |m\rangle\langle n| e^{-iH_S t} = e^{i(E_m - E_n)t} |m\rangle\langle n|,$$

so substituting into (168) gives

$$\Theta_\alpha[\rho_S] = \sum_{m,n} \rho_{mn} \left(\int_{-\infty}^{\infty} dt e^{i(\alpha+E_m-E_n)t} \right) |m\rangle\langle n|. \quad (170)$$

Using the Fourier representation of the Dirac delta,

$$\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dt e^{ixt},^7$$

we obtain (up to the overall factor 2π , depending on convention)

$$\int_{-\infty}^{\infty} dt e^{i(\alpha+E_m-E_n)t} = 2\pi \delta(\alpha + E_m - E_n), \quad (171)$$

and therefore

$$\Theta_\alpha[\rho_S] = 2\pi \sum_{m,n} \rho_{mn} \delta(\alpha + E_m - E_n) |m\rangle\langle n|. \quad (172)$$

Hence Θ_α filters ρ_S to retain only matrix elements satisfying

$$E_n - E_m = \alpha.$$

Θ_α isolates the coherences between levels separated by an energy gap α ; these are often called *modes of coherence*.⁸

Energy conservation, the classicality-of-work assumption, and the fact that the initial bath state commutes with its Hamiltonian imply that the reduced system map Γ_S is covariant under time translations generated by H_S . Equivalently, it commutes with the coherence-mode projectors Θ_α :

$$\Theta_\alpha \circ \Gamma_S = \Gamma_S \circ \Theta_\alpha. \quad (173)$$

Here

$$\Gamma_S(\rho_S) := \text{Tr}_{BW} \left(U(\rho_S \otimes \rho_B \otimes \rho_W) U^\dagger \right) \quad (174)$$

is the reduced transformation on the system.

Proof, Step 1: Time-translation covariance of Γ_S . Define the system time-translation channel

$$\mathcal{U}_t^S(\rho) := e^{-iH_S t} \rho e^{+iH_S t}. \quad (175)$$

⁷Wikipedia, “Dirac delta function” (oldid: 1335765949).

⁸Lostaglio, Korzekwa, Jennings & Rudolph, “Quantum coherence, time-translation symmetry, and thermodynamics” (2015).

We claim that

$$\Gamma_S(\mathcal{U}_t^S(\rho_S)) = \mathcal{U}_t^S(\Gamma_S(\rho_S)) \quad \forall t \in \mathbb{R}. \quad (176)$$

Starting from the left-hand side,

$$\Gamma_S(\mathcal{U}_t^S(\rho_S)) = \text{Tr}_{BW} \left(U \left((e^{-iH_S t} \rho_S e^{+iH_S t}) \otimes \rho_B \otimes \rho_W \right) U^\dagger \right). \quad (177)$$

Introduce the global time-translation unitary

$$V_t := e^{-i(H_S + H_B + H_W)t} = e^{-iH_S t} \otimes e^{-iH_B t} \otimes e^{-iH_W t}.$$

Then one can rewrite

$$(e^{-iH_S t} \rho_S e^{+iH_S t}) \otimes \rho_B \otimes \rho_W = V_t \left(\rho_S \otimes e^{+iH_B t} \rho_B e^{-iH_B t} \otimes e^{+iH_W t} \rho_W e^{-iH_W t} \right) V_t^\dagger.$$

Now use the assumptions $[H_B, \rho_B] = 0$ (the bath is initially stationary, e.g. thermal) and $[H_W, \rho_W] = 0$ (classicality of work: the weight starts diagonal in the energy basis). These imply

$$e^{+iH_B t} \rho_B e^{-iH_B t} = \rho_B, \quad e^{+iH_W t} \rho_W e^{-iH_W t} = \rho_W,$$

so the expression simplifies to

$$(e^{-iH_S t} \rho_S e^{+iH_S t}) \otimes \rho_B \otimes \rho_W = V_t (\rho_S \otimes \rho_B \otimes \rho_W) V_t^\dagger.$$

Hence

$$\Gamma_S(\mathcal{U}_t^S(\rho_S)) = \text{Tr}_{BW} \left(U V_t (\rho_S \otimes \rho_B \otimes \rho_W) V_t^\dagger U^\dagger \right). \quad (178)$$

Energy conservation, $[U, H_S + H_B + H_W] = 0$, is equivalent to $UV_t = V_t U$. Therefore,

$$\Gamma_S(\mathcal{U}_t^S(\rho_S)) = \text{Tr}_{BW} \left(V_t U (\rho_S \otimes \rho_B \otimes \rho_W) U^\dagger V_t^\dagger \right). \quad (179)$$

Writing $V_t = e^{-iH_S t} \otimes e^{-i(H_B + H_W)t}$ and using invariance of the partial trace under unitary conjugation on the traced subsystem,

$$\text{Tr}_{BW} \left((I_S \otimes V) X (I_S \otimes V^\dagger) \right) = \text{Tr}_{BW}(X),$$

we obtain

$$\Gamma_S(\mathcal{U}_t^S(\rho_S)) = e^{-iH_S t} \text{Tr}_{BW} \left(U (\rho_S \otimes \rho_B \otimes \rho_W) U^\dagger \right) e^{+iH_S t} = \mathcal{U}_t^S(\Gamma_S(\rho_S)), \quad (180)$$

which proves the covariance relation (176).

Step 2: Commutation with Θ_α . Recall the definition

$$\Theta_\alpha(\rho) = \int_{\mathbb{R}} dt \, e^{i\alpha t} e^{+iH_S t} \rho e^{-iH_S t} = \int_{\mathbb{R}} dt \, e^{i\alpha t} \mathcal{U}_{-t}^S(\rho).$$

Using covariance (176),

$$\begin{aligned} (\Theta_\alpha \circ \Gamma_S)(\rho) &= \int dt \, e^{i\alpha t} \mathcal{U}_{-t}^S(\Gamma_S(\rho)) \\ &= \int dt \, e^{i\alpha t} \Gamma_S(\mathcal{U}_{-t}^S(\rho)) \\ &= \Gamma_S \left(\int dt \, e^{i\alpha t} \mathcal{U}_{-t}^S(\rho) \right) \\ &= (\Gamma_S \circ \Theta_\alpha)(\rho), \end{aligned} \tag{181}$$

where we used linearity of Γ_S to pull it outside the integral.

In summary, under these assumptions Γ_S is time-translation covariant and therefore cannot mix different coherence modes: each α -sector is mapped into itself.

Note the assumption that the initial state of system, bath, and weight is uncorrelated. Setting $\alpha = 0$ singles out the energy-incoherent part of the state. In particular, one can show that if the initial system state commutes with H_S , then the final system state does as well. Starting from Eq. (168), for $\alpha = 0$ we have

$$\Theta_0[\rho_S] = \int_{\mathbb{R}} dt \, e^{iH_S t} \rho_S e^{-iH_S t}.$$

From this expression it is clear that

$$\Theta_0[\rho_S] = \rho_S \iff [\rho_S, H_S] = 0,$$

because ρ_S is unchanged by averaging over all time-translations iff it is invariant under the unitary generated by H_S , which is equivalent to commuting with H_S .

Now use the commutation (covariance) identity at $\alpha = 0$:

$$\Theta_0(\Gamma_S(\rho_S)) = \Gamma_S(\Theta_0(\rho_S)).$$

If the initial state commutes with H_S , then $\Theta_0(\rho_S) = \rho_S$, and therefore

$$\Theta_0(\Gamma_S(\rho_S)) = \Gamma_S(\rho_S).$$

This means the output is a fixed point of Θ_0 , i.e. it has no coherence between different energy eigenspaces. Equivalently,

$$[\Gamma_S(\rho_S), H_S] = 0. \tag{182}$$

For the converse direction, assume that the *final* state commutes with H_S . Then

$$\Theta_0(\Gamma_S(\rho_S)) = \Gamma_S(\rho_S).$$

Using covariance again gives

$$\Gamma_S(\rho_S) = \Theta_0(\Gamma_S(\rho_S)) = \Gamma_S(\Theta_0(\rho_S)).$$

So we can conclude that ρ_S and its dephased version $\Theta_0(\rho_S)$ lead to the *same* final state under Γ_S ; in other words, any initial coherences are irrelevant for the system output in this case.

Missing assumption The step

$$\Gamma_S(\rho_S) = \Gamma_S(\Theta_0(\rho_S)) \Rightarrow \Theta_0(\rho_S) = \rho_S$$

does *not* hold in general, because Γ_S need not be injective (many different inputs can produce the same output after tracing out BW). If Γ_S were injective (for example, if Γ_S were unitary on S), then the above equality would indeed imply $\Theta_0(\rho_S) = \rho_S$, and hence $[\rho_S, H_S] = 0$.

In this paper the authors only consider processes in which both states are diagonal. For example, optimal work extraction from an arbitrary initial state is one such process. We write the initial and final states as $\rho_S = \sum_S P(s) |s\rangle \langle s|$ and $\rho'_S = \sum_{s'} P(s') |s'\rangle \langle s'|$, respectively, where $|s\rangle$ and $|s'\rangle$ are the initial and final energy eigenstates. Note that we allow initial and final Hamiltonians H_S and H'_S to be not necessarily equal.

2.3.2 Corrections to the second law

In this section we analyze the transformation power of the operations defined in the previous section and study the effects of not having an infinite heat bath. First, we present a generalization to the second law (128) to the case of arbitrary heat bath. This necessary and sufficient condition has an important limitation in that it involves the final state of the bath. However, it precisely articulates the effect of having a finite bath. Later, more practical bounds that are independent from the state of the bath are provided.

Theorem 1. A necessary and sufficient condition for the possibility of a transformation after a thermal operation is

$$\langle W \rangle \leq -\Delta F - \frac{1}{\beta} [I(E' : S') + D(P(E') \| p_G(E'))], \quad (183)$$

where $D(p \| q) = \sum_{x,y} p(x) \log(p(x)/q(x))$ is the relative entropy, $p_G(E')$ is the probability distribution of the bath with energy E' , $P(E')$ is the final probability distribution of the bath, and $I(E' : S')$ is the mutual information of system and bath after the operation.

Proof of Theorem 1 We begin by going through preliminaries that will be useful later.

Preliminaries Let us define the stochastic matrix

$$t(s', E' | s, E) = \text{Tr} \left((|s'\rangle\langle s'| \otimes Q_{E'} \otimes \mathbb{I}) U \left(|s\rangle\langle s| \otimes \frac{Q_E}{\Omega_B(E)} \otimes \rho_W \right) U^\dagger \right), \quad (184)$$

where Q_E is the projector onto the eigenspace of H_B with energy E and

$$\Omega_B(E) = \text{Tr}(Q_E),$$

is the density of states.

Operationally, $t(s', E' | s, E)$ is the probability of the following experiment. Prepare the system in the energy eigenstate $|s\rangle\langle s|$, prepare the bath in the maximally mixed state on its energy- E subspace, $Q_B^E/\Omega_B(E)$, and prepare the weight in some state ρ_W . Evolve the joint state with the global unitary $U(\cdot)U^\dagger$. Finally, measure the system in the $\{|s'\rangle\}$ basis and measure the bath energy (projectors $\{Q_B^{E'}\}$), while discarding the weight (implemented by $\mathbb{I}_W$ and the trace).

By construction, for fixed (s, E) these transition probabilities are normalized:

$$\sum_{s', E'} t(s', E' | s, E) = 1,$$

since $\sum_{s'} |s'\rangle\langle s'| = \mathbb{I}_S$ and $\sum_{E'} Q_{E'} = \mathbb{I}_B$.

It is straightforward to verify that the map defined in (184) satisfies the four properties listed below.

Property 1 (Marginalization property): Here $p(s)$ denotes the given initial distribution of the system over the energy eigenstates $\{|s\rangle\}$, and

$$p_b(E) = \frac{1}{Z} \Omega_B(E) e^{-\beta E}$$

is the probability that the bath initially occupies the energy- E subspace (with $\Omega_B(E) = \text{Tr}(Q_E)$ the corresponding degeneracy). With these definitions, we can derive Eq. (185).

$$\sum_{E, E', s} p(s) p_b(E) t(E', s' | E, s) = q(s') \quad \forall s'. \quad (185)$$

To understand (185), start from the initial joint state

$$\rho_{SBW} = \sum_{s, E} p(s) p_b(E) \left(|s\rangle\langle s| \otimes \frac{Q_E}{\Omega_B(E)} \otimes \rho_W \right), \quad (186)$$

$$\rho'_S = \text{Tr}_{BW}(\rho'_{SBW}) = \text{Tr}_{BW}(U \rho_{SBW} U^\dagger). \quad (187)$$

By the Born rule applied to the reduced system state,⁹ the probability of obtaining outcome s' when measuring S in the $\{|s'\rangle\}$ basis is

$$q(s') = \text{Tr}(|s'\rangle\langle s'| \otimes \mathbb{I}_{BW}) \rho'_{SBW}. \quad (188)$$

Now substitute $\rho'_{SBW} = U \rho_{SBW} U^\dagger$ and expand the mixture in ρ_{SBW} :

$$q(s') = \sum_{s,E} p(s) p_b(E) \text{Tr} \left((|s'\rangle\langle s'| \otimes \mathbb{I}_{BW}) U \left(|s\rangle\langle s| \otimes \frac{\mathcal{Q}_E}{\Omega_B(E)} \otimes \rho_W \right) U^\dagger \right).$$

Next, use the bath resolution of the identity $\sum_{E'} \mathcal{Q}_{E'} = \mathbb{I}_B$, which implies

$$|s'\rangle\langle s'| \otimes \mathbb{I}_{BW} = |s'\rangle\langle s'| \otimes \mathbb{I}_B \otimes \mathbb{I}_W = \sum_{E'} (|s'\rangle\langle s'| \otimes \mathcal{Q}_{E'} \otimes \mathbb{I}_W).$$

Inserting this into the expression for $q(s')$ yields

$$q(s') = \sum_{s,E,E'} p(s) p_b(E) \text{Tr} \left((|s'\rangle\langle s'| \otimes \mathcal{Q}_{E'} \otimes \mathbb{I}_W) U \left(|s\rangle\langle s| \otimes \frac{\mathcal{Q}_E}{\Omega_B(E)} \otimes \rho_W \right) U^\dagger \right).$$

The trace inside the sum is exactly the definition of the transition probability $t(s', E' | s, E)$, so we arrive at (185).

Property 2 (Normalization property):

$$\sum_{E',s'} t(E', s' | E, s) = 1 \quad \forall E, s. \quad (189)$$

This follows from the fact that $\{|s'\rangle\langle s'|\}_{s'}$ is a complete set of orthogonal projectors on S , and that $\{\mathcal{Q}_{E'}\}_{E'}$ resolves the identity on the bath Hilbert space. Hence,

$$\sum_{E',s'} (|s'\rangle\langle s'| \otimes \mathcal{Q}_{E'} \otimes \mathbb{I}_W) = \mathbb{I}_S \otimes \mathbb{I}_B \otimes \mathbb{I}_W.$$

Therefore, using the definition of $t(\cdot)$,

$$\begin{aligned} \sum_{s',E'} t(s', E' | s, E) &= \text{Tr} \left(\left(\sum_{s',E'} |s'\rangle\langle s'| \otimes \mathcal{Q}_{E'} \otimes \mathbb{I}_W \right) U \left(|s\rangle\langle s| \otimes \frac{\mathcal{Q}_E}{\Omega_B(E)} \otimes \rho_W \right) U^\dagger \right) \\ &= \text{Tr} \left(U \left(|s\rangle\langle s| \otimes \frac{\mathcal{Q}_E}{\Omega_B(E)} \otimes \rho_W \right) U^\dagger \right) \\ &= \text{Tr} \left(|s\rangle\langle s| \otimes \frac{\mathcal{Q}_E}{\Omega_B(E)} \otimes \rho_W \right) \\ &= 1. \end{aligned} \quad (190)$$

⁹Wikipedia, Born rule

In the third line we used invariance of the trace under unitary conjugation, and the final equality holds because $|s\rangle\langle s|$, $Q_E/\Omega_B(E)$, and ρ_W are normalized states on their respective subsystems.

Property 3 (Positivity):

$$t(E', s' | E, s) \geq 0 \quad \forall E, E', s, s'. \quad (191)$$

This is straightforward from the definition in (184). The operator

$$|s'\rangle\langle s'| \otimes Q_{E'} \otimes \mathbb{I}_W$$

is positive because $|s'\rangle\langle s'|$ and $Q_{E'}$ are projectors, $\mathbb{I}_W$ is positive, and the tensor product of positive operators is positive. Moreover,

$$U\left(|s\rangle\langle s| \otimes \frac{Q_E}{\Omega_B(E)} \otimes \rho_W\right)U^\dagger$$

is also positive, since it is a unitary conjugation of a valid density operator.

Finally, for positive semidefinite operators $A, B \geq 0$ one has $\text{Tr}(AB) \geq 0$ (e.g. write $A^{1/2}BA^{1/2} \geq 0$, so its trace is nonnegative). Applying this with

$$A = |s'\rangle\langle s'| \otimes Q_{E'} \otimes \mathbb{I}_W, \quad B = U\left(|s\rangle\langle s| \otimes \frac{Q_E}{\Omega_B(E)} \otimes \rho_W\right)U^\dagger,$$

immediately gives (191).

Property 4 (Microreversibility constraint):

$$\sum_{E,s} t(E' s' | E s) \frac{\Omega_B(E)}{\Omega_B(E')} = 1 \quad \forall s' E'. \quad (192)$$

From (166), we can see that this channel is a unitary channel

$$\Phi_{SB}(\mathbb{I}_{SB}) = \mathbb{I}_{SB}. \quad (193)$$

Starting from Eq. (184) and using the defining property of the partial trace

$$\text{Tr}_{SBW}((A_{SB} \otimes \mathbb{I}_W)Y_{SBW}) = \text{Tr}_{SB}(A_{SB} \text{Tr}_W(Y_{SBW})), \quad (194)$$

with $A_{SB} = |s'\rangle\langle s'| \otimes Q_{E'}$ and $Y_{SBW} = U\left(|s\rangle\langle s| \otimes \frac{Q_E}{\Omega_B(E)} \otimes \rho_W\right)U^\dagger$, we obtain

$$t(E' s' | E s) = \text{Tr}_{SB}\left(\left(|s'\rangle\langle s'| \otimes Q_{E'}\right) \text{Tr}_W\left(U\left(|s\rangle\langle s| \otimes \frac{Q_E}{\Omega_B(E)} \otimes \rho_W\right)U^\dagger\right)\right). \quad (195)$$

Using the reduced map on SB defined in (166), Eq. (195) can be rewritten as

$$t(E's'|Es) = \text{Tr}_{SB} \left(\left(|s'\rangle\langle s'| \otimes Q_{E'} \right) \Phi_{SB} \left(|s\rangle\langle s| \otimes \frac{Q_E}{\Omega_B(E)} \right) \right). \quad (196)$$

Multiply by $\Omega(E)$ and sum over E, s :

$$\begin{aligned} \sum_{E,s} \Omega_B(E) t(E's'|Es) &= \sum_{E,s} \text{Tr}_{SB} \left(\left(|s'\rangle\langle s'| \otimes Q_{E'} \right) \Phi_{SB} (|s\rangle\langle s| \otimes Q_E) \right) \\ &= \text{Tr}_{SB} \left(\left(|s'\rangle\langle s'| \otimes Q_{E'} \right) \Phi_{SB} \left(\sum_s |s\rangle\langle s| \otimes \sum_E Q_E \right) \right) \\ &= \text{Tr}_{SB} \left(\left(|s'\rangle\langle s'| \otimes Q_{E'} \right) \Phi_{SB} (\mathbb{I}_S \otimes \mathbb{I}_B) \right). \end{aligned} \quad (197)$$

Using unitality (193), $\Phi_{SB}(\mathbb{I}_S \otimes \mathbb{I}_B) = \mathbb{I}_S \otimes \mathbb{I}_B$, hence

$$\begin{aligned} \sum_{E,s} \Omega_B(E) t(E's'|Es) &= \text{Tr}_{SB} \left(\left(|s'\rangle\langle s'| \otimes Q_{E'} \right) (\mathbb{I}_S \otimes \mathbb{I}_B) \right) \\ &= \text{Tr}_S(|s'\rangle\langle s'|) \text{Tr}_B(Q_{E'}) \\ &= 1 \cdot \Omega_B(E') = \Omega_B(E'). \end{aligned} \quad (198)$$

Dividing both sides of (198) by $\Omega(E')$ yields the claimed constraint

$$\sum_{E,s} t(E's'|Es), \frac{\Omega_B(E)}{\Omega_B(E')} = 1 \quad \forall, E', s',$$

which is Eq. (192). This can be interpreted as the map t being one-to-one on a set of joint system-bath states, where $\sum_{E,s} \Omega(E) t(E's'|Es)$ is the number of states mapped to joint energy subspace (E', s') , which has degeneracy $\Omega_B(E')$.

To express the average work in terms of the transition probabilities $t(E's'|Es)$, we note that in each realisation the system changes energy from ϵ_s to $\epsilon_{s'}$ while the bath changes energy from E to E' . By overall energy conservation, the work delivered to the weight is therefore

$$W(E's'|Es) = (E - E') + (\epsilon_s - \epsilon_{s'}). \quad (199)$$

Averaging over the initial distribution of the system, $p(s)$, the initial distribution of bath energies, $p_b(E)$, and the transition probabilities $t(E's'|Es)$, yields the mean extracted work

$$\langle W \rangle = \sum_{E,E',s,s'} p(s) p_b(E) t(E's'|Es) \left[(E - E') + (\epsilon_s - \epsilon_{s'}) \right]. \quad (200)$$

With these preliminaries in place, we can now proceed to the proof. For each thermal

operation $t(E'S'|Es)$, we define the following two functions:

$$R(E', s'|s) = \sum_E p_G(E) t(E's'|Es), \quad (201)$$

$$Q(E', s', s) = \frac{\sum_E p_G(E) t(E', s'|E, s) (E - E')}{R(E', s'|s)}. \quad (202)$$

Given the system starts in level s and the bath starts Gibbs-distributed over E , $R(E'|s)$ is the probability tha the bath ends in energy shell E' . Eq. (202) is the conditional expectation

$$Q(E'|s) = \mathbb{E}[E - E'|E', s],$$

i.e. the average bath energy drop (energy given to $S + W$) conditioned on the bath finishing in shell E' and the system starting in s .

The average work extracted under the transition $t(E', s' | E, s)$ can be written in terms of $R(\cdot)$ and $Q(\cdot)$ as follows:

$$\begin{aligned} \langle W \rangle &= \sum_{E, E', s, s'} p(s) p_G(E) t(E', s' | E, s) (E - E' + \epsilon_s - \epsilon_{s'}) \\ &= \sum_{E', s, s'} p(s) R(E', s' | s) Q(E', s', s) + \sum_{E, E', s, s'} p(s) p_G(E) t(E', s' | E, s) (\epsilon_s - \epsilon_{s'}). \end{aligned} \quad (203)$$

For the second term we use the identity

$$\begin{aligned} &\sum_{E, E', s, s'} p(s) p_G(E) t(E', s' | E, s) (\epsilon_s - \epsilon_{s'}) \\ &= \sum_{E, E', s, s'} p(s) p_G(E) t(E', s' | E, s) \epsilon_s - \sum_{E, E', s, s'} p(s) p_G(E) t(E', s' | E, s) \epsilon_{s'}. \end{aligned} \quad (204)$$

Consider the first of these two contributions. Summing over E' and s' and using the normalization $\sum_{E', s'} t(E', s' | E, s) = 1$ together with $\sum_E p_G(E) = 1$, we obtain

$$\begin{aligned} \sum_{E, E', s, s'} p(s) p_G(E) t(E', s' | E, s) \epsilon_s &= \sum_s p(s) \epsilon_s \sum_E p_G(E) \sum_{E', s'} t(E', s' | E, s) \\ &= \sum_s p(s) \epsilon_s. \end{aligned} \quad (205)$$

For the second contribution, we apply Property 1,

$$\sum_{E, E', s} p(s) p_G(E) t(E', s' | E, s) = q(s') \quad \forall s',$$

to get

$$\begin{aligned} \sum_{E,E',s,s'} p(s) p_G(E) t(E', s' | E, s) \epsilon_{s'} &= \sum_{s'} \epsilon_{s'} \sum_{E,E',s} p(s) p_G(E) t(E', s' | E, s) \\ &= \sum_{s'} q(s') \epsilon_{s'} . \end{aligned} \quad (206)$$

Substituting (205) and (206) into (204) yields

$$\sum_{E,E',s,s'} p(s) p_G(E) t(E', s' | E, s) (\epsilon_s - \epsilon_{s'}) = \sum_s p(s) \epsilon_s - \sum_{s'} q(s') \epsilon_{s'} . \quad (207)$$

The right-hand side is the difference between the initial and final average system energies; in particular,

$$\sum_s p(s) \epsilon_s - \sum_{s'} q(s') \epsilon_{s'} = U_{\text{in}} - U_{\text{out}} = \Delta U,$$

if we use the convention $\Delta U := U_{\text{in}} - U_{\text{out}}$.

Putting everything together, (203) becomes

$$\langle W \rangle = \sum_{E',s,s'} p(s) R(E', s' | s) Q(E', s', s) + \Delta U. \quad (208)$$

The microscopic reversibility condition (192) (Property 4) translates into a useful constraint on the quantities $R(\cdot)$ and $Q(\cdot)$. Using

$$p_G(E) = \frac{1}{\mathcal{Z}} \Omega_B(E) e^{-\beta E} \iff \frac{\Omega_B(E)}{\Omega_B(E')} = \frac{p_G(E)}{p_G(E')} e^{\beta(E-E')},$$

Property 4 can be rewritten as

$$\begin{aligned} 1 &= \sum_{E,s} t(E', s' | E, s) \frac{\Omega_B(E)}{\Omega_B(E')} \\ &= \sum_{E,s} t(E', s' | E, s) \frac{p_G(E)}{p_G(E')} e^{\beta(E-E')} \\ &= \sum_{E,s} \frac{R(E', s' | s)}{R(E', s' | s)} t(E', s' | E, s) \frac{p_G(E)}{p_G(E')} e^{\beta(E-E')} \\ &= \sum_s \frac{R(E', s' | s)}{p_G(E')} \sum_E \frac{p_G(E) t(E', s' | E, s)}{R(E', s' | s)} e^{\beta(E-E')}. \end{aligned} \quad (209)$$

(In the last line we have simply rearranged the sums and pulled out factors that are independent of E .)

At this point it is convenient to view the coefficients

$$\mu_{E|E',s',s} := \frac{p_G(E) t(E', s' | E, s)}{R(E', s' | s)}$$

as a probability distribution over E (for each fixed triple (E', s', s)). Indeed, by definition of $R(E', s' | s)$ we have $\mu_{E|E', s', s} \geq 0$ and $\sum_E \mu_{E|E', s', s} = 1$ whenever $R(E', s' | s) > 0$. (If $R(E', s' | s) = 0$, then the corresponding term in (209) vanishes and can be ignored.) Now we use Jensen's inequality.

If X is a random variable such that $X \in \text{dom } f$ with probability one and f is convex, then

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)], \quad (210)$$

provided the expectations exist.

-Boyd & Vandenberghe, *Convex Optimization* (Cambridge University Press, 2004), Section 3.1.

In particular, since $f(x) = e^{\beta x}$ is convex for any $\beta \in \mathbb{R}$, we have

$$\sum_i p_i e^{\beta x_i} \geq \exp\left(\beta \sum_i p_i x_i\right). \quad (211)$$

We now apply (211) for each fixed (E', s', s) , with the identifications

$$p_i \rightarrow \mu_{E|E', s', s} = \frac{p_G(E) t(E', s' | E, s)}{R(E', s' | s)}, \quad x_i \rightarrow (E - E').$$

This yields

$$\sum_E \frac{p_G(E) t(E', s' | E, s)}{R(E', s' | s)} e^{\beta(E-E')} \geq \exp\left(\beta \sum_E \frac{p_G(E) t(E', s' | E, s)}{R(E', s' | s)} (E - E')\right). \quad (212)$$

Finally, using the definition of $Q(\cdot)$ from (202), the exponent on the right-hand side is precisely $\beta Q(E', s', s)$, so we obtain

$$\sum_E \frac{p_G(E) t(E', s' | E, s)}{R(E', s' | s)} e^{\beta(E-E')} \geq e^{\beta Q(E', s', s)}. \quad (213)$$

Substituting (213) into (209) gives the bound, valid for all (E', s') ,

$$1 \geq \sum_s \frac{R(E', s' | s)}{p_G(E')} e^{\beta Q(E', s', s)}. \quad (214)$$

This bound shows that, for each final label (E', s') and each s , the average bath energy drop Q cannot be chosen freely: microscopic reversibility imposes an exponential constraint linking Q to the marginal probability $R(E', s' | s)$ and the Gibbs weight $p_G(E')$.

To optimize the average extracted work, we may in principle optimize over $p(s)$, $R(E', s' |$

s), or $Q(E', s', s)$. The first two are initial/outcome probabilities; changing them amounts to changing the process. We therefore fix R (and hence $p(s)$) and optimize over Q . The optimal Q is found by maximizing the Lagrangian

$$\mathcal{L} = \sum_{E', s, s'} p(s) R(E', s' | s) Q(E', s', s) + \sum_{E', s'} \lambda_{E', s'} \left(p_G(E') - \sum_s R(E', s' | s) e^{\beta Q(E', s', s)} \right). \quad (215)$$

Note that ΔU is independent of Q , which is why it is not included in the Lagrangian. Taking the derivative with respect to $Q(E', s', s)$ and setting it to zero gives

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial Q(E', s', s)} &= \sum_{E', s, s'} p(s) R(E', s' | s) - \beta \sum_{E', s', s} \lambda_{E', s'} R(E', s' | s) e^{\beta Q(E', s', s)} = 0 \\ \implies e^{\beta Q(E', s', s)} &= \frac{p(s) R(E', s' | s)}{\beta \lambda_{E', s'} R(E', s' | s)} = \frac{p(s)}{\beta \lambda_{E', s'}} \\ \implies \beta Q(E', s', s) &= \log \left(\frac{p(s)}{\beta \lambda_{E', s'}} \right) \implies \boxed{Q(E', s', s) = \frac{1}{\beta} \log \left(\frac{p(s)}{\beta \lambda_{E', s'}} \right)}. \end{aligned} \quad (216)$$

Substituting this back into the Lagrangian gives

$$\mathcal{L} = \frac{1}{\beta} \sum_{E', s, s'} p(s) R(E', s' | s) \log \left(\frac{p(s)}{\beta \lambda_{E', s'}} \right) + \sum_{E', s'} \lambda_{E', s'} p_G(E') - \sum_{E', s', s} R(E', s' | s) \frac{p(s)}{\beta}. \quad (217)$$

Differentiating with respect to $\lambda_{E', s'}$ and setting the result to zero gives the constraint

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \lambda_{E', s'}} &= \frac{1}{\beta} \sum_s p(s) R(E', s' | s) \left(\frac{\beta \lambda_{E', s'}}{p(s)} \right) \left(-\frac{p(s)}{\beta \lambda_{E', s'}^2} \right) + \sum_s p_G(E') = 0 \\ \implies \sum_s \frac{p(s) R(E', s' | s)}{\beta \lambda_{E', s'}} &= p_G(E') \implies \frac{1}{\beta \lambda_{E', s'}} = \frac{p_G(E')}{\sum_s p(s) R(E', s' | s)}. \end{aligned} \quad (218)$$

Since $R(E', s' | s) = \Pr(E', s' | s)$, we have

$$p(E', s') = \Pr(E', s') = \sum_s p(s) R(E', s' | s) \implies \boxed{\frac{1}{\beta \lambda_{E', s'}} = \frac{p_G(E')}{p(E', s')}}. \quad (219)$$

This gives the optimal Q

$$Q(E', s', s) = \frac{1}{\beta} \log \left(\frac{p(s) p_G(E')}{p(E', s')} \right), \quad (220)$$

and thus the optimal work

$$\langle W \rangle \leq \frac{1}{\beta} \sum_{E', s, s'} p(E', s') \log \left(\frac{p(s) p_G(E')}{p(E', s')} \right) + \Delta U. \quad (221)$$

Split the logarithm by multiplying and dividing by $p(s')$:

$$\frac{p(s)p_G(E')}{p(E', s')} = \frac{p(s)}{p(s')} \cdot \frac{p(s')p_G(E')}{p(E', s')}.$$

So

$$\log\left(\frac{p(s)p_G(E')}{p(E', s')}\right) = \log\left(\frac{p(s)}{p(s')}\right) + \log\left(\frac{p(s')p_G(E')}{p(E', s')}\right).$$

Plug this into Eq. (221) and separate the sum into two terms:

$$\langle W \rangle \leq \Delta U + \frac{1}{\beta} (T_1 + T_2),$$

where

$$\begin{aligned} T_1 &= \sum_{E', s, s'} p(s) R(E', s' | s) \log\left(\frac{p(s)}{p(s')}\right), \\ T_2 &= \sum_{E', s, s'} p(s) R(E', s' | s) \log\left(\frac{p(s')p_G(E')}{p(E', s')}\right). \end{aligned}$$

Evaluate T_2 : In T_2 , the logarithm depends only on (E', s') , not on s . Therefore,

$$\begin{aligned} T_2 &= \sum_{E', s'} \left(\sum_s p(s) R(E', s' | s) \right) \log\left(\frac{p(s')p_G(E')}{p(E', s')}\right) \\ &= - \sum_{E', s'} p(E', s') \log\left(\frac{p(E', s')}{p(s')p_G(E')}\right). \end{aligned}$$

By the definition of the relative entropy, $D(P||Q) = \sum_x P(x) \log\left(\frac{P(x)}{Q(x)}\right)$, we conclude

$$T_2 = -D(p(E', s') \parallel p(s')p_G(E')). \quad (222)$$

Evaluate T_1 : Write T_1 as two pieces:

$$T_1 = \sum_{E', s, s'} p(s) R(E', s' | s) \log(p(s)) - \sum_{E', s, s'} p(s) R(E', s' | s) \log(p(s')).$$

For the first piece, $\log(p(s))$ depends only on s , and $\sum_{E', s'} R(E', s' | s) = 1$, so

$$\sum_{E', s, s'} p(s) R(E', s' | s) \log(p(s)) = \sum_s p(s) \log(p(s)).$$

For the second piece, $\log(p(s'))$ depends only on s' , and $\sum_{E',s} p(s)R(E', s'|s) = p(s')$. Hence,

$$\sum_{E',s,s'} p(s) R(E', s' | s) \log(p(s')) = \sum_{s'} p(s') \log(p(s')).$$

Therefore,

$$T_1 = \sum_s p(s) \log(p(s)) - \sum_{s'} p(s') \log(p(s')) = -S(S) + S(S'), \quad (223)$$

where we used $S(X) = -\sum_x p(x) \log(p(x))$. Putting Eqs. (222) and (223) together gives

$$\langle W \rangle \leq \Delta U + \frac{1}{\beta} (-S(S) + S(S')) - \frac{1}{\beta} D(p(E', s') \| p(s') p_G(E')). \quad (224)$$

Now use the paper's free energy convention,

$$-\Delta F := \Delta U + \frac{1}{\beta} (-S(S) + S(S')), \quad (225)$$

so Eq. (224) becomes

$$\langle W \rangle \leq -\Delta F - \frac{1}{\beta} D(p(E', s') \| p(s') p_G(E')). \quad (226)$$

Now we rewrite the relative entropy term in terms of mutual information. Expanding the logarithm,

$$\begin{aligned} D(p(E', s') \| p(s') p_G(E')) &= \sum_{E',s'} p(E', s') \log(p(E', s')) \\ &\quad - \sum_{E',s'} \sum_s p(s) R(E', s'|s) \log(p_G(E')) \\ &\quad - \sum_{E',s'} \sum_s p(s) R(E', s'|s) \log(p(s')). \end{aligned} \quad (227)$$

Using the marginals,

$$\begin{aligned} \sum_{E',s'} \sum_s p(s) R(E', s'|s) \log(p_G(E')) &= \sum_{E'} p(E') \log(p_G(E')) \\ \sum_{E',s'} \sum_s p(s) R(E', s'|s) \log(p(s')) &= \sum_{s'} p(s') \log(p(s')), \end{aligned}$$

and the definition of relative entropy for the bath marginal,

$$D(p(E') \| p_G(E')) = \sum_{E'} p(E') \log p(E') - \sum_{E'} p(E') \log p_G(E'),$$

we can rewrite

$$-\sum_{E'} p(E') \log p_G(E') = -\sum_{E'} p(E') \log p(E') + D(p(E') \| p_G(E')). \quad (228)$$

Inserting these identities into (227) and converting $-\sum P \log P$ into entropies yields

$$D(p(E', s') \| p(s') p_G(E')) = \underbrace{[S(E') + S(s') - S(E', s')]}_{I(E': s')} + D(p(E') \| p_G(E')). \quad (229)$$

Finally substitute (229) into the work bound gives

$$\boxed{\langle W \rangle \leq -\Delta F - \frac{1}{\beta} [I(E' : s') + D(p(E') \| p_G(E'))]} \quad (230)$$

This shows that the full free energy can be extracted as work only if both of the following conditions are satisfied:

1. the bath remains thermal and its temperature is unchanged, i.e. $p(E') = p_G(E')$,
2. the system and bath are uncorrelated at the end of the process, i.e. $p(E', s') = p_G(E') p(s')$.

In other words, the equality $W = -\Delta F$ requires the bath to remain Gibbs and the final joint distribution to factorize; if either condition fails, the extracted work is strictly smaller in magnitude than the free-energy change. In the finite-bath setting considered here, this idealized situation is generally unattainable: optimal work-extraction transformations leave the bath slightly athermal and correlated with the system.

Corollary 2. A necessary condition for the possibility of a transformation is

$$\langle W \rangle \leq -\Delta F - \frac{1}{2\beta C} \Delta S^2 + O\left(\frac{1}{C^2}\right), \quad (231)$$

where ΔS is the change in entropy of the system.

Proof of Corollary 2 First we address the case of infinite bath ($\gamma = 0$).

Lemma 1. When $\gamma = 0$, any thermal operation that saturates the work bound (i.e. achieves $\langle W \rangle = -\Delta F$) must be of the form

$$t(E', s' | E, s) = f(E', s', s) \delta(E - E' - f_{s'} + f_s), \quad (232)$$

where the function $f(E', s', s)$ satisfies

$$\sum_{s'} f(E - f_{s'} + f_s, s', s) = 1 \quad \forall s, \quad (233)$$

$$\sum_s p(s) f(E', s', s) = p(s') \quad \forall s', \quad (234)$$

$$0 \leq f(E', s', s) \leq 1 \quad \forall E', s', s, \quad (235)$$

and

$$f_s := -\frac{1}{\beta} \log p(s). \quad (236)$$

(Equivalently, $-\log p(s)$ is the fine-grained entropy, while f_s is the corresponding β^{-1} -rescaled quantity.)

In the reversible case, we will later obtain a condition of the form

$$Q(E', s', s) = \frac{1}{\beta} \log \left(\frac{p(s)}{p(s')} \right).$$

Thus, the key quantity is the logarithmic ratio of probabilities. Introducing

$$f_s := -\frac{1}{\beta} \log p(s), \quad f_{s'} := -\frac{1}{\beta} \log p(s'),$$

allows us to rewrite this ratio as a simple difference,

$$\frac{1}{\beta} \log \left(\frac{p(s)}{p(s')} \right) = f_{s'} - f_s.$$

This is one reason for introducing f_s : it turns the reversible condition into a compact, transparent expression. It also connects directly to Shannon entropy, since in information theory the quantity $-\log p(s)$ is called the *surprisal* and measures how unlikely a particular outcome s is. Moreover,

$$\sum_s p(s) f_s = -\frac{1}{\beta} \sum_s p(s) \log p(s) = \frac{1}{\beta} S(S), \quad (237)$$

so the Shannon entropy is the average of these fine-grained entropies. In this sense, f_s represents the entropy content of a specific microstate s , rescaled by $1/\beta$ so that it carries units of energy.

$f(E', s', s)$ is a nonnegative probability kernel that encodes the remaining freedom in an optimal (reversible, $\gamma = 1$) thermal operation: the delta function in (232) fixes the required deterministic bath energy shift for each $s \rightarrow s'$, and $f(E', s', s)$ assigns the probabilities of the allowed outcomes subject to normalization and marginal constraints (233)-(235).

Proof of Lemma 1. In order to achieve $\langle W \rangle = -\Delta F$, a thermal operation $t(E', s' | E, s)$ must saturate the upper bound derived in Theorem 1, and the right-hand side of the bound must be equal to $-\Delta F$. First note that in Theorem 1 the optimal $Q(E', s', s)$ is given uniquely by

$$Q(E', s', s) = \frac{1}{\beta} \log \left(\frac{p(s)p_G(E')}{p(E', s')} \right).$$

By Theorem 1, as the relative entropy is a distance measure for probability distributions the upper bound is $-\Delta F$ if and only if $p(E', s') = p_G(E')p(s')$ and, therefore, for all optimal thermal operations

$$Q(E', s', s) = \frac{1}{\beta} \log \left(\frac{p(s)}{p(s')} \right) = f_s - f_{s'}. \quad (238)$$

Finally, note that the thermal operation only constructs an optimal Q if the reversibility constraint (214) is saturated. This requires that

$$\begin{aligned} & \sum_{E,s} \frac{R(E', s' | s)}{p_G(E')} \frac{p_G(E)t(E', s' | E, s)}{R(E', s' | s)} e^{\beta(E-E')} \\ &= \sum_s \frac{R(E', s' | s)}{p_G(E')} \exp \left(\beta \frac{\sum_E p_G(E)t(E', s' | E, s)(E-E')}{R(E', s' | s)} \right). \end{aligned} \quad (239)$$

Define the conditional distribution over the initial bath energy E ,

$$\pi(E | E', s', s) := \frac{p_G(E)t(E', s' | E, s)}{R(E', s' | s)}, \quad (240)$$

so that $\pi(E | E', s', s) \geq 0$ and $\sum_E \pi(E | E', s', s) = 1$. With this notation, Eq. (239) can be written as

$$\sum_E \pi(E | E', s', s) e^{\beta(E-E')} = \exp \left(\beta \sum_E \pi(E | E', s', s) (E-E') \right). \quad (241)$$

Now define the positive numbers

$$y_E := e^{\beta(E-E')} > 0. \quad (242)$$

Then the right-hand side of (241) becomes

$$\exp \left(\beta \sum_E \pi(E)(E-E') \right) = \exp \left(\sum_E \pi(E) \log(y_E) \right) = \exp \left(\log \left(\prod_E y_E^{\pi(E)} \right) \right) = \prod_E y_E^{\pi(E)}.$$

Hence (241) is equivalent to

$$\sum_E \pi(E) y_E = \prod_E y_E^{\pi(E)}. \quad (243)$$

Now we use the condition for weighted AM–GM inequality.

Weighted AM–GM inequality.

Let $x_1, \dots, x_n > 0$ be positive real numbers. Then their geometric mean is no greater than their arithmetic mean, i.e.

$$(x_1, \dots, x_n)^{1/n} \leq \frac{x_1 + \dots + x_n}{n}.$$

Moreover, the equality holds iff all the numbers are equal to each other, $x_1 = \dots = x_n$.

–Hardy–Littlewood–Polya, *Inequalities*, the AM–GM inequality and its equality condition.

Applying this condition to (243) implies that the equality holds if and only if y_E is constant on the support of π , i.e. for all E with $\pi(E | E', s', s) > 0$. Since $y_E = e^{\beta(E-E')}$ is a strictly increasing function of E (for $\beta > 0$), the equality can hold only if all contributing terms have the same exponent. Equivalently,

$$E - E' = \text{constant} \quad \text{whenever} \quad p_G(E) t(E', s' | E, s) > 0. \quad (244)$$

(Here we use that $R(E', s' | s) = \sum_E p_G(E) t(E', s' | E, s)$ is nonnegative, so only values of E with positive weight can contribute.) Equivalently, for fixed (E', s', s) there exists a unique value $E^\star(E', s', s)$ such that $t(E', s' | E, s) = 0$ for all $E \neq E^\star$. In the continuum limit, this can be written as a Dirac-delta constraint,

$$t(E', s' | E, s) = f(E', s', s) \delta(E - E^\star(E', s', s)). \quad (245)$$

From the definition of $Q(E', s', s)$ in (202), if only a single value $E = E^\star$ contributes to the sum, then the conditional average collapses to that value:

$$\begin{aligned} Q(E', s', s) &= \frac{\sum_E p_G(E) t(E', s' | E, s) (E - E')}{R(E', s' | s)} \\ &= \frac{p_G(E^\star) t(E', s' | E^\star, s) (E^\star - E')}{p_G(E^\star) t(E', s' | E^\star, s)} \\ &= E^\star - E'. \end{aligned}$$

Hence,

$$E^\star = E' + Q(E', s', s). \quad (246)$$

Using Eq. (238), i.e. $Q(E', s', s) = f_s - f_{s'}$, we obtain

$$E^\star = E' + f_s - f_{s'}. \quad (247)$$

Therefore, Eq. (245) becomes

$$t(E', s' | E, s) = f(E', s', s) \delta(E - E' - f_{s'} + f_s). \quad (248)$$

The transition matrix $t(E', s' | E, s)$ is induced by the global energy-conserving unitary

acting on the system, bath, and weight, and is therefore a dynamical object. However, not every stochastic matrix is admissible: in particular, microscopic reversibility (Property 4) imposes the constraint

$$\sum_{E,s} t(E', s' | E, s) \frac{\Omega_B(E)}{\Omega_B(E')} = 1 \quad \forall (E', s').$$

Thus, although t is not *defined* in terms of Ω_B , the set of allowed transition matrices depends explicitly on the bath degeneracies $\Omega_B(E)$.

In previous sections we expressed $\Omega_B(E, V)$ in terms of β and γ (see (153)). Using that approximation, we have

$$\frac{\Omega_B(E)}{\Omega_B(E')} \approx \exp\left(\beta(E - E') - \frac{\gamma}{2}(E^2 - E'^2)\right).$$

Plugging this into (192) yields the explicit γ -dependent form

$$1 = \sum_{E,s} t(E', s' | E, s) \exp\left(\beta(E - E') - \frac{\gamma}{2}(E^2 - E'^2)\right).$$

When $\gamma = 0$, this reduces to the infinite-bath constraint

$$1 = \sum_{E,s} t(E', s' | E, s) e^{\beta(E - E')}.$$

We already obtained a solution t_0 satisfying the $\gamma = 0$ constraint (Eq. (248)). For small γ , we can treat the finite-bath effects perturbatively around this solution:

$$t(E', s' | E, s) = t_0(E', s' | E, s) + \gamma t_1(E', s' | E, s) + O(\gamma^2). \quad (249)$$

Here t_1 is the first-order correction (equivalently $t_1 = \partial t / \partial \gamma |_{\gamma=0}$). Substituting this expansion into Properties (185)–(192) produces first-order constraints on t_1 .

Insert the expansion into the normalization property (189):

$$\sum_{E',s'} \left(t_0(E', s' | E, s) + \gamma t_1(E', s' | E, s) \right) = 1.$$

Matching orders in γ shows that

$$\sum_{E',s'} t_1(E', s' | E, s) = 0 \quad \forall E, s. \quad (250)$$

Next, inserting the expansion into Eq. (185) gives

$$\sum_{E,E',s} p(s) p_G(E) \left(t_0(E', s' | E, s) + \gamma t_1(E', s' | E, s) \right) = q(s').$$

Again matching orders in γ yields

$$\sum_{E, E', s} p(s) p_G(E) t_1(E', s' | E, s) = 0 \quad \forall s'. \quad (251)$$

Now take Eq. (250), multiply by $p_G(E)$, and sum over E :

$$\sum_{E', s'} \sum_E p_G(E) t_1(E', s' | E, s) = 0 \quad \forall s. \quad (252)$$

Define the E -marginalized first-order correction

$$t_1(E', s' | s) := \sum_E p_G(E) t_1(E', s' | E, s), \quad (253)$$

so (252) becomes

$$\boxed{\sum_{E', s'} t_1(E', s' | s) = 0 \quad \forall s}. \quad (254)$$

Now regroup the terms in Eq. (251) by first summing over E (and using the definition above for the E -marginal):

$$\sum_{E'} \left(\sum_{E, s} p(s) p_b(E) t_1(E', s' | E, s) \right) = 0 \quad \forall s' \implies \boxed{\sum_{E', s} p(s) t_1(E', s' | s) = 0 \quad \forall s'}.$$

Using these constraints, we now find an expression for the optimal work in terms of $f(E', s', s)$ only.

Lemma 2. For any given transformation $p(s) \rightarrow p(s')$, there exists a map $t(E', s' | E, s)$ which extracts work

$$\langle W \rangle = -\Delta F - \frac{\gamma^2}{2\beta} \sum_{E', s'} p(s') p_G(E') E'^2 \left(\frac{\sum_s p(s) f(E', s', s) (f_{s'} - f_s)}{p(s')} \right)^2 + O(\gamma^2), \quad (255)$$

up to first order in γ .¹⁰

Proof of Lemma 2. The upper bound to the work in Theorem 1 can be written as

$$W \leq -\Delta F - \beta^{-1} \sum_{E', s'} p(E', s') \log \left(\frac{p(E', s')}{p(s') p_G(E')} \right). \quad (256)$$

¹⁰Although the prefactor appears as γ^2 , note that $p_G(E') E'^2$ scales as $O(1/\gamma)$, so the net correction is $O(\gamma)$.

To first order in γ , optimal thermal operations take the form

$$t(E', s' | E, s) = t_0(E', s' | E, s) + \gamma t_1(E', s' | E, s) + O(\gamma^2).$$

We now compute the joint final distribution $p(E', s')$ to first order in γ . First,

$$\begin{aligned} p(E', s') &= \sum_{E, s} p(s) p_G(E) t(E', s' | E, s) \\ &= \sum_{E, s} p(s) p_G(E) f(E', s', s) \delta(E - E' - f_{s'} + f_s) + \gamma \sum_s p(s) p_G(E') t_1(E', s' | E, s) \\ &= \sum_{E, s} p(s) p_G(E' - f_{s'} - f_s) f(E', s', s) + \gamma \sum_s p(s) p_G(E') t_1(E', s' | E, s). \end{aligned} \quad (257)$$

Second, we expand $p_G(E' + f_{s'} - f_s)$ in γ . From

$$\Omega_B(E) \propto \exp\left(\beta E - \frac{\gamma}{2} E^2\right) \implies p_G(E) = \frac{1}{Z} \Omega_B(E) e^{-\beta E} \propto \exp\left(-\frac{\gamma}{2} E^2\right), \quad (258)$$

we get

$$\frac{p_G(E' + f_{s'} - f_s)}{p_G(E')} = \exp\left(-\frac{\gamma}{2} ((E' + f_{s'} - f_s)^2 - E'^2)\right).$$

Expanding the square gives

$$(E' + f_{s'} - f_s)^2 - E'^2 = 2E'(f_{s'} - f_s) + (f_{s'} - f_s)^2,$$

so,

$$p_G(E' + f_{s'} - f_s) = p_G(E') \exp\left(-\gamma E'(f_{s'} - f_s) - \frac{\gamma}{2} (f_{s'} - f_s)^2\right). \quad (259)$$

Now Taylor expand with

$$x = -\gamma E'(f_{s'} - f_s) - \frac{\gamma}{2} (f_{s'} - f_s)^2, \quad e^x = 1 + x + \frac{x^2}{2} + O(x^3).$$

Compute the pieces:

$$\begin{aligned} \text{Linear term:} & \quad -\gamma E'(f_{s'} - f_s) - \frac{\gamma}{2} (f_{s'} - f_s)^2 \\ \text{Quadratic term:} & \quad \frac{1}{2} \left(-\gamma E'(f_{s'} - f_s) - \frac{\gamma}{2} (f_{s'} - f_s)^2 \right)^2, \\ & \quad = \frac{1}{2} \left(\gamma^2 E'^2 (f_{s'} - f_s)^2 + \gamma^2 E' (f_{s'} - f_s)^3 + \frac{\gamma^2}{4} (f_{s'} - f_s)^4 \right). \end{aligned} \quad (260)$$

We want to keep contributions up to the order relevant for a first-order ($O(\gamma)$) correction

to the work bound. For this, we need the typical scale of E' . Normalize

$$p_G(E) = \frac{1}{Z} \exp\left(-\frac{\gamma}{2}E^2\right), \quad Z = \int_{-\infty}^{\infty} \exp\left(-\frac{\gamma}{2}E^2\right) dE.$$

Using the Gaussian integral $\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}$ for $a > 0$ with $a = \gamma/2$, we obtain

$$Z = \sqrt{\frac{2\pi}{\gamma}}, \quad p_G(E) = \sqrt{\frac{\gamma}{2\pi}} \exp\left(-\frac{\gamma}{2}E^2\right).$$

By definition,

$$\langle E^2 \rangle = \int_{-\infty}^{\infty} E^2 p_G(E) dE = \sqrt{\frac{\gamma}{2\pi}} \int_{-\infty}^{\infty} E^2 e^{-\frac{\gamma}{2}E^2} dE.$$

To compute the remaining integral, differentiate $J(a) = \int_{-\infty}^{\infty} e^{-aE^2} dE = \sqrt{\frac{\pi}{a}}$:

$$\frac{dJ}{da} = \int_{-\infty}^{\infty} \frac{\partial}{\partial a} (e^{-aE^2}) dE = \int_{-\infty}^{\infty} (-E^2) e^{-aE^2} dE = \frac{d}{da} \sqrt{\frac{\pi}{a}} = -\frac{1}{2} \sqrt{\pi} a^{-3/2}.$$

Hence

$$\int_{-\infty}^{\infty} E^2 e^{-aE^2} dE = \frac{1}{2} \sqrt{\pi} a^{-3/2}.$$

With $a = \gamma/2$,

$$\langle E^2 \rangle \int_{-\infty}^{\infty} E^2 e^{-\frac{\gamma}{2}E^2} dE = \frac{1}{2} \sqrt{\pi} \left(\frac{\gamma}{2}\right)^{-3/2}, \quad (261)$$

so,

$$\langle E^2 \rangle = \sqrt{\frac{\gamma}{2\pi}} \cdot \frac{1}{2} \sqrt{\pi} \left(\frac{\gamma}{2}\right)^{-3/2} = \frac{1}{\gamma}.$$

Therefore,

$$\boxed{E \sim \sqrt{\langle E^2 \rangle} \sim \gamma^{-1/2}}. \quad (262)$$

We can now decide which terms in (260) must be retained:

$$\begin{aligned} \frac{1}{2} \gamma^2 E'^2 (f_{s'} - f_s)^2 &\sim \frac{1}{2} (\gamma^2) (\gamma^{-1}) (f_{s'} - f_s)^2 \sim O(\gamma), \\ \frac{1}{2} \gamma^2 E' (f_{s'} - f_s)^3 &\sim \frac{1}{2} (\gamma^2) (\gamma^{-1/2}) (f_{s'} - f_s)^3 \sim O(\gamma^{3/2}), \\ \frac{1}{2} \cdot \frac{\gamma^2}{4} (f_{s'} - f_s)^4 &\sim O(\gamma^2). \end{aligned}$$

Thus, to the required accuracy,

$$p_G(E' + f_{s'} - f_s) = p_G(E') \left[1 - \gamma E' (f_{s'} - f_s) + \frac{\gamma}{2} (\gamma E'^2 - 1) (f_{s'} - f_s)^2 \right] + O(\gamma^{3/2}). \quad (263)$$

Substituting (263) into (257) gives

$$\begin{aligned} p(E', s') &= p_G(E') \sum_s p(s) f(E', s', s) \left[1 - \gamma E' (f_{s'} - f_s) + \frac{\gamma}{2} (\gamma E'^2 - 1) (f_{s'} - f_s)^2 \right] \\ &\quad + \gamma \sum_{E,s} p(s) p_G(E) t_1(E', s' | E, s) + O(\gamma^{3/2}). \end{aligned} \quad (264)$$

Expanding the bracket and separating the zeroth order gives

$$\begin{aligned} p(E', s') &= p_G(E') \sum_s p(s) f(E', s', s) \\ &\quad - \gamma p_G(E') E' \sum_s p(s) f(E', s', s) (f_{s'} - f_s) \\ &\quad + \frac{\gamma}{2} p_G(E') (E'^2 \gamma - 1) \sum_s p(s) f(E', s', s) (f_{s'} - f_s)^2 \\ &\quad + \gamma \sum_{s,E} p(s) p_G(E) t_1(E', s' | E, s) + O(\gamma^{3/2}). \end{aligned}$$

For the $\gamma \rightarrow 0$ optimal map, $f(E', s', s)$ satisfies Eq. (234),

$$\sum_s p(s) f(E', s', s) = p(s').$$

Applying this to the first line yields

$$\begin{aligned} p(E', s') &= p_G(E') p(s') \\ &\quad - \gamma p_G(E') E' \sum_s p(s) f(E', s', s) (f_{s'} - f_s) \\ &\quad + \frac{\gamma}{2} p_G(E') (E'^2 \gamma - 1) \sum_s p(s) f(E', s', s) (f_{s'} - f_s)^2 \\ &\quad + \gamma \sum_{s,E} p(s) p_G(E) t_1(E', s' | E, s) + O(\gamma^{3/2}). \end{aligned}$$

We want to expand the term $\log \left(\frac{p(E', s')}{p(s') p_G(E')} \right)$ appearing in the work bound. To streamline the expansion, we introduce

$$x(E', s') := \frac{p(E', s')}{p(s') p_G(E')} - 1.$$

As discussed in the previous section, the ideal (uncorrelated) situation is characterized by a factorized final distribution,

$$p(E', s') = p_G(E') p(s'),$$

which is equivalent to $x(E', s') = 0$. Therefore, $x(E', s')$ serves as a dimensionless measure of how far the actual final joint distribution deviates from this ideal factorized form. Using the expansion of $p(E', s')$ derived above, we obtain

$$\begin{aligned}
 x(E', s') = \frac{p(E', s')}{p(s')p_G(E')} - 1 = & \underbrace{-\frac{\gamma E'}{p(s')} \sum_s p(s) f(E', s', s) (f_{s'} - f_s)}_{(A)} \\
 & + \underbrace{\frac{\gamma}{2p(s')} (\gamma E'^2 - 1) \sum_s p(s) f(E', s', s) (f_{s'} - f_s)^2}_{(B)} \\
 & + \underbrace{\frac{\gamma}{p(s')p_G(E')} \sum_{s,E} p(s)p_G(E) t_1(E', s' | E, s)}_{(C)}. \quad (265)
 \end{aligned}$$

Since $x(E', s') \ll 1$ for small γ , we use

$$\log(1+x) = x - \frac{1}{2}x^2 + O(x^3).$$

Keeping terms up to x^2 , the correction term in (256) becomes

$$\begin{aligned}
 -\frac{1}{\beta} \sum_{E', s'} p(E', s') \log\left(\frac{p(E', s')}{p(s')p_G(E')}\right) = & -\frac{1}{\beta} \sum_{E', s'} p(E', s') \left(\frac{p(E', s')}{p(s')p_G(E')} - 1\right) \\
 & + \frac{1}{2\beta} \sum_{E', s'} p(E', s') \left(\frac{p(E', s')}{p(s')p_G(E')} - 1\right)^2. \quad (266)
 \end{aligned}$$

First, we show that the contribution from term (C) in (265) is zero. Since (C) is $O(\gamma)$, its contribution appears in the linear term $-\frac{1}{\beta} \sum p(E', s') x(E', s')$. To zeroth order in γ ,

$$p(E', s') = \sum_s p(s) f(E', s', s) p_G(E') = p(s') p_G(E').$$

Substituting this and (C) into the linear term gives

$$\begin{aligned}
 & -\frac{1}{\beta} \sum_{E', s'} p(s') p_G(E') \frac{\gamma}{p(s') p_G(E')} \sum_{s,E} p(s) p_G(E) t_1(E', s' | E, s) \\
 & = -\frac{\gamma}{\beta} \sum_{E', s', E, s} p(s) p_G(E) t_1(E', s' | E, s) = -\frac{\gamma}{\beta} \sum_{E, s} p(s) p_G(E) \underbrace{\sum_{E', s'} t_1(E', s' | E, s)}_{=0 \text{ by (250)}},
 \end{aligned}$$

and hence the (C) term does not contribute to the linear term. Note that $E' \sim \gamma^{-1/2}$, so

$$(A) \sim \gamma E' \sim \sqrt{\gamma}, \quad (B) \sim \gamma, \quad (C) \sim \gamma.$$

Therefore, products involving (C) contribute only at higher order and can be dropped at the accuracy considered here.

We now substitute (A) and (B) into Eq. (266). As before, we may replace $p(E', s')$ by its $\gamma = 0$ value inside the prefactor to extract the first-order correction.

Linear term

$$-\frac{1}{\beta} \sum_{E', s'} p(E', s') x \approx -\frac{1}{\beta} \sum_{E', s'} p_G(E') p(s') ((A) + (B)). \quad (267)$$

The (A) contribution is

$$\begin{aligned} -\frac{1}{\beta} \sum p_G(E') p(s') (A) &= -\frac{1}{\beta} \sum p_G(E') p(s') \left(-\frac{\gamma E'}{p(s')} \sum_s p(s) f(E', s', s) (f_{s'} - f_s) \right), \\ &= \frac{\gamma}{\beta} \sum_{E', s'} p_G(E') E' \left(\sum_s p(s) f(E', s', s) (f_{s'} - f_s) \right). \end{aligned} \quad (268)$$

Since $p_G(E')$ is even, $\sum_{E'} p_G(E') E' = 0$. Thus the (A) contribution vanishes whenever the bracketed term in (268) is independent of E' .

According to Lemma 1, in the reversible limit $\gamma = 0$ the function $f(E', s', s)$ must satisfy

$$\sum_s p(s) f(E', s', s) = p(s') \quad \forall s'.$$

The key observation is that the right-hand side does not depend on E' . Hence, for each fixed s' , the vector $\{f(E', s', s)\}_s$ must satisfy the *same* linear constraint for every value of E' . This constraint does not require any explicit E' -dependence, so a simple and valid choice is to take f independent of E' :

$$f(E', s', s) \equiv f(s', s).$$

With this choice, the bracketed term in (268) becomes independent of E' , and since $p_G(E')$ is even we have $\sum_{E'} p_G(E') E' = 0$. Therefore the entire (A) contribution vanishes. Thus, the (A) contribution becomes

$$-\frac{1}{\beta} \sum p_G(E') p(s') (A) = \underbrace{\frac{\gamma}{\beta} \sum_{E', s'} p_G(E') E'}_{=0} \left(\sum_s p(s) f(s', s) (f_{s'} - f_s) \right), \quad (269)$$

and therefore (A) has does not make a linear term and is equal to zero.

For the (B) part we obtain

$$-\frac{1}{\beta} \sum_{E',s'} p_G(E') p(s')(B) = -\frac{\gamma}{2\beta} \sum_{E',s'} p_G(E') (\gamma E'^2 - 1) \left(\sum_s p(s) f(E', s', s) \frac{(f_{s'} - f_s)^2}{p(s')} \right). \quad (270)$$

Quadratic term Now consider

$$\frac{1}{2\beta} \sum_{E',s'} p(E', s') x^2 \approx \frac{1}{2\beta} \sum_{E',s'} p_G(E') P(s') [(A) + (B)]^2.$$

Expanding,

$$[(A) + (B)]^2 = (A)^2 + 2(A)(B) + (B)^2.$$

and using $(A) \sim \sqrt{\gamma}$ and $(B) \sim \gamma$, we have

$$(A)^2 \sim \gamma, \quad 2(A)(B) \sim \gamma^{3/2}, \quad (B)^2 \sim \gamma^2,$$

so to the required accuracy we keep only $(A)^2$. Thus,

$$\begin{aligned} \frac{1}{2\beta} \sum_{E',s'} p(E', s') x^2 &\approx \frac{1}{2\beta} \sum_{E',s'} p_G(E') P(s') (A)^2 \\ &\approx \frac{1}{2\beta} \sum_{E',s'} p_G(E') p(s') \gamma^2 E'^2 \left(\frac{\sum_s p(s) f(E', s', s) (f_{s'} - f_s)}{p(s')} \right)^2. \end{aligned} \quad (271)$$

Putting the linear and quadratic terms together gives

$$\begin{aligned} \langle W \rangle &= -\Delta F - \frac{\gamma^2}{2\beta} \sum_{E',s'} p_G(E') p(s') E'^2 \left(\frac{\sum_s p(s) f(E', s', s) (f_{s'} - f_s)}{p(s')} \right)^2 \\ &\quad + \frac{\gamma}{2\beta} \sum_{E',s'} p_G(E') p(s') (E'^2 \gamma - 1) \left(\frac{\sum_s p(s) f(E', s', s) (f_{s'} - f_s)}{p(s')} \right) + O(\gamma^2) \end{aligned} \quad (272)$$

We now show that the second term in (272) is zero to first order in γ . First note that $\sum_{E',s',E,s} p(s) p_G(E') t(E', s' | E, s) = 1$ and therefore

$$1 = \sum_{E',s',s} p(s) p_G(E') (E' - f_{s'} + f_s) f(E', s', s) + \underbrace{\gamma \sum_{E',s',E,s} p(s) p_G(E') t_1(E', s' | E, s)}_{=0 \text{ by (251)}}.$$

Substituting the expansion (263) gives

$$\begin{aligned} 1 &= \sum_{E',s',s} p(s) p_G(E') \left[1 - \gamma E' (f_{s'} - f_s) + \frac{\gamma}{2} (\gamma E'^2 - 1) (f_{s'} - f_s)^2 \right] f(E', s', s) \\ &= 1 - \gamma \sum_{E',s',s} p(s) p_G(E') E' f(E', s', s) (f_{s'} - f_s) + \frac{\gamma}{2} \sum_{E',s',s} p(s) p_G(E') (E'^2 \gamma - 1) (f_{s'} - f_s)^2 f(E', s', s), \end{aligned}$$

where we used Eq. (233).

As above, taking $f(E', s', s) \equiv f(s', s)$ makes the first-order term proportional to $\sum_{E'} p_G(E')E' = 0$, since $p_G(E')$ is an even function of E' . Then,

$$0 = \gamma \sum_{E', s', s} p(s) p_G(E') E' f(s', s) (f_{s'} - f_s) + \frac{\gamma}{2} \sum_{E'} p_G(E') (E'^2 \gamma - 1) \sum_{s', s} p(s) (f_{s'} - f_s)^2 f(s', s). \quad (273)$$

The first term vanishes because $\sum_{E'} p_G(E')E' = 0$. For the second term we use $\langle E'^2 \rangle = \gamma^{-1}$, which implies

$$\sum_{E'} p_G(E') (\gamma E'^2 - 1) = \gamma \langle E'^2 \rangle - 1 = \gamma(\gamma^{-1}) - 1 = 0.$$

Note that this step works precisely because we removed the E' -dependence of $f(E', s', s)$, allowing the E' -sum to factor out. Hence the second term in (272) vanishes at the required order, and we obtain

$$\langle W \rangle = -\Delta F - \frac{\gamma^2}{2\beta} \sum_{E', s'} p_G(E') p(s') E'^2 \left(\sum_s \frac{p(s) f(E', s', s) (f_{s'} - f_s)}{p(s')} \right)^2 + O(\gamma^2). \quad (274)$$

This completes the proof of Lemma 2.

Corollary 2. In any process, work is bounded from above by

$$\langle W \rangle \leq -\Delta F - \frac{1}{2\beta C} \Delta S^2 + O(\gamma^2). \quad (275)$$

Proof of corollary 2. Starting with the equation for the optimal work derived in Lemma 2, (274), the upper bound can be derived through simple convexity arguments. We use the Jensen's inequality for a convex function,

$$\phi \left(\sum_i p_i x_i \right) \leq \sum_i p_i \phi(x_i),$$

to get

$$\begin{aligned} & -\frac{\gamma^2}{2\beta} \sum_{E'} p_G(E') E'^2 \sum_{s'} p(s') \left(\frac{\sum_s p(s) f(E', s', s) (f_{s'} - f_s)}{p(s')} \right)^2 \\ & \leq -\frac{\gamma^2}{2\beta} \sum_{E'} p_G(E') E'^2 \sum_{s'} \left(p(s') \cdot \frac{\sum_s p(s) f(E', s', s) (f_{s'} - f_s)}{p(s')} \right)^2. \end{aligned}$$

Using Eq. (237), we can relate the left side to entropy

$$\sum_s p(s) f_s = -\frac{1}{\beta} \sum_s p(s) \log p(s) = \frac{1}{\beta} S(S).$$

Simplify the right side of the bound:

$$\begin{aligned} & \underbrace{\sum_{s,s'} p(s) f(E', s', s) f_{s'}}_{(I)} - \underbrace{\sum_{s,s'} p(s) f(E', s', s) f_s}_{(II)} \\ (I) &= \sum_{s'} f_{s'} \underbrace{\sum_s p(s) f(E', s', s)}_{=p(s') \text{ By (234)}} = \sum_{s'} f_{s'} p(s'), \\ (II) &= \sum_s p(s) f_s \underbrace{\sum_{s'} f(E', s', s)}_{=1 \text{ by (233)}} = \sum_s p(s) f_s. \end{aligned} \tag{276}$$

(277)

Therefore,

$$\boxed{\sum_{s,s'} P(s) f(E', s', s) (f_{s'} - f_s) = \frac{\Delta S}{\beta}}. \tag{278}$$

Thus the right term is equal to

$$-\frac{\gamma^2}{2\beta} \underbrace{\sum_{E'} p_G(E') E'^2}_{=\langle E'^2 \rangle = \gamma^{-1}} \cdot \frac{\Delta S^2}{\beta^2} = -\frac{\gamma}{2\beta^3} \Delta S^2$$

and the bound becomes

$$-\frac{\gamma^2}{2\beta} \sum_{E', s'} p_G(E') p(s') E'^2 \left(\sum_s \frac{p(s) f(E', s', s) (f_{s'} - f_s)}{p(s')} \right)^2 \leq -\frac{\gamma}{2\beta^3} \Delta S^2. \tag{279}$$

Replace γ with $\gamma = \beta^2/C$ and substitute this into the work bound to get the statement of corollary 2:

$$\langle W \rangle \leq -\Delta F - \frac{\beta^2/C}{2\beta^3} \Delta S^2 + O(\gamma^2) = -\Delta F - \frac{1}{2\beta C} \Delta S^2 + O(\gamma^2).$$

Thus, our proof is now complete.

There are some particular cases of interest in which a necessary and sufficient condition that only depends on the initial and final states of the system can be shown (this sentence makes no sense).

Theorem 3. When the final state of the system is maximally mixed, the necessary and sufficient condition is

$$\langle W \rangle \leq -\Delta F - \frac{\beta^2/C}{2\beta^3} \Delta S^2 + O(\gamma^2) = -\Delta F - \frac{1}{2\beta C} \Delta S^2 + O(\gamma^2), \quad (280)$$

up to first order in $1/C$. When the initial state is maximally mixed, the necessary and sufficient condition is

$$\langle W \rangle \leq -\Delta F - \frac{1}{2\beta C} (\Delta S^2 + \text{Var } p(s')) + O\left(\frac{1}{C^2}\right), \quad (281)$$

up to first order in $1/C$. The varentropy is defined as

$$\text{Var}(p(s)) = \sum_s p(s) \log^2(p(s)) - \left(\sum_s p(s) \log(p(s)) \right)^2, \quad (282)$$

and is always positive.

In general, the exact feasibility conditions in this framework depend not only on the system's initial and final states, but also on additional details of the process, most importantly how the bath and system become correlated and how far the bath is driven from equilibrium. However, the authors point out that there are special cases where this extra information becomes unnecessary: one can give a criterion that is both necessary and sufficient using only the system's endpoint states. This simplification occurs when either the initial or the final system state is maximally mixed, i.e. $p(s) = \text{const}$. In that situation the target (or the starting) state has no bias or structure in the chosen basis, which removes otherwise relevant constraints and allows the optimal work condition to be characterized purely in terms of the system's initial and final states.

Proof of Theorem 3. We have already shown that these bounds are necessary. It remains to prove that, under the additional assumption that one of the endpoint states is maximally mixed, the bounds are also sufficient, that is, the upper bound can be saturated by a physically allowed process.

To make the proof more transparent, we first restate the theorem in an equivalent form. We also have shown before that these Consider transformations between an arbitrary state (ρ, H_s) and the maximally mixed state with a trivial Hamiltonian $(1/d\mathbb{I}, \mathbb{I})$, noting that a trivial Hamiltonian implies the thermal state is maximally mixed at any temperature. For the work extraction process $(\rho, H_s) \rightarrow (1/d\mathbb{I}, \mathbb{I})$, the maximal achievable average work is

$$\langle W \rangle = -\Delta F - \frac{\beta^2/C}{2\beta^3} \Delta S^2 + O(\gamma^2) = -\Delta F - \frac{1}{2\beta C} \Delta S^2 + O(1/C^2).$$

For the reverse process $(\rho, H_S) \leftarrow (1/d\mathbb{I}, \mathbb{I})$, interpreted as the Landauer erasure, the optimal work includes an additional second-order term

$$\langle W \rangle = -\Delta F - \frac{1}{2\beta C} (\Delta S^2 + \text{Var } p(s')) + O\left(\frac{1}{C^2}\right),$$

where $\text{Var}(\rho)$ is the varentropy of the final state. For the Landauer erasure process, note that $p(s') = 1/d$, which is independent of s' . Simply choosing $f(E', s', s) = p(s') = 1/d$ and substituting this into the equation derived in Lemma 2, (272),

$$f_s = -\frac{1}{\beta} \log(p(s)),$$

gives

$$\begin{aligned} \langle W \rangle &= -\Delta F - \frac{\gamma^2}{2\beta} \sum_{E', s'} p(s') p_G(E') E'^2 \left(\frac{\sum_s p(s) p(s')}{p(s')} \cdot \frac{1}{\beta} (\log(d) + \log(p(s))) \right)^2 \\ &\quad + \frac{\gamma}{2\beta} \sum_{E', s'} p(s') p_G(E') (E'^2 \gamma - 1) \left(\frac{\sum_s p(s) p(s')}{p(s')} \cdot \frac{1}{\beta^2} (\log(d) + \log(p(s)))^2 \right) \\ &= -\Delta F - \frac{\gamma^2}{2\beta^3} \sum_{E', s'} p(s') p_G(E') E'^2 \left(\sum_s p(s) (\log(d) + \log(p(s))) \right)^2 \\ &\quad + \frac{\gamma}{2\beta} \sum_{E', s'} p_G(E') (E'^2 \gamma - 1) \left(\sum_s p(s) p(s') \cdot \frac{1}{\beta^2} (\log(d) + \log(p(s)))^2 \right). \end{aligned}$$

We previously showed that the underlined term is equal to zero. Using the von Neumann entropy, we therefore have

$$\begin{aligned} \langle W \rangle &= -\Delta F - \frac{\gamma^2}{2\beta^3} \sum_{E', s'} p(s') p_G(E') E'^2 \left(\log(d) \underbrace{\sum_s p(s)}_{=1} + \sum_s p(s) \log(p(s)) \right)^2 + 0 \\ &\quad - \Delta F - \frac{\gamma^2}{2\beta^3} \cdot \underbrace{\sum_{s'} p(s')}_{=1} \underbrace{\sum_{E'} p_G(E') E'^2}_{=\langle E'^2 \rangle = \gamma^{-1}} \left(\log(d) + S(\rho) \right)^2 \end{aligned} \tag{283}$$

$$\begin{aligned} &= -\Delta F - \frac{\gamma}{2\beta^3} \cdot \left(\log(d) + S(\rho) \right)^2 = -\Delta F - \frac{\gamma}{2\beta^3} \Delta S^2 \\ \implies \langle W \rangle &= -\Delta F - \frac{1}{2\beta C} \Delta S^2 + O\left(\frac{1}{C^2}\right). \end{aligned} \tag{284}$$

In this special case the desired final distribution is uniform ($p(s') = 1/d$). Hence, the simplest way to ensure the system ends up with that distribution is to make the conditional distribution independent of the input, $f(E', s', s) = p(s')$.

For the reverse process, note that

$$f_{s'} - f_s = -\log(d) - \log(p(s')),$$

which is independent of s . Using Eq. (234), the equality in Lemma 2 becomes

$$\begin{aligned} \langle W \rangle &= -\Delta F - \frac{\gamma^2}{2\beta^3} \sum_{E', s'} p(s') p_G(E') E'^2 \left(\sum_s \frac{p(s) f(E', s', s) [\log(d) + \log(p(s'))]}{p(s')} \right)^2 \\ &= -\Delta F - \frac{\gamma^2}{2\beta^3} \sum_{E', s'} p(s') p_G(E') E'^2 \left(\frac{p(s') [\log(d) + \log(p(s'))]}{p(s')} \right)^2 \\ &= -\Delta F - \frac{\gamma^2}{2\beta^3} \sum_{E'} p_G(E') E'^2 \sum_{s'} p(s') (\log^2(d) + \log^2(p(s')) + 2 \log(d) \log(p(s'))) \\ &= -\Delta F - \frac{1}{2\beta C} \left(\left[\log(d) - \sum_{s'} p(s') \log(p(s')) \right]^2 + \underbrace{\sum_{s'} p(s') \log^2(p(s')) - \left[\sum_{s'} p(s') \log(p(s')) \right]^2}_{=\text{Var}(p(s'))} \right) \\ &= -\Delta F - \frac{1}{2\beta C} \left([\log(d) - S(\rho)]^2 + \text{Var}(p(s')) \right) \\ &= -\Delta F - \frac{1}{2\beta C} (\Delta S^2 + \text{Var}(p(s'))) + O(1/C^2). \end{aligned} \tag{285}$$

We note that for this theorem to hold we do not need to make any assumptions about the Hamiltonian, only about the final state of the system.

Thus for these cases we can refine the work upper bound in (275) to a tight bound and show that it is achievable through operations that are independent of the bath energy.

Let us now discuss the case in which the system Hamiltonian is trivial, $H_S = 0$. (Equivalently, one may take H_S to be proportional to the identity, $H_S = c\mathbb{I}$, since adding a constant shift does not affect the physics.) In this situation the system has no energy structure, and its thermal state at any temperature is simply the maximally mixed state. This places us in the special regime of Theorem 3, where the finite work bounds become tight. In particular, Eq. (280) gives the maximum average work extractable when transforming an arbitrary state to the maximally mixed state. Conversely, Eq. (281) gives the minimal work cost to prepare a state from equilibrium. In these cases, the optimal work protocols are precisely those that saturate the corresponding inequalities, meaning the bounds are achievable and therefore equal to the true optimal work values. We see that the minimal work cost for preparing an arbitrary state $p(s')$ minus the maximal work extractable from it is

$$\frac{1}{2\beta C} (\Delta S^2 + \text{Var}(p(s'))) + O\left(\frac{1}{C^2}\right),$$

which is always positive. Hence the cyclic of applying first one operation and then the other

circular process, despite being optimal, is not reversible, unlike in the presence of an infinite bath. Therefore, we conclude that **thermodynamic reversibility requires an infinite heat bath**. Indeed, this is already implied by (275) for all processes that change the entropy of the system $\Delta S \neq 0$.

The varentropy (281) can be interpreted as giving the variance of the "surprise" or "fine-grained entropy", $\log(p(s))$, whereas the Shannon/von Neumann entropy gives the average of this. For example, for a thermal distribution $p(s) \propto e^{-\beta \varepsilon_s}$, the varentropy coincides with the system heat capacity (Eq. (145)). To see this, recall that the partition function is

$$\mathcal{Z}(\beta) = \text{Tr} \left(e^{-\beta H} \right), \quad \langle H \rangle = \frac{\text{Tr} (H e^{-\beta H})}{\mathcal{Z}(\beta)}.$$

Differentiating $\ln(\mathcal{Z})$ with respect to β gives

$$\frac{d}{d\beta} \ln \mathcal{Z} = \frac{1}{\mathcal{Z}} \frac{d\mathcal{Z}}{d\beta} = -\frac{\text{Tr}(H e^{-\beta H})}{\mathcal{Z}(\beta)} = -\langle H \rangle.$$

Therefore,

$$\frac{d \langle H \rangle}{d\beta} = -\frac{\partial^2}{\partial \beta^2} \ln \mathcal{Z}.$$

Writing this second derivative explicitly in terms of $\mathcal{Z}$,

$$\frac{d \langle H \rangle}{d\beta} = -\left(\frac{\mathcal{Z}''}{\mathcal{Z}} - \left(\frac{\mathcal{Z}'}{\mathcal{Z}} \right)^2 \right).$$

Now observe that

$$\frac{\mathcal{Z}''}{\mathcal{Z}} = \frac{\text{Tr}(H^2 e^{-\beta H})}{\mathcal{Z}(\beta)} = \langle H^2 \rangle, \quad \left(\frac{\mathcal{Z}'}{\mathcal{Z}} \right)^2 = \langle H \rangle^2.$$

Substituting these identities yields

$$\frac{d \langle H \rangle}{d\beta} = -(\langle H^2 \rangle - \langle H \rangle^2) = -\text{Var}(E).$$

For a thermal state, the fine-grained entropy is $-\log p(s) = \beta \varepsilon_s + \log \mathcal{Z}$, so its variance (the varentropy) satisfies

$$\text{Var}(p(s)) = \beta^2 \text{Var}(E).$$

Hence, using the standard thermodynamic identity $C_S = \beta^2 \text{Var}(E_S)$, we conclude

$$C_S = \beta^2 \text{Var}(E_S) = \text{Var}(p(s)). \quad (286)$$

As an example, consider preparing the thermal state of an arbitrary Hamiltonian H'_S starting from the thermal state of a trivial Hamiltonian. In that case, the term $\text{Var}(p(s'))$ is exactly the heat capacity of the prepared system, so the finite-bath correction contains the ratio of

the system and bath heat capacities:

$$-\frac{1}{2\beta C_B} (\Delta S^2 + C_S) = -\frac{\Delta S^2}{2\beta C_B} - \frac{1}{2\beta} \frac{C_S}{C_B}.$$

2.4 Deterministic work

For some applications it is preferable that the work extracted from a system does not fluctuate. This has led some authors to consider a more restrictive definition of work for the quantum regime, namely deterministic work, which consists of raising or lowering off a system from one energy level to another with a very high probability. We now explore in which way deterministic work can appear when one does not have an infinite bath. We shall focus on joint transformations of the system and weight such as

$$\rho \otimes |0\rangle\langle 0| \rightarrow \sigma \otimes |W\rangle\langle W|,$$

where $|0\rangle\langle 0|$ and $|W\rangle\langle W|$ are energy eigenstates of the weight with definite energies 0 and W . The relevant optimization problem is therefore to determine, for the class of allowed operations introduced earlier, the largest deterministic work W that can be guaranteed. Because the weight's final energy is no longer distributed over different outcomes, it does not introduce additional stochastic degrees of freedom; one can therefore treat the weight as an additional subsystem included in the overall system description, and analyze deterministic work extraction as a state transformation on the enlarged system.

The joint system-bath weight has a well-defined total energy E_{tot} with probability distribution $p(E_{\text{tot}})$. Because the joint transformation conserves the total energy, a transition is possible in general if it is also possible separately in each subspace of fixed E_{tot} (as the unitary acts separately and independently on each subspace). This is different from the previous section, where we were able to consider optimal averages over all total energies. In other words, you cannot compensate for a poor choice in one energy sector by doing better in another, because the work value W must be the same regardless of which total energy sector the process occupies.

In this section we address the issue of extracting and expending deterministic work, free from fluctuations. As a first step, we shall describe what is the power of thermal operations with a finite bath when no work is involved and only system-bath interactions occur. We then extend the analysis to deterministic work extraction and work cost by enlarging the system to include the weight degrees of freedom: work is represented by a definite energy shift of the weight, so transformations involving work can be treated as ordinary system-bath state transformations on the composite $S \otimes W$.

We will be making the approximation that the energy scales of the system are small compared to the energy fluctuations of the bath. This is accurate provided the bath is sufficiently large.

In the present context, out of the assumptions in Sec. 2.3, only the first two are relevant, since they have to do with the weight and work. The energy conservation constraint implies that the unitaries follow

$$[U, H_S + H_B] = 0,$$

as the weight is not involved for now. We append a system S with Hamiltonian $H_S = \sum_s \epsilon_s |s\rangle\langle s|$ and assume the system state is diagonal in the energy eigenbasis, $\rho = \sum_s p(s) |s\rangle\langle s|$, while the finite bath B is initially thermal, τ_β . Because $[U, H_S + H_B] = 0$ implies that U is block-diagonal in the eigenspaces of the total Hamiltonian $H_S + H_B$, the joint Hilbert space decomposes into a direct sum of fixed total energy subspaces labeled by E_{tot} , and the dynamics within each such subspace is independent of all others. Consequently, the analysis of state transitions reduces to analyzing each projected state $\Pi_{E_{\text{tot}}}(\rho \otimes \tau_\beta)\Pi_{E_{\text{tot}}}$ separately. $\Pi_{E_{\text{tot}}}$ denotes the projector onto the total-energy eigenspace with energy E_{tot} . To see its action explicitly, expand $\rho \otimes \tau_\beta$ in the joint energy eigenbasis. Since both ρ and τ_β are diagonal in $\{|s\rangle\}$ and $\{|b\rangle\}$, we have

$$\rho \otimes \tau_\beta = \left(\sum_s p(s) |s\rangle\langle s| \right) \otimes \left(\frac{1}{\mathcal{Z}_B} \sum_b e^{-\beta E} |b\rangle\langle b| \right) = \frac{1}{\mathcal{Z}_B} \sum_{s,b} p(s) e^{-\beta E} |s\rangle\langle s| \otimes |b\rangle\langle b|.$$

The total energy of the product eigenstate $|s\rangle \otimes |b\rangle$ is $\epsilon_s + E$. Hence the projector onto total energy E_{tot} can be written as

$$\Pi_{E_{\text{tot}}} = \sum_{s,b: \epsilon_s + E = E_{\text{tot}}} |s\rangle\langle s| \otimes |b\rangle\langle b|.$$

Because the state is already diagonal in this basis, applying the projection simply keeps the terms satisfying $\epsilon_s + E = E_{\text{tot}}$:

$$\Pi_{E_{\text{tot}}}(\rho \otimes \tau_\beta)\Pi_{E_{\text{tot}}} = \frac{1}{\mathcal{Z}_B} \sum_s \sum_{b \in (E_{\text{tot}} - \epsilon_s)} p(s) e^{-\beta E} |s\rangle\langle s| \otimes |b\rangle\langle b|. \quad (287)$$

In contrast to this, looking at each particular total energy subspace is not necessary in the case of fluctuating work because one can average over it (or, more precisely, over E and ϵ_s , the energies of the bath and system) in the final results.

We assume that $\epsilon_s^2 \ll 1/\gamma$ for all s . Effectively this means we can approximate the density of states from Eq. (153) as

$$\begin{aligned} \Omega(E_{\text{tot}} - \epsilon_s) &\simeq \exp\left(\beta(E_{\text{tot}} - \epsilon_s) - \frac{\gamma}{2}(E_{\text{tot}} - \epsilon_s)^2\right) \\ &= \exp\left(\beta E_{\text{tot}} - \beta \epsilon_s - \frac{\gamma}{2} \epsilon_s^2 + \gamma E_{\text{tot}} \epsilon_s - \frac{\gamma}{2} E_{\text{tot}}^2\right) \\ &\quad \underbrace{\phantom{-\frac{\gamma}{2} \epsilon_s^2 + \gamma E_{\text{tot}} \epsilon_s}}_{\ll 1} \\ &= \exp\left(\beta E_{\text{tot}} - \frac{\gamma}{2} E_{\text{tot}}^2\right) \exp(-\beta \epsilon_s + \gamma E_{\text{tot}} \epsilon_s) = \Omega(E_{\text{tot}}) e^{-(\beta - \gamma E_{\text{tot}}) \epsilon_s}. \end{aligned} \quad (288)$$

This gives the size of the subspace with the same E_{tot} and ϵ_s that appears in the expansion

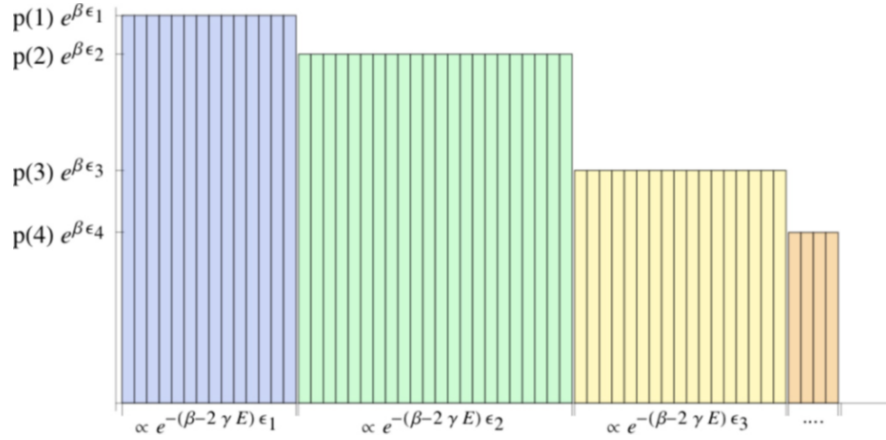


Figure 7: Majorization diagram of a particular subspace of total energy E of the joint system-bath product state, where the system is state $\rho = \sum_s p(s) |s\rangle \langle s|$ diagonal in energy, and with the energy levels β ordered such that $p(k)e^{\beta E_k} \geq p(k+1)e^{\beta E_{k+1}}$.

(287). In each of these subspaces, the projectors have weights $\propto \frac{1}{Z_B} e^{-\beta E_{\text{tot}}} p(s) e^{\beta \epsilon_s}$. We can order these weights in decreasing magnitude as

$$p(1)e^{\beta \epsilon_1} \geq p(2)e^{\beta \epsilon_2} \geq \dots \geq p(n)e^{\beta \epsilon_n},$$

which is usually referred to as β order. This order allows us to draw a majorization diagram for the diagram. An example is shown in Fig. 7.

Now we look at the effect of an arbitrary energy-conserving unitary U to the joint system-bath pair. Within each fixed total-energy subspace E_{tot} , the global energy-conserving constraint $[U, H_S + H_B] = 0$ implies that the dynamics is confined to that subspace and may be described by an arbitrary unitary acting on it. Allowing classical randomness, one can equivalently consider arbitrary convex mixtures of such unitaries (random-unitary channels) applied independently in each E_{tot} block. For states that are diagonal in the relevant energy/microstate basis, these random unitary channels induce bistochastic (doubly stochastic) maps on the corresponding probability vectors, i.e., linear maps represented by matrices with nonnegative entries whose rows and columns each sum to 1, so they preserve normalization and implement pure mixing without creating bias. By the Birkhoff-von Neumann theorem, any bistochastic map can be decomposed as a convex combination of permutation matrices, which correspond to basis-permuting unitaries. Consequently, a transition between two diagonal block states is achievable in a given E_{tot} subspace if and only if the initial block probability distribution majorizes the final one. Since different total-energy sectors are dynamically independent, this majorization condition must hold separately for each E_{tot} .

★ *A full proof of the equivalences random unitaries $\leftrightarrow$ bistochastic maps $\leftrightarrow$ majorization is standard and not the main focus of the paper's finite-bath analysis; accordingly, we follow the paper's level of detail and use these results as established tools.*

Given two probability distributions $\lambda(s)$ and $\eta(s)$ with n elements, ordered such that

$\lambda(s+1) \geq \lambda(s)$ and $\eta(s+1) \geq \eta(s)$, we say $\lambda(s)$ majorizes $\eta(s)$ if it is true that

$$\sum_{s=1}^k \lambda(s) \geq \sum_{s=1}^k \eta(s) \quad \forall k \in \{1, n\}. \quad (289)$$

A transition from $\rho \otimes \tau_\beta$ to $\sigma \otimes \tau_\beta$ is possible exactly when the following holds: for every fixed total-energy sector E_{tot} , if we restrict both states to that sector and look at their probability weights on the corresponding energy-eigenbasis, then the initial sector distribution can be converted into the final one by mixing within that sector. Since energy-conserving unitaries act independently inside each E_{tot} subspace, this sector-by-sector convertibility is required for the overall transformation. In the majorization-diagram language, this means that in each total-energy subspace the distribution associated with $\rho \otimes \tau_\beta$ must majorize the distribution associated with $\sigma \otimes \tau_\beta$; only then can the required bistochastic mixing be implemented in every sector, and hence globally.

We may represent the state restricted to a fixed total-energy subspace E_{tot} by a concave thermomajorization-like curve in the plane. Starting at the origin $(0,0)$, we construct the curve by consecutively joining the points

$$\left(\frac{\sum_{s=1}^k p(s) e^{\gamma E_{\text{tot}} \varepsilon_s}}{\sum_{s=1}^n p(s) e^{\gamma E_{\text{tot}} \varepsilon_s}}, \sum_{s=1}^k e^{-(\beta - \gamma E_{\text{tot}}) \varepsilon_s} \right), \quad k \in 1, \dots, n, \quad (290)$$

where the energy eigenstates $|s\rangle$ are first relabeled in β -order, i.e. so that the relevant weights (equivalently, the slopes of the segments) appear in decreasing order. The vertical coordinate is the cumulative probability weight accumulated after including the first k energy levels in this ordering (normalized so that the final height is 1), while the horizontal coordinate is the cumulative Gibbs weight associated with those same levels, with the effective inverse temperature $\beta - \gamma E_{\text{tot}}$ that arises in a fixed- E_{tot} sector for a finite bath. Each segment of the curve corresponds to a single system energy level s and is obtained by adding, one level at a time, the total probability weight of all compatible joint microstates (s, b) in the majorization diagram. Hence, we have that for each energy E_{tot} there is a slightly different thermomajorization-like criteria, and for a transition to be possible with complete certainty all of them apply.

At $E_{\text{tot}} = 0$, i.e. at the average total energy (after shifting the zero of energy so that $\langle E_{\text{tot}} \rangle = 0$), the finite-bath criterion reduces to the usual thermomajorization criterion of [M. Horodecki and J. Oppenheim, Nat. Commun. 4, 2059 \(2013\)](#). In this sector one has $\beta_{\text{eff}} = \beta - \gamma E_{\text{tot}} = \beta$, so the corresponding diagram coincides with the standard thermomajorization curve. The finite-bath conditions are therefore *strictly stronger*: for a transition to be possible with complete certainty one must satisfy the thermomajorization-like criterion for *every* relevant value of E_{tot} , not only at the typical sector $E_{\text{tot}} = 0$.

For an infinite bath, standard thermomajorization provides a necessary and sufficient condition for state transformations under thermal operations. Equivalently, in the diagonal (classical) setting this can be phrased as follows: access to an ideal infinite bath allows one to implement any Gibbs-preserving stochastic map on the system. With the finite-bath re-

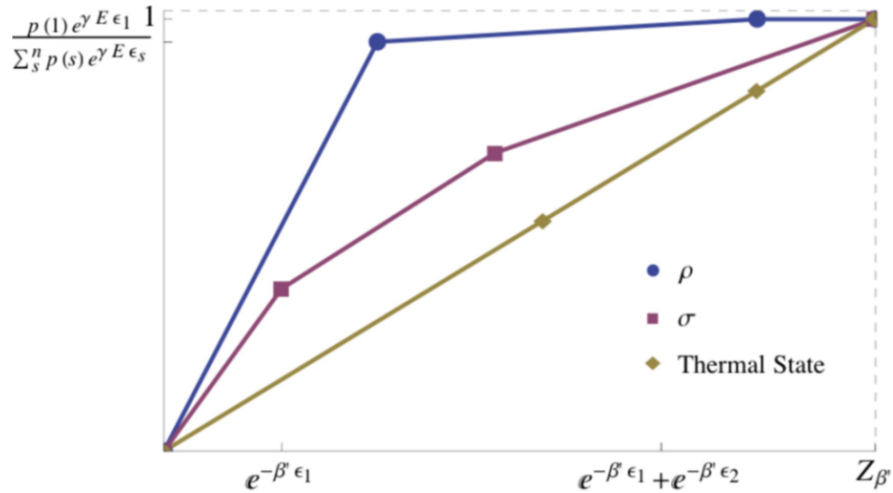


Figure 8: Example of a thermomajorization diagram for three different states of a three-level system, ρ , σ , and the thermal state. the β order in each is different, and at this particular energy the transition $\rho \rightarrow \sigma$ is possible, as the diagram of ρ lies above that of σ .

striction considered here, the admissible transformations still correspond to Gibbs-preserving stochastic maps (since the thermal state remains a fixed point of the dynamics), but the set of implementable maps is *smaller*: not every Gibbs-preserving stochastic matrix can be realized using a finite bath alone. For further explicit examples of analogous restrictions arising from a different finite-bath model, see [J. Scharlau and M. P. Mueller, Quantum 2, 54 \(2018\)](#).

Because each thermomajorization diagram is associated with a particular total energy E_{tot} , a fully conclusive answer requires verifying the condition over the entire range of E_{tot} values that can occur. In general this may be difficult, since for a large bath E_{tot} effectively takes values in a continuous interval. Nevertheless, the analysis simplifies in certain special cases. For instance, for qubit systems the thermomajorization curve has only two segments, and when the β -ordering is the same for the initial and final states the feasibility check often reduces to comparing the initial slope; in many such cases this comparison does not depend on the particular value of E_{tot} . Another simplification occurs for trivial system Hamiltonians ($H_S = 0$): the thermomajorization construction collapses to ordinary majorization because all Gibbs weights are equal. Both simplifications are consistent with the results of [J. Scharlau and M. P. Mueller, Quantum 2, 54 \(2018\)](#), which show that (i) a large class of maps can be implemented on qubits using a finite bath, and (ii) for trivial Hamiltonians (“noisy operations”) it suffices to have a bath of comparable size to the system.

In any case, considering the full range of total energies may not be necessary in particular cases. We know from Sec. 2.2 that the energy distribution Eq. (153) is a Gaussian with tails that cause the range of energies we have to check to be very large. However, the further the energy is from the mean, the less likely it is to occur. Hence this criteria may give a definite answer only if we are willing to make the transition with probability slightly less than 1, and just consider a certain range of total energies. The next subsection is devoted to analyzing the trade-off between the range of energies that need to be checked and the probability of

failure.

2.4.1 Total energy distribution

Given that we have the energy distribution of both bath and system, we can obtain the total energy distribution.

$$p_B(E) \propto \Omega_B(E) e^{-\beta E} = \exp\left(-\frac{\gamma E^2}{2}\right)$$

The probability density of E_{tot} is

$$\begin{aligned} p(E_{\text{tot}}) &= \sum_s p(s) p_B(E_{\text{tot}} - \epsilon_s) = \sum_s p(s) \exp\left(-\frac{\gamma}{2}(E_{\text{tot}} - \epsilon_s)^2\right) \\ &= \sum_s p(s) \exp\left(-\frac{\gamma}{2}(E_{\text{tot}}^2 - E_{\text{tot}} \epsilon_s)\right), \end{aligned} \quad (291)$$

which is a convex mixture of Gaussians with width $1/\sqrt{\gamma}$, each with its center offset by ϵ_s . If we adopt the slightly stronger approximation (compared to Eq. (288)) that

$$\frac{1}{\sqrt{\gamma}} \gg \epsilon_s,$$

then the system energies are negligible on the scale of typical bath-energy fluctuations. In this regime, the total-energy distribution is well approximated by the bath energy distribution alone. Hence, $E_{\text{tot}} \simeq E \simeq \gamma^{-1/2}$ and

$$p(E_{\text{tot}}) \approx p_B(E) \propto e^{-\frac{\gamma}{2}}. \quad (292)$$

Note that the previous approximation was $1/\gamma \gg \epsilon_s^2$. Normalize the distribution to get

$$p(E_{\text{tot}}) = \sqrt{\frac{\gamma}{2\pi}} e^{-\frac{\gamma}{2} E_{\text{tot}}^2}. \quad (293)$$

Now say we want a transformation to happen with probability $1 - \epsilon$. The range of energies $\{E_{\text{min}}, E_{\text{max}}\}$ that have to be checked is then given by the integral equation

$$1 - \epsilon = \int_{E_{\text{min}}}^{E_{\text{max}}} \sqrt{\frac{\gamma}{2\pi}} e^{-\frac{\gamma}{2} E^2} dE = 2\sqrt{\frac{\gamma}{2\pi}} \int_0^{E_{\text{max}}} e^{-\frac{\gamma}{2} E^2} dE. \quad (294)$$

We may take $-E_{\text{min}} = E_{\text{max}} = E^*$; then

$$1 - \epsilon = 1 - 2\sqrt{\frac{\gamma}{2\pi}} \int_{E^*}^{\infty} e^{-\frac{\gamma}{2} E^2} dE \implies \epsilon = 2\sqrt{\frac{\gamma}{2\pi}} \int_{E^*}^{\infty} e^{-\frac{\gamma}{2} E^2} dE. \quad (295)$$

Substitute

$$x = \sqrt{\frac{\gamma}{2}} E \implies dE = \frac{\sqrt{2}}{\sqrt{\gamma}} dx.$$

Then

$$\epsilon = \frac{2}{\sqrt{\pi}} \int_{\sqrt{\gamma/2}E^*}^{\infty} e^{-x^2} dx.$$

Using the standard definition

$$\text{erf}(a) = \frac{2}{\sqrt{\pi}} \int_0^a e^{-t^2} dt \implies 1 - \text{erf}(a) = \frac{2}{\sqrt{\pi}} \int_a^{\infty} e^{-t^2} dt, \quad (296)$$

we get

$$\epsilon = 1 - \text{erf}\left(\sqrt{\frac{\gamma}{2}}E^*\right). \quad (297)$$

For large x we can approximate the error function as

$$\text{erf}(x) \simeq 1 - \frac{e^{-x^2}}{\sqrt{\pi}x}. \quad (298)$$

Hence we can approximate

$$\epsilon \simeq \sqrt{2} \frac{e^{-\frac{\gamma}{2}E^{*2}}}{\sqrt{\pi\gamma}E^*}. \quad (299)$$

With the definitions and normalization used in Eq. (294), the correctly derived tail expression gives (299), which is not consistent with the paper's (C11)-(C13) as printed (the lower limit/prefactors imply a different scaling).

We can also define two minimum and maximum effective temperatures given by

$$\beta_+ = \beta - \gamma E_{\max}, \quad (300)$$

$$\beta_- = \beta - \gamma E_{\min}. \quad (301)$$

So we showed that, for an energy range $E_{\text{tot}} \in [-E^*, E^*]$, the probability of failure ϵ is approximately given by Eq. (299).

Even with this restriction we cannot give final answers for general state transitions, but the problem becomes manageable when either the initial or the final state is thermal. In that case, the condition simplifies because we only need to check the extreme ends of the total-energy distribution, namely, the two endpoints of the energy window we keep after allowing a small probability of failure. With this allowed failure probability, we can then compute (i) the maximum deterministic work that can be extracted when taking $\rho = \sum_s p(s) |s\rangle \langle s|$ to the thermal state, and (ii) the minimum deterministic work needed for the reverse process, i.e., creating $\rho = \sum_s p(s) |s\rangle \langle s|$ from a thermal state. Therefore, we focus on these two specific cases.

2.4.2 Deterministic work extraction and expenditure

Denote the system Hamiltonian by

$$H_S = \sum_s \varepsilon_s |s\rangle\langle s|,$$

the thermal (Gibbs) state by

$$\tau_\beta = \frac{1}{\mathcal{Z}_\beta} e^{-\beta H_S},$$

and the partition function by

$$\mathcal{Z}_\beta = \sum_s e^{-\beta \varepsilon_s}.$$

With this notation in place, the authors state two theorems, whose derivations we present after.

Theorem 4. (Work extraction) The maximal deterministic work that one can extract from state $P(s)$ is, up to error ϵ , given by $W_{\text{ext}}^\epsilon = F_{\min}^{\beta_-}(\rho)$, where $\beta_- = \beta - \gamma E^*$ and

$$F_{\min}^\beta(\rho) = \frac{1}{\beta} \log(\mathcal{Z}_\beta) - \frac{1}{\beta} \log\left(\sum_s e^{-\beta \varepsilon_s} P(s)^0\right), \quad (302)$$

and the relation between ϵ and E^* is given by (299). Here $x^0 = 1$ if $x > 1$ and $x^0 = 0$ if $x = 0$, as the paper notes.

Proof of Theorem 4. In Appendix C the authors introduce an effective inverse temperature for each fixed total-energy block E ,

$$\beta' := \beta - \gamma E, \quad (303)$$

which comes from approximating ratios of bath degeneracies around that block energy. Concretely, one uses an approximation of the form

$$\frac{\Omega(E - x)}{\Omega(E)} \approx e^{-\beta' x},$$

so inside a given E -block the bath behaves as if it had inverse temperature β' .

Fix a block energy E . The single-block work extraction task is: find the largest W such that the transition

$$\rho \otimes |0\rangle\langle 0| \rightarrow \tau \otimes |W\rangle\langle W| \quad (304)$$

is possible *within that E -block*. Here $|0\rangle$ and $|W\rangle$ are weight energy eigenstates. Since only these two weight levels matter for this transition, we ignore all other weight energies and treat the weight as an effective two-level system with gap W . The target state τ is thermal;

in the block picture the relevant temperature is β' , so the natural choice is

$$\tau = \tau_{\beta'} := \frac{e^{-\beta' H_S}}{\mathcal{Z}_{\beta'}}.$$

(If the paper writes τ without the prime, it is using the same symbol but the block calculation is still done at β' .)

Now we use a standard thermomajorization fact: when the final state is thermal, its thermomajorization curve is a straight line. Because the initial curve is concave, checking feasibility reduces to comparing only the endpoints. In this setting, the endpoint is captured by the *rank* (dimension) of the restriction of the global state to the fixed total-energy block.

For the initial global state $\rho \otimes |0\rangle\langle 0| \otimes \tau_\beta$, a system level s contributes in the E -block only if $p(s) > 0$; we write this support indicator as $p(s)^0$ (equal to 1 if $p(s) > 0$, and 0 otherwise). For each such s , the bath must have energy $E - \epsilon_s$, giving $\Omega(E - \epsilon_s)$ bath microstates. Hence the initial block rank is

$$d_{\text{in}}(E) = \sum_s \Omega(E - \epsilon_s) p(s)^0. \quad (305)$$

Using (288), $\Omega(E - \epsilon_s) \approx \Omega(E) e^{-(\beta - \gamma E)\epsilon_s} = \Omega(E) e^{-\beta' \epsilon_s}$, so

$$d_{\text{in}}(E) \approx \Omega(E) \sum_s e^{-\beta' \epsilon_s} p(s)^0. \quad (306)$$

For the final global state $\tau \otimes |W\rangle\langle W| \otimes \tau_\beta$, if the weight stores energy W , then in the same total-energy block the bath energy must be lower by W . Thus for each system level s the bath contributes $\Omega(E - \epsilon_s - W)$, and

$$d_{\text{fin}}(E) = \sum_s \Omega(E - \epsilon_s - W). \quad (307)$$

Applying the same approximation,

$$\Omega(E - \epsilon_s - W) \approx \Omega(E) e^{-\beta'(\epsilon_s + W)} = \Omega(E) e^{-\beta' W} e^{-\beta' \epsilon_s},$$

gives

$$d_{\text{fin}}(E) \approx \Omega(E) e^{-\beta' W} \sum_s e^{-\beta' \epsilon_s} p(s)^0 = \Omega(E) e^{-\beta' W} \mathcal{Z}_{\beta'}. \quad (308)$$

Feasibility requires $d_{\text{in}}(E) \leq d_{\text{fin}}(E)$. The maximum work in that block is obtained when this is tight (endpoint saturation), so

$$\sum_s e^{-\beta' \epsilon_s} p(s)^0 = e^{-\beta' W} \sum_s e^{-\beta' \epsilon_s} = e^{-\beta' W} \mathcal{Z}_{\beta'}.$$

Solving for W yields

$$e^{-\beta' W} = \frac{\sum_s e^{-\beta' \epsilon_s} p(s)^0}{\mathcal{Z}_{\beta'}}, \quad W = \frac{1}{\beta'} \left(\log \mathcal{Z}_{\beta'} - \log \sum_s e^{-\beta' \epsilon_s} p(s)^0 \right). \quad (309)$$

The paper denotes this expression using a min-free-energy quantity $F_{\min}^{\beta'}(\rho)$.

$$W = F_{\min}^{\beta'}(\rho). \quad (310)$$

Equation (309) gives the maximum work *if we knew* we were in the block E . For deterministic work extraction, however, the same work value must be produced regardless of which total-energy block actually occurs. This means we must choose a single W that works for all relevant E , so the achievable deterministic work is limited by the worst block, i.e. by the minimum of the blockwise value (309) over the allowed range of E (set by the failure probability ϵ in (299)). The paper then assumes (or argues in the regime of interest) that the relevant expression is monotone in E , so that the minimum is reached at the smallest allowed E , equivalently at $\beta' = \beta_-$ as in (301). Under that assumption, the optimal deterministic extractable work is

$$W_{\text{ext}} = F_{\min}^{\beta_-}(\rho), \quad (311)$$

with failure probability ϵ determined by (299). Hence the proof is complete.

The next result involves the cost of creating an arbitrary state from a thermal one.

Theorem 5. (Work of formation) In the transition

$$\tau_{\beta} \otimes |W\rangle \langle W| \rightarrow \rho \otimes |0\rangle \langle 0|, \quad (312)$$

the minimum possible value of W is given by

$$W_{\text{for}}^{\epsilon} = \frac{\beta}{\beta_+} F_{\max}^{\beta}(\rho) - \frac{1}{\beta_+} \log \left(\frac{\mathcal{Z}_{\beta}}{\mathcal{Z}_{\beta_+}} \sum_s P(s) e^{(\beta - \beta_+) \epsilon_s} \right), \quad (313)$$

where $\beta_+ = \beta + \gamma E^*$ and

$$F_{\max}^{\beta}(\rho) = \frac{1}{\beta} \log \left(\max_s P(s) e^{\beta \epsilon_s} \right). \quad (314)$$

Again in the infinite limit where $\gamma \rightarrow 0$, we have that $\beta_+ \rightarrow \beta$, and thus we have $W_{\text{for}}^{\epsilon} \rightarrow F_{\max}^{\beta}(\rho)$. Unlike in Eq. (302), the finite-size expression of Eq. (314) does not exactly correspond to a fluctuating temperature, as an extra term is present.

Proof of Theorem 5. Now, for a given total energy E , we want the lowest W for which the transition

$$\tau \otimes |W\rangle \langle W| \rightarrow \rho \otimes |0\rangle \langle 0|, \quad (315)$$

is possible. From Eq. (290), the thermomajorization curve in the E -block is built from segments whose horizontal length is

$$\Delta x_s = e^{-\beta' \epsilon_s},$$

and whose vertical increment is

$$\Delta y_s = \frac{p(s)e^{\gamma E \epsilon_s}}{\sum_l p(l)e^{\gamma E \epsilon_l}}.$$

Hence the slope of the s -th segment is

$$m_s = \frac{\Delta x_s}{\Delta y_s} = \frac{p(s)e^{\gamma E \epsilon_s}}{\sum_l p(l)e^{\gamma E \epsilon_l}} \cdot e^{\beta' \epsilon_s} = \frac{p(s)e^{\beta \epsilon_s}}{\sum_l p(l)e^{\gamma E \epsilon_l}}, \quad (316)$$

because $\beta' = \beta - \gamma E$. The β -ordering is therefore the ordering by decreasing $p(s)e^{\beta \epsilon_s}$. Now take the initial state in the fixed- E block:

$$\tau \beta \otimes |W\rangle \langle W|.$$

Treat system+weight as the effective system. Its energy levels are

$$\epsilon_s + W,$$

and its probabilities are

$$q_s = \frac{e^{-\beta \epsilon_s}}{\mathcal{Z}_\beta}.$$

Apply the same themomajorization recipe to this composite system. The vertical increment for segment s is

$$\Delta y_s^{\text{in}} = \frac{q_s e^{\gamma E (\epsilon_s + W)}}{\sum_l q_l e^{\gamma E (\epsilon_l + W)}}.$$

Substitute $q_s = e^{-\beta \epsilon_s} / \mathcal{Z}_\beta$:

$$\Delta y_s^{\text{in}} = \frac{e^{-\beta \epsilon_s} e^{\gamma E (\epsilon_s + W)}}{\sum_l e^{-\beta \epsilon_l} e^{\gamma E (\epsilon_l + W)}} = \frac{e^{-(\beta - \gamma E) \epsilon_s}}{\sum_l e^{-(\beta - \gamma E) \epsilon_s}} = \frac{e^{-\beta' \epsilon_s}}{\mathcal{Z}_{\beta'}}.$$

The horizontal increment is

$$\Delta x_s^{\text{in}} = e^{-\beta' (\epsilon_s + W)} = e^{-\beta' W} e^{-\beta' \epsilon_s}.$$

Therefore the slope of every segment is

$$m_{\text{in}} = \frac{\Delta y_s^{\text{in}}}{\Delta x_s^{\text{in}}} = \frac{e^{\beta' W}}{\mathcal{Z}_{\beta'}}. \quad (317)$$

This is independent of s , so the initial thermomajorization curve is a straight line.

For the final state $\rho \otimes |0\rangle \langle 0|$, the weight contributes zero energy, so the effective system

energies are just ϵ_s with probabilities $p(s)$. From the general slope formula above,

$$m_s^{\text{fin}} = \frac{p(s)e^{\beta\epsilon_s}}{\sum_l p(l)e^{\gamma E_{\epsilon_l}}}. \quad (318)$$

Because the final curve is β -ordered, its first segment is the steepest one, so

$$m_{\text{fin}}^{(1)} = \frac{\max_s p(s)e^{\beta\epsilon_s}}{\sum_l p(l)e^{\gamma E_{\epsilon_l}}}. \quad (319)$$

Since the initial curve is a straight line and the final curve is concave with nonincreasing slopes, the transition is possible iff the straight line lies above the final curve. For a concave piecewise-linear curve, that is equivalent to demanding that the straight line slope be at least the slope of the first segment. At the minimum work cost, this condition is tight, so one sets

$$m_{\text{in}} = m_{\text{fin}}^{(1)}.$$

$$\frac{e^{\beta'W}}{Z_{\beta'}} = \frac{\max_s p(s)e^{\beta\epsilon_s}}{\sum_l p(l)e^{\gamma E_{\epsilon_l}}}. \quad (320)$$

Take logs:

$$\beta'W - \log(Z_{\beta'}) = \log\left(\max_s p(s)e^{\beta\epsilon_s}\right) - \log\left(\sum_l p(l)e^{\gamma E_{\epsilon_l}}\right).$$

So

$$W(E) = \frac{1}{\beta'} \left[\log(Z_{\beta'}) + \log\left(\max_s p(s)e^{\beta\epsilon_s}\right) - \log\left(\sum_l p(l)e^{\gamma E_{\epsilon_l}}\right) \right]. \quad (321)$$

Now use the paper's definition

$$F_{\text{max}}^{\beta}(\rho) = \frac{1}{\beta} \log \max_s p(s)e^{\beta\epsilon_s}.$$

Substitute this into the expression above:

$$W(E) = \frac{1}{\beta'} \left[\log(Z_{\beta'}) + \beta F_{\text{max}}^{\beta}(\rho) - \log\left(\sum_l p(l)e^{\gamma E_{\epsilon_l}}\right) \right].$$

Equivalently,

$$W(E) = \frac{\beta}{\beta'} F_{\text{max}}^{\beta}(\rho) - \frac{1}{\beta'} \log\left(\frac{Z_{\beta}}{Z_{\beta'}} \sum_l p(l)e^{\gamma E_{\epsilon_l}}\right) + \frac{1}{\beta'} \log(Z_{\beta}).$$

Since $\beta - \beta' = \gamma E$, this is

$$W(E) = \frac{\beta}{\beta'} F_{\max}^{\beta}(\rho) - \frac{1}{\beta'} \log \left(\frac{\mathcal{Z}_{\beta}}{\mathcal{Z}_{\beta'}} \sum_l p(l) e^{(\beta - \beta') \epsilon_l} \right) + \frac{1}{\beta'} \log(\mathcal{Z}_{\beta}). \quad (322)$$

There is a clear notational mismatch in the derivation of Theorem 5. Starting from Eqs. (C20)-(C22) of the paper, one obtains a term $\log \mathcal{Z}_{\beta'}$, and rewriting it as $\log(\mathcal{Z}_{\beta'}/\mathcal{Z}_{\beta})$ necessarily leaves an additional $+\log \mathcal{Z}_{\beta}$ term. Therefore, the theorem as printed follows only if this extra $\log \mathcal{Z}_{\beta}$ is implicitly absorbed into the definition of $F_{\max}^{\beta}(\rho)$. Since Eq. (27) of the paper does not include such a factor, the derived expression is not fully consistent with the formula stated in the paper.

We will proceed using the paper's equation and assume that it holds.

As Theorem 4, one keeps only total energies $E \in [E_{\min}, E_{\max}]$ that contain probability $(1 - \epsilon)$, with the window chosen via Eq. (294). Since the work must succeed for every kept block, the deterministic formation cost is

$$W_{\text{for}}^{\epsilon} = \max_{E \in [E_{\min}, E_{\max}]} W(E). \quad (323)$$

At this point the appendix simply states that $W(E)$ is monotonically increasing in E , which is not generally true. But still using this therefore the maximum is achieved at $E = E_{\max}$. So using the stated claim, put

$$\beta_{+} = \beta' \big|_{E=E_{\max}},$$

and obtain

$$W_{\text{for}}^{\epsilon} = \frac{\beta}{\beta_{+}} F_{\max}^{\beta}(\rho) - \frac{1}{\beta_{+}} \log \left(\frac{\mathcal{Z}_{\beta}}{\mathcal{Z}_{\beta_{+}}} \sum_s p(s) e^{(\beta - \beta_{+}) \epsilon_s} \right). \quad (324)$$

Therefore, the proof is complete.

In this case because the bath is finite, the total energy E_{tot} of system+bath+weight fluctuates and has nonzero Gaussian tails; energy conservation then forces the protocol to work separately in each fixed- E_{tot} sector. Demanding perfectly deterministic work would require the same work value to be achievable for all sectors, including arbitrarily rare tail events, which effectively pushes the model toward an infinite bath / infinite time limit. This is precisely the kind of only attainable in a limit behavior associated with the third law (unattainability): perfect determinism would require accessing degrees of freedom that lie arbitrarily far in the tails and are, operationally, only reachable with unbounded resources or time. Therefore the authors introduce a small failure probability ϵ by cutting off the tails of $p(E_{\text{tot}})$, and analyze deterministic work only on the high probability energy window that remains.

In the standard smoothed one-shot framework, the small error does not come from the bath's energy tails. Instead, you allow the target state ρ to be approximated by a nearby

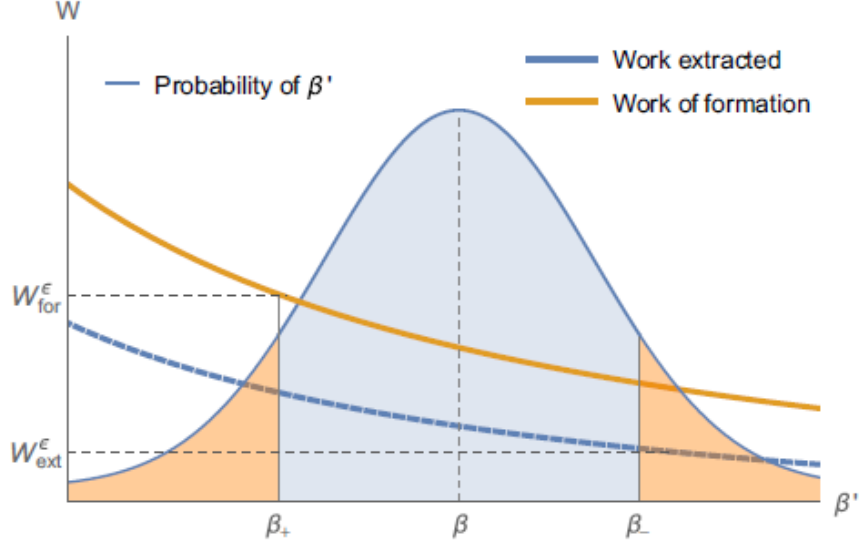


Figure 9: Curves for the maximum work extracted and minimum work of formation for a given state ρ , given each particular total energy. The distribution of total energies (and hence the distribution of $\beta' = \beta - \gamma E$) is the superposed Gaussian curve, for which a cut in the tails gives the values of effective temperatures β_- and β_+ that determine the work of formation and the extractable work given some error probability, whose value is equal to the orange region.

state $\rho_{\epsilon'}$ within trace distance $\|\rho_{\epsilon'} - \rho\|_1 \leq \epsilon'$, and then you optimize the relevant quantity over the ball. Both effects can be included at once: with energy-tail error ϵ (fixing $\beta_{\pm}$) and state-smoothing error ϵ' , the optimal deterministic work becomes

$$W_{\epsilon, \epsilon'} = \frac{1}{\beta_-} \sup_{\rho_{\epsilon'}} F_{\min}^{\beta_-}(\rho_{\epsilon'}), \quad (325)$$

for the extractable one and

$$W_{\epsilon, \epsilon'} = \frac{1}{\beta_+} \log\left(\frac{\mathcal{Z}_{\beta_+}}{\mathcal{Z}_{\beta}}\right) + \frac{1}{\beta_+} \inf_{\rho_{\epsilon'}} \left(F_{\max}^{\beta}(\rho_{\epsilon'}) + \log\left(\frac{1}{\sum_s p_{\epsilon'}(s) e^{(\beta - \beta_+) \epsilon_s}}\right) \right), \quad (326)$$

for the work of formation. In Fig 9 they show the maximum extractable work and the minimum work of formation for a particular example as a function of the modifies temperature β' .

2.5 Conclusion

This paper shows that the usual free-energy form of the second law is exact only in the ideal limit of an infinite heat bath. For a bath with finite heat capacity C , the maximum

extractable work is reduced by a correction term, so that the leading-order bound becomes

$$\langle W \rangle \leq -\Delta F - \frac{(\Delta S)^2}{2\beta C} + O(C^{-2}).$$

The main strength of the paper is that this correction is derived in a model-independent way: at leading order, the bath enters only through its temperature and heat capacity, without any need to assume a particular microscopic structure. This makes the result more general and, in many physically relevant cases, more meaningful than bounds formulated in terms of bath dimension alone.

A second important conclusion is that finite baths destroy ideal thermodynamic reversibility. In the infinite-bath limit, work extraction and work formation are governed by the same free-energy quantity and can coincide in the reversible limit. Here this symmetry is lost: finite heat capacity introduces unavoidable dissipation, and the work cost of creating a state is strictly larger than the work that can be recovered from it. The paper makes this especially clear in Landauer-type processes, where second order information quantities, in particular the varentropy, appear naturally in the optimal work expressions. This is one of the most interesting conceptual points of the paper, since it shows that once one moves away from the asymptotic regime, ordinary entropy alone is no longer sufficient.

The deterministic work part of the paper is more limited. The analysis is restricted to states diagonal in the energy basis, it requires introducing a small failure probability by discarding the tails of the total energy distribution, and it yields closed form results only in special cases, mainly when either the initial or the final state is thermal. Thus, while the paper does provide finite bath corrections to the usual single-shot $F_{\min}$ and $F_{\max}$ expressions, it does not give a complete characterization of arbitrary deterministic state transformations with a finite bath. In this sense, the fluctuating work results are stronger and more general than the deterministic ones.

Overall, the paper makes a strong and useful contribution: it identifies finite heat capacity as the key thermodynamic parameter controlling deviations from the standard second law, and it shows that irreversibility is not merely a technical imperfection but a fundamental feature of finite bath thermodynamics. At the same time, the work is best viewed as a leading-order and partial theory rather than a complete framework. Important open questions remain, including the role of coherence, the characterization of general deterministic transitions, the implications for finite time heat engines, and the form of higher order corrections that depend on the microscopic details of the bath. For these reasons, the paper is most convincing as a general account of why finite baths necessarily modify the second law, and less convincing as a final answer to finite bath thermodynamics in full generality.